

Parameter List

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10.1 Section Safety

DANGER

Heed the safety messages in this manual. Failure to comply will result in death or serious injury.

The operating company is responsible for any injuries or equipment damage resulting from failure to heed the warnings in this manual.

10.2 How to Read the Parameter List

◆ Icons and Terms Used to Represent Control Modes

The following table lists terms and symbols used in this section.

Icon	Description
	Parameter is available when operating the drive with V/f Control.
	Parameter is available when operating the drive with Closed Loop V/f Control.
	Parameter is available when operating the drive with Open Loop Vector Control.
	Parameter is available when operating the drive with Closed Loop Vector Control.
	Parameter is available when operating the drive with Advanced Open Loop Vector Control.
	Parameter is available when operating the drive with Open Loop Vector Control for PM motors.
	Parameter is available when operating the drive with Advanced Open Loop Vector Control for PM motors.
	Parameter is available when operating the drive with Closed Loop Vector Control for PM motors.
	Parameter is available when operating the drive with EZ Open Loop Vector Control.
Hex.	MEMOBUS addresses used to change parameters over network communication are represented in hexadecimal numbers.
RUN	Parameter can be changed during run.
EXP	Parameters that are enabled in Expert Mode only. <i>*1</i>

*1 When $A1-01 = 3$ [Access Level Selection = Expert Level], Expert Mode parameters can be displayed on and set with the keypad. Refer to [Table 10.1](#) for more information on parameters available in Expert Mode.

Note:

Icons in gray indicate that the parameter is not available in control mode.

10.3 Parameter Groups

Represents the type of product parameters.

Parameters	Name
A1	Initialization
A2	User Parameters
b1	Operation Mode Selection
b2	DC Injection Braking and Short Circuit Braking
b3	Speed Search
b4	Timer Function
b5	PID Control
b6	Dwell Function
b7	Droop Control
b8	Energy Saving
b9	Zero Servo
C1	Accel and Decel Times
C2	S-Curve Characteristics
C3	Slip Compensation
C4	Torque Compensation
C5	Automatic Speed Regulator Automatic Speed Regulator)
C6	Carrier Frequency
d1	Frequency Reference
d2	Reference Limits
d3	Jump Frequency
d4	Frequency Reference Hold and Up/Down 2 Function
d5	Torque control
d6	Field Weakening and Field Forcing
d7	Offset Frequency
E1	V/f Pattern for Motor 1
E2	Motor Parameters
E3	V/f Pattern for Motor 2
E4	Motor 2 Parameters
E5	PM Motor Settings
E9	Motor Setting
F1	PG Speed Control Card Encoder
F2	Analog Input Option
F3	Digital Input Option
F4	Analog Monitor Option
F5	Digital Output Option
F6	Communication Options
F7	Communication Options
H1	Multi-function digital input

Parameters	Name
H2	Multi-function digital output
H3	Multi-Function Analog In
H4	Analog Outputs
H5	MEMOBUS/Modbus communications
H6	Pulse Train Input/Output
H7	Virtual Multi-Function I/O
L1	Motor Protection
L2	Momentary Power Loss Ride-Thru
L3	Stall Prevention
L4	Speed Detection
L5	Auto Restart
L6	Detection of Overtorque/Undertorque
L7	Torque Limit
L8	Drive Protection
L9	Drive Protection 2
n1	Hunting prevention function
n2	Speed feedback detection control
n3	High Slip Braking (HSB) and Overexcitation Braking
n4	Adv Vect Tune
n5	Feed Forward Control
n6	Online Tuning
n7	EZ Drive
n8	PM Motor Control Tuning
o1	Keypad Display Selection
o2	Keypad Operation
o3	Copy Function
o4	Maintenance Mon Settings
o5	Log Function
q	DriveWorksEZ Parameters
r	DWEZ Connection 1-20
T0	Tuning Mode Selection
T1	Induction Motor Auto-Tuning
T2	PM Motor Auto-Tuning
T3	ASR and Inertia Tuning
T4	EZ Tuning
U1	Operation Status Monitor
U2	Fault Trace
U3	Fault History
U4	Maintenance Monitors

Parameters	Name
U5	PID Monitors
U6	Operation Status Monitors

Parameters	Name
U8	DriveWorksEZ Monitors

◆ List of Parameters Enabled in Expert Mode

Table 10.1 Parameters Enabled in Expert Mode

No.	Name	No.	Name
A1-11	Firmware update lock	b8-21	PM E-Save Search Gain
b1-21	CLV Start Selection	b8-22	PM E-Save Search LPF Cutoff Freq
b1-35	Multi-Function DI dead band time	b8-23	PM E-Save Search Limit
b3-06	Speed Estimation Current Level 1	b8-24	PM E-Save High Freq ACR Gain
b3-07	Speed Estimation Current Level 2	b8-25	PM E-Save Search Start level
b3-08	Speed Estimation ACR P Gain	b8-26	PM E-Save Power Setpoint
b3-09	Speed Estimation ACR I Time	b8-28	Save Energy Priority Function
b3-10	Speed Estimation Detection Gain	C3-16	Vout Modulation Limit Start Lvl
b3-17	Speed Est. Retry Current Level	C3-17	Vout Modulation Limit Max Level
b3-18	Speed Est. Retry Detection Time	C3-18	Output Voltage Limit Level
b3-25	Speed Search Wait Time	C3-28	Adaptive Slip Contrl Mode Select
b3-26	Direction Determining Level	C4-19	T-ripple suppress freq
b3-27	Start Speed Search Select	C4-20	vol compensation adjust 1
b3-29	Speed Search Back-EMF Threshold	C4-21	vol compensation adjust 2
b3-31	Search current Level 1	C5-17	Motor Inertia
b3-32	Search current Level 2	C5-18	Load Inertia Ratio
b3-33	Spd Search during UV Selection	C5-29	Speed Control Response
b3-35	Voltage Detection Low Level	C5-37	Motor 2 Inertia
b3-36	Wait Restart Level	C5-38	Motor 2 Load Inertia Ratio
b3-55	Current Increment Time	C5-50	Notch Filter Frequency
b4-03	H2-01 ON Delay Time	C5-51	Notch Filter Bandwidth
b4-04	H2-01 OFF Delay Time	E1-11	Mid Point B Frequency
b4-05	H2-02 ON Delay Time	E1-12	Mid Point B Voltage
b4-06	H2-02 OFF Delay Time	E1-13	Base Voltage
b4-07	H2-03 ON Delay Time	E2-09	Motor Mechanical Loss
b4-08	H2-03 OFF Delay Time	E3-11	Motor 2 Mid Point B Frequency
b5-08	PID Primary Delay Time Constant	E3-12	Motor 2 Mid Point B Voltage
b8-02	Energy Saving Gain	E3-13	Motor 2 Base Voltage
b8-03	Energy Saving Filter Time	E4-09	Motor 2 Mechanical Loss
b8-04	Energy Saving Coefficient Value	E5-25	Polarity Estimation Reversal
b8-05	Power Detection Filter Time	F3-10	Terminal D0 Function Selection
b8-06	Search Operation Voltage Limit	F3-11	Terminal D1 Function Selection
b8-16	PM E-Save Coefficient Ki	F3-12	Terminal D2 Function Selection
b8-17	PM E-Save Coefficient Kt	F3-13	Terminal D3 Function Selection
b8-18	E-Save d-axis Current FilterTime	F3-14	Terminal D4 Function Selection
b8-19	E-Save Search Injection Freq	F3-15	Terminal D5 Function Selection
b8-20	PM E-Save Search Width	F3-16	Terminal D6 Function Selection

10.3 Parameter Groups

No.	Name
F3-17	Terminal D7 Function Selection
F3-18	Terminal D8 Function Selection
F3-19	Terminal D9 Function Selection
F3-20	Terminal DA Function Selection
F3-21	Terminal DB Function Selection
F3-22	Terminal DC Function Selection
F3-23	Terminal DD Function Selection
F3-24	Terminal DE Function Selection
F3-25	Terminal DF Function Selection
H2-60	Terminal M1-M2 Function B Select
H2-61	Terminal M1-M2 Logical Operation
H2-62	Terminal M1-M2 Delay Time
H2-63	Terminal M3-M4 Function B Select
H2-64	Terminal M3-M4 Logical Operation
H2-65	Terminal M3-M4 Delay Time
H2-66	Terminal M5-M6 Function B Select
H2-67	Terminal M5-M6 Logical Operation
H2-68	Terminal M5-M6 Delay Time
H2-69	Digital Output4 Secondary Func
H2-70	DigitalOutput4 Logical Operation
H2-71	Digital Output 4 Delay Time
H2-72	Digital Output 5 Secondary Func
H2-73	DigitalOutput5 Logical Operation
H2-74	Digital Output 5 Delay Time
H5-17	Busy Enter Selection
H5-20	Comm. Parameters Activation Sel
H7-00	Virtual Multi-Function I/O
H7-01	Virtual Multi-Function Input 1
H7-02	Virtual Multi-Function Input 2
H7-03	Virtual Multi-Function Input 3
H7-04	Virtual Multi-Function Input 4
H7-10	Virtual Multi-Function Output 1
H7-11	Virtual Output 1 Delay Time
H7-12	Virtual Multi-Function Output 2
H7-13	Virtual Output 2 Delay Time
H7-14	Virtual Multi-Function Output 3
H7-15	Virtual Output 3 Delay Time
H7-16	Virtual Multi-Function Output 4
H7-17	Virtual Output 4 Delay Time
H7-30	Virtual Analog Input Selection
H7-31	Virtual Analog Input Gain
H7-32	Virtual Analog Input Bias
L2-06	KEB Deceleration Time

No.	Name
L2-07	KEB Acceleration Time
L2-08	Frequency Gain at KEB Start
L2-09	KEB Minimum Frequency Level
L2-10	KEB Detect Time (Min KEB Time)
L2-11	DC Bus Vol Setpoint during KEB
L2-29	KEB Method Selection
L2-30	KEB Zero Speed Operation
L2-31	KEB Start Voltage Offset Level
L3-20	DC Bus Voltage Adjustment Gain
L3-21	Accel/Decel Rate Calculate Gain
L3-24	Motor Accel Time for Inertia Cal
L3-25	Load Inertia Ratio
L3-26	Additional DC Bus Capacitors
L3-34	Torque Limit Delay Time
L3-35	IntDecSpdAgrWdth
L3-37	Current Limit P Gain duringAccel
L3-38	Current Limit I Time duringAccel
L7-35	DeratingTrqLimForLowFreq&Regratn
L7-36	Ope Freq band for deratingTrqLim
L8-20	CF/STPo Fault Detection Select
L8-51	STPo I Detection Level
L8-52	STPo Integration Level
L8-53	STPo Integration Time
L8-54	STPo Id Diff Detection
L8-56	Stl Act Time
L8-57	Stl Retry Count
L8-90	STPo Detection Level
L8-93	LSo Detection Time at Low Speed
L8-94	LSo Detection Level at Low Speed
L8-95	Average LSo Freq at Low Speed
L9-16	FAN1 Detect Time
n1-02	Hunting Prevention Gain Setting
n1-03	Hunting Prevention Time Constant
n1-05	HuntingPreventionGain whileinRev
n1-08	Leak cur antivib
n1-13	DC Bus Stabilization Control
n1-14	DC Bus Stabilization Time
n1-15	Voltage Calibration Select
n1-17	Hunt Prev Time
n3-01	HSB Deceleration Frequency Width
n3-02	High-Slip Braking Current Limit
n3-03	HSB Dwell Time at Stop
n3-04	High-Slip Braking Overload Time

No.	Name
n3-14	HarmInj@HiFlxBrk
n4-74	Limit of Flux loop
n5-04	Spd Response F
n6-05	Online Tuning Gain
n6-11	online resister tuning
n7-01	Flux Estimation Cut-off Freq
n7-05	Torque Control Response Gain
n7-07	PLL response 1
n7-08	PLL response 2
n7-10	Sensorless Switchover StartSpeed
n8-01	Init Rotor Position Est Current
n8-02	Pole Attraction Current
n8-04	Polar Attraction Time
n8-11	Induction Volt Estimation Gain 2
n8-14	Polarity Compensation Gain 3
n8-15	Polarity Compensation Gain 4
n8-21	Motor Ke Gain
n8-37	High Freq Injection Amplitude
n8-41	HFI Overlap Pole Detection Pgain
n8-42	HFI Overlap Pole Detection iTime
n8-49	d-Axis Cur forHighEfficiencyCont
n8-54	VoltErrorCompensationTime Const
n8-62	Output Voltage Limit (for PM)
n8-65	SpdFdbkDetectCtrlGainduringOVSup
n8-69	Speed Calculation Gain
n8-72	Speed Estimation Method Select

No.	Name
n8-74	LghtLoadCurLvl 1
n8-75	LghtLoadCurLvl 2
n8-77	IPM HiEffCtrLev2
n8-78	MedLoad Id Level
n8-84	InitPolarityEstimationTimeoutCur
n8-94	Selection of Recognition Criteria
n8-95	Observer Estimation TimeConstant
U1-50	Virtual Ai
U1-91	Output Voltage
U2-21	STO Det Sts
U5-21	AutoCalEnSav Coef Ki
U5-22	AutoCalEnSav Coef Kt
U6-07	q-Axis ACR Output
U6-08	d-Axis ACR Output
U6-09	AdvPhase Compen $\Delta\theta$
U6-10	ContAxisDeviation $\Delta\theta$
U6-13	FluxPosDetect sensor
U6-14	FluxPosEstimObserver
U6-17	ESav Coeff Value
U6-25	Fdbk Control Output
U6-26	Feed Fwd Cont Output
U6-27	FF Estimate SPD
U6-36	Comm Errors-HOST
U6-37	Comm Errors-HOST
U6-48	ASIC Comm Errors

10.4 A: Initialization Parameters

◆ A1: Initialization

No. (Hex.)	Name	Description	Default (Range)	Ref.
A1-00 (0100) RUN	Language Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the language for the LCD keypad. Note: This parameter is not reset when the drive is initialized using parameter <i>A1-03 [Initialize Parameters]</i>. 0 : English 1 : Japanese 2 : Deutsch 3 : Français 4 : Italiano 5 : Español 6 : Português 7 : Chinese 8 : Czech 9 : Russian 10 : Turkish 11 : Polish 12 : Greek	0 (0 - 12)	573
A1-01 (0101) RUN	Access Level Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Restricts user access to parameter settings. The set access level restricts what parameters the keypad will display, and what parameters the user can set. 0 : Operation Only 1 : User Parameters 2 : Advanced Level 3 : Expert Level	2 (0 - 3)	573
A1-02 (0102)	Control Method Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the control method for the drive application and the motor. 0 : V/f Control 1 : V/f Control w/ PG 2 : Open Loop Vector 3 : Closed Loop Vector 4 : Advanced Open Loop Vector 5 : PM Open Loop Vector 6 : PM Advanced Open Loop Vector 7 : PM Closed Loop Vector 8 : EZ Vector Control	0 (0 - 8)	574
A1-03 (0103)	Initialize Parameters	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Resets parameters to default values. 0 : Keypad or Multi-Speed Selection 1110 : User Initialization 2220 : 2-Wire initialization 3330 : 3-Wire initialization	0 (0 - 3330)	575
A1-04 (0104)	Password	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Enter the correct password set in <i>A1-05 [Password Setting]</i> to unlock parameters. The user can still view parameter settings while they are locked without entering the password. Enter the password in <i>A1-04 [Password]</i> to unlock and change the settings.	0000 (0000 - 9999)	576

No. (Hex.)	Name	Description	Default (Range)	Ref.
A1-05 (0105)	Password Setting	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The drive locks the following parameters once the password has been set. The following parameters can be changed if the user enters the correct password in A1-04 [Password] that matches the password set in A1-05. • A1-01 [Access Level Selection] • A1-02 [Control Method Selection] • A1-03 [Initialize Parameters] • A1-06 [Application Preset] • A1-07 [DriveWorksEZ Function Selection] • A2-01 to A2-32 [User Parameters 1 to 32]	0000 (0000 - 9999)	577
A1-06 (0127)	Application Preset	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV This parameter conveniently sets up the drive for certain applications. 0 : Disabled 1 : Water supply pump 2 : Conveyor 3 : Exhaust fan 4 : HVAC 5 : Air compressor 6 : Hoist 7 : Traveling	0 (0 - 7)	577
A1-07 (0128)	DriveWorksEZ Function Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the drive for operation with a program created in DriveWorksEZ. 0 : DWEZ Disabled 1 : DWEZ Enabled 2 : Digital input	0 (0 - 2)	592
A1-11 (111D)	Firmware update lock	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Enables and disables the firmware update function via cloud service. 0 : Disable 1 : Enable	0 (0, 1)	592

◆ A2: User Parameters

No. (Hex.)	Name	Description	Default Setting (Range)	Ref.
A2-01 to A2-32 (0106 - 0125)	User Parameters 1 to 32	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The user can select up to 32 parameters for the drive and assign them to parameters A2-01 through A2-32. Registered parameters can be displayed in [User Parameters] under the main menu. The user can immediately access necessary parameters. Note: Settings for A2-01 to A2-32 vary depending on the value selected for A1-06 [Application Preset].	Parameters in General-Purpose Setup Mode (Determined by A1-07)	593
A2-33 (0126)	User Parameter Auto Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Determines if the drive automatically saves the history of parameter changes to A2-17 to A2-32 [User Parameters 17 to 32]. 0 : Disabled 1 : Enabled	Determined by A1-06 (0, 1)	593

10.5 b: Application

◆ b1: Operation Mode Selection

No. (Hex.)	Name	Description	Default (Range)	Ref.
b1-01 (0180)	Frequency Reference Selection 1	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects frequency reference input method. 0 : Keypad or Multi-Speed Selection 1 : Analog Input 2 : MEMOBUS/Modbus communications 3 : Option PCB 4 : Pulse Train Input	1 (0 - 4)	594
b1-02 (0181)	Run Command Selection 1	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the Run command input method. 0 : Keypad 1 : Digital Input 2 : Memobus/Modbus Communications 3 : Option PCB	1 (0 - 3)	596
b1-03 (0182)	Stopping Method Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects how the drive stops the motor after removing a Run command or entering a Stop command. Note: The setting range is 0, 1, and 3 when A1-02 = 3, 4, 5, 6, 7, or 8 [Control Method Selection = CLV, AOLV, OLV/PM, AOLV/PM, CLV/PM, or EZOLV]. 0 : Ramp to Stop 1 : Coast to Stop 2 : DC Injection Braking to Stop 3 : Coast to Stop with Timer 9 : Stop with Constant Distance	0 (0 - 3, 9)	596
b1-04 (0183)	Reverse Operation Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Enables and disables reverse operation. Used in applications such as fans or pumps where reverse motor rotation is potentially problematic. 0 : Reverse Enabled 1 : Reverse Disabled	0 (0, 1)	600
b1-05 (0184)	Operation Below Minimum Freq	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the operation to perform when the frequency reference falls below the value set in E1-09 [Minimum Output Frequency]. 0 : Operate at frequency reference 1 : Baseblock (motor coasts) 2 : Operate at minimum frequency 3 : Operate at zero speed	0 (0 - 3)	600
b1-06 (0158)	Digital Input Reading	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects if the drive should read the sequence input (FWD/REV, multi-function input) command once or twice to prevent problems from noise. 0 : Input status is read once 1 : Input status is read twice	1 (0, 1)	602
b1-07 (0186)	LOCAL/REMOTE Run Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV When switching from LOCAL to REMOTE, or between External reference 1 and External reference 2, the Run command may already be present at the location to which the source is being switched. In this case, use parameter b1-07 to determine how the Run command is treated. 0 : Cycle existing RUN command 1 : Accept existing RUN command	0 (0, 1)	602

No. (Hex.)	Name	Description	Default (Range)	Ref.
b1-08 (0187)	Run Command Select in PRG Mode	V/f CL-V/f OLV CLV AOLV OLVP/PM AOLV/PM CLV/PM EZOLV Sets the conditions for the drive to accept a Run command entered from an external source when using the keypad to set parameters. 0 : Do not accept RUN at Programming 1 : Accept RUN while Programming 2 : Allow Programming only at Stop	0 (0 - 2)	602
b1-14 (01C3)	Phase Order Selection	V/f CL-V/f OLV CLV AOLV OLVP/PM AOLV/PM CLV/PM EZOLV Sets the phase order for output terminals U/T1, V/T2, and W/T3. This parameter is useful for making sure the Forward Run command from the drive and the forward direction of the motor match, without needing to change any wiring. 0 : Standard 1 : Switch phase order	0 (0, 1)	603
b1-15 (01C4)	Frequency Reference Selection 2	V/f CL-V/f OLV CLV AOLV OLVP/PM AOLV/PM CLV/PM EZOLV Selects the input method for frequency reference 2. 0 : Keypad 1 : Analog Input 2 : Memobus/Modbus Communications 3 : Option PCB 4 : Pulse Train Input	0 (0 - 4)	603
b1-16 (01C5)	Run Command Selection 2	V/f CL-V/f OLV CLV AOLV OLVP/PM AOLV/PM CLV/PM EZOLV Sets the input method for Run Command 2 when changing the source of the Run command by switching the control circuit terminals ON/OFF. 0 : Keypad 1 : Analog Input 2 : Memobus/Modbus Communications 3 : Option PCB	0 (0 - 3)	605
b1-17 (01C6)	Run Command at Power Up	V/f CL-V/f OLV CLV AOLV OLVP/PM AOLV/PM CLV/PM EZOLV Selects what action to take when the power supply is switched ON and the Run command is input from an external source. It is necessary to set this parameter in applications where the Run command is already enabled when the power supply switches ON/OFF. 0 : Disregard existing RUN command 1 : Accept existing RUN command	0 (0, 1)	606
b1-21 (0748)	CLV Start Selection	V/f CL-V/f OLV CLV AOLV OLVP/PM AOLV/PM CLV/PM EZOLV Selects the condition used to accept the Run command when $A1-02 = 3$ or 7 [Control Method Selection = Closed Loop Vector Control or PM Closed Loop Vector Control]. Normally there is no need to change this setting. 0 : Reject Run if $b2-01 < Nfdbk < E1-09$ 1 : Accept Run command at any speed	0 (0, 1)	606
b1-35 (1117)	Multi-Function DI dead band time	V/f CL-V/f OLV CLV AOLV OLVP/PM AOLV/PM CLV/PM EZOLV Sets the dead band time for multi-function digital inputs.	0.0 ms (0.0 to 100.0 ms)	607

◆ b2: DC Injection Braking and Short Circuit Braking

No. (Hex.)	Name	Description	Default Setting (Range)	Ref.
b2-01 (0189)	DC Injection/Zero SpeedThreshold	V/f CL-V/f OLV CLV AOLV OLVP/PM AOLV/PM CLV/PM EZOLV Sets the frequency to begin DC Injection Braking, Short Circuit Braking, and Zero Servo. Note: This parameter is available when $b1-03 = 0$ [Stopping Method Selection = Ramp to Stop].	Determined by A1-02 (0.0 - 10.0 Hz)	607
b2-02 (018A)	DC Injection Braking Current	V/f CL-V/f OLV CLV AOLV OLVP/PM AOLV/PM CLV/PM EZOLV Sets the DC Injection Braking current as a percentage of the drive rated current.	50% (0 - 100%)	608

No. (Hex.)	Name	Description	Default Setting (Range)	Ref.
b2-03 (018B)	DC Inject Braking Time at Start	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the DC Inject Braking Time at start. Configures the zero speed control at start when using Closed Loop Vector Control, Advanced Open Loop Vector Control, or Closed Loop Vector Control for PM.	A1-02 = 4: 0.03 s Other than A1-02 = 4: 0.00 s (0.00 - 10.00 s)	608
b2-04 (018C)	DC Inject Braking Time at Stop	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the DC Inject Braking Time at stop. Configures the zero speed control at stop when using Closed Loop Vector Control, Advanced Open Loop Vector Control, or Closed Loop Vector Control for PM.	Determined by A1-02 (0.00 - 10.00 s)	609
b2-08 (0190)	Magnetic Flux Compensation Value	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the amount of current injected in the beginning of DC Injection Braking at start (initial excitation) as a percentage of the value set in E2-03 [Motor No-Load Current].	0% (0 - 1000%)	609
b2-12 (01BA)	Short Circuit Brake Time @ Start	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the Short Circuit Braking time at start.	0.00 s (0.00 - 25.50 s)	609
b2-13 (01BB)	Short Circuit Brake Time @ Stop	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the Short Circuit Braking time at stop.	A1-02 = 8: 0.00 s Other than A1-02 = 8: 0.50 s (0.00 - 25.50 s)	610
b2-18 (0177)	Short Circuit Braking Current	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the Short Circuit Braking current as a percentage of the motor rated current.	100.0% (0.0 - 200.0%)	610

◆ b3: Speed Search

No. (Hex.)	Name	Description	Default (Range)	Ref.
b3-01 (0191)	Speed Search Selection at Start	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets whether to execute Speed Search or not with each Run command. 0 : Disabled 1 : Enabled	Determined by A1-02 (0, 1)	613
b3-02 (0192)	Speed Search Deactivation Current	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the current level that ends Speed Search as a percentage of the drive rated output current. Normally there is no need to change this setting.	Determined by A1-02 (0 - 200%)	614
b3-03 (0193)	Speed Search Deceleration Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Determines the deceleration time during Speed Search operation. Set the time it takes to decelerate from the maximum output frequency to the minimum output frequency.	2.0 s (0.1 - 10.0 s)	614
b3-04 (0194)	V/f Gain during Speed Search	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the ratio used to reduce the V/f during searches to reduce the output current during speed searches.	Determined by o2-04 (10 - 100)	614
b3-05 (0195)	Speed Search Delay Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV If a magnetic contactor is installed between the drive and the motor, this parameter sets a delay time to activate the magnetic contactor by delaying Speed Search.	0.2 s (0.0 - 100.0 s)	614
b3-06 (0196)	Speed Estimation Current Level 1	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configure the amount of current that flows to the motor as a coefficient of the motor rated current when executing the Speed Estimation Speed Search. Normally there is no need to change this setting.	Determined by o2-04 (0.0 - 2.0)	614
b3-07 (0197)	Speed Estimation Current Level 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The amount of current that flows to the motor when Speed Estimation Speed Searches are executed is configured as a coefficient of E2-03 [Motor No-Load Current] or E4-03 [Motor 2 Rated No-Load Current]. Normally there is no need to change this setting.	1.0 (0.0 - 3.0)	615

No. (Hex.)	Name	Description	Default (Range)	Ref.
b3-08 (0198)	Speed Estimation ACR P Gain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the proportional gain for the current controller used when executing the Speed Estimation Speed Search. Adjusts the speed search responsiveness. Normally there is no need to change this setting.	A1-02 = 0 through 4: Determined by $\alpha 2-04$, A1-02 = 5, 6, or 8: Determined by A1-02 (0.00 - 6.00)	615
b3-09 (0199)	Speed Estimation ACR I Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the integral time for the current controller used when executing the Speed Estimation Speed Search. Adjusts the speed search responsiveness. Normally there is no need to change this setting.	Determined by A1-02 (0.0 - 1000.0 ms)	615
b3-10 (019A)	Speed Estimation Detection Gain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the gain used to correct frequencies estimated by the Speed Estimation Speed Search.	1.05 (1.00 - 1.20)	615
b3-14 (019E)	Bi-Direction Speed Search Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects which direction to start Speed Estimation Speed Search, either in the direction of the frequency reference, or the direction that the drive detected the motor is rotating. 0 : Disabled 1 : Enabled	Determined by A1-02 (0, 1)	615
b3-17 (01F0)	Speed Est. Retry Current Level	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the current level used to operate the search retry function for Speed Estimation Speed Search, as a percentage, on the basis that the drive rated current is the 100% value.	150% (0 - 200%)	616
b3-18 (01F1)	Speed Est. Retry Detection Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the time until the speed search is reexecuted (retried) when a speed search is interrupted due to significant current flowing during Speed Estimation Speed Search.	0.10 s (0.00 - 1.00 s)	616
b3-19 (01F2)	Number of Speed Search Restarts	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the number of times to restart Speed Search if Speed Search fails.	3 times (0 - 10 times)	616
b3-24 (01C0)	Speed Search Method Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the type of Speed Search to perform when starting the motor, or when power is restored following a momentary power loss. 1 : Speed Estimation 2 : Current Detection 2	2 (1, 2)	616
b3-25 (01C8)	Speed Search Wait Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the wait time used until the speed search retry function is executed.	0.5 s (0.0 - 30.0 s)	616
b3-26 (01C7)	Direction Determining Level	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the level used to determine the direction of motor rotation. Increase the setting value if determination fails.	1000 (40 to 60000)	617
b3-27 (01C9)	Start Speed Search Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the conditions used to start the speed search. 0 : SS only if RUN applied before BB 1 : SS regardless of RUN/BB sequence	0 (0, 1)	617
b3-29 (077C)	Speed Search Back-EMF Threshold	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the induced voltage of motors for which speed searches are performed. Speed searches are performed when the level of the motor induced voltage reaches the setting value. Normally there is no need to change this setting.	10% (0 - 10%)	617
b3-31 (0BC0)	Search current Level 1	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the current level used to restrict the output current during the Current Detection Speed Search.	1.50 (1.50 - 3.50)	617
b3-32 (0BC1)	Search current Level 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the current level used to complete motor speed searches.	1.20 (0.00 - 1.49)	617

No. (Hex.)	Name	Description	Default (Range)	Ref.
b3-33 (0B3F)	Spd Search during UV Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects whether or not to execute the speed search at start-up when the Run command is input while <i>Uv</i> [DC Bus Undervoltage] is detected. 0 : Disabled 1 : Enabled	1 (0, 1)	618
b3-35 (0BC3)	Voltage Detection Low Level	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The speed search is enabled when the detected induced voltage of motors $\geq$ b3-35.	10% (5 - 50%)	618
b3-36 (0BC4)	Wait Restart Level	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The restart process is prohibited and the drive enters the standby state when the detected induced voltage of the motor $\geq$ power supply voltage $\times$ b3-36. The restart process is executed when the detected induced voltage of the motor $<$ power supply voltage $\times$ b3-36. Normally there is no need to change this setting.	0.970 (0.500 - 1.000)	618
b3-54 (3123)	Search Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the speed search time.	400 ms (10 - 2000 ms)	618
b3-55 (3124)	Current Increment Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the time used to increase the current value from zero current until the setting value of b3-06 [Speed Estimation Current Level 1].	10 ms (10 - 2000 ms)	618

◆ b4: Timer Function

No. (Hex.)	Name	Description	Default (Range)	Ref.
b4-01 (01A3)	Timer Function On-Delay Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the ON-delay time for the timer input.	0.0 s (0.0 - 3000.0 s)	619
b4-02 (01A4)	Timer Function Off-Delay Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the OFF-delay time for the timer input.	0.0 s (0.0 - 3000.0 s)	619
b4-03 (0B30)	H2-01 ON Delay Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the delay time until the contact is turned on after the function selected with H2-01 turns on.	0 ms (0 - 65000 ms)	620
b4-04 (0B31)	H2-01 OFF Delay Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the delay time until the contact is turned off after the function selected with H2-01 turns off.	0 ms (0 - 65000 ms)	620
b4-05 (0B32)	H2-02 ON Delay Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the delay time until the contact is turned on after the function selected with H2-02 turns on.	0 ms (0 - 65000 ms)	620
b4-06 (0B33)	H2-02 OFF Delay Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the delay time until the contact is turned off after the function selected with H2-02 turns off.	0 ms (0 - 65000 ms)	620
b4-07 (0B34)	H2-03 ON Delay Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the delay time until the contact is turned on after the function selected with H2-03 turns on.	0 ms (0 - 65000 ms)	620
b4-08 (0B35)	H2-03 OFF Delay Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the delay time until the contact is turned off after the function selected with H2-03 turns off.	0 ms (0 - 65000 ms)	620

◆ b5: PID Control

No. (Hex.)	Name	Description	Default (Range)	Ref.
b5-01 (01A5)	PID Function Setting	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the type of PID control. 0 : Disabled 1 : Enabled D=Fdbk 2 : Enabled D=Fdfwd 3 : Fref+PID D=Fdbk 4 : Fref+PID D=Fdfwd 5 : Enabled D=Fdbk2 6 : Enabled D=Fdfwd2 7 : Fref+PID D=Fdbk2 8 : Fref+PIDD=Fdfwd2 Note: Use settings 5 to 8 instead of settings 1 to 4 if retrofitting the drive with a Varispeed series drive, or a similar product from a previous product line.	0 (0 - 8)	626
b5-02 (01A6) RUN	Proportional Gain Setting (P)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the proportional gain (P) applied to PID input.	1.00 (0.00 - 25.00)	627
b5-03 (01A7) RUN	Integral Time Setting (I)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the integral time (I) in seconds that is applied to PID input.	1.0 s (0.0 - 360.0 s)	627
b5-04 (01A8) RUN	Integral Limit Setting	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the upper limit for I control as a percentage of <i>E1-04 [Maximum Output Frequency]</i>.	100.0% (0.0 - 100.0%)	627
b5-05 (01A9) RUN	Derivative Time (D)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the derivative time (D) for PID control. This parameter adjusts system responsiveness.	0.00 s (0.00 - 10.00 s)	627
b5-06 (01AA) RUN	PID Output Limit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the maximum output possible from the entire PID controller as a percentage of <i>E1-04 [Maximum Output Frequency]</i>.	100.0% (0.0 - 100.0%)	628
b5-07 (01AB) RUN	PID Offset Adjustment	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the offset for the PID control output as a percentage of <i>E1-04 [Maximum Output Frequency]</i>.	0.0% (-100.0 - +100.0%)	628
b5-08 (01AC)	PID Primary Delay Time Constant	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the primary delay time constant for the PID control output. Normally there is no need to change this setting.	0.00 s (0.00 - 10.00 s)	628
b5-09 (01AD)	PID Output Level Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Reverses the polarity of the PID output. 0 : Normal output (direct acting) 1 : Reverse output (reverse acting)	0 (0, 1)	628
b5-10 (01AE) RUN	PID Output Gain Setting	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Adjusts the degree of compensation by multiplying the gain by the PID output.	1.00 (0.00 - 25.00)	628
b5-11 (01AF)	PID Output Reverse Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets whether or not the motor should rotate in reverse when the PID control output is negative. 0 : Disabled: 0 lower limit 1 : Enabled: Negative lower limit	0 (0, 1)	628

No. (Hex.)	Name	Description	Default (Range)	Ref.
b5-12 (01B0)	Feedback Loss Detection Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Enables or disables PID feedback loss detection. Sets operation after the drive detects PID feedback loss. 0 : No alarm_Always detected_DO only 1 : Alarm_Always detected 2 : Fault_Always detected 3 : No alarm_only @ PID enbl_DO only 4 : Alarm_only detected @ PID Enbl 5 : Fault_only detected @ PID Enbl	0 (0 - 5)	629
b5-13 (01B1)	PID Feedback Loss Detection Lvl	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the level that triggers <i>PID Feedback Loss [FbL]</i> as a percentage of the maximum output frequency.	0% (0 - 100%)	630
b5-14 (01B2)	PID Feedback Loss Detection Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the time that the PID feedback has to fall below <i>b5-13 [PID Feedback Loss Detection Lvl]</i> before <i>PID Feedback Loss [FbL]</i> is detected.	1.0 s (0.0 - 25.5 s)	630
b5-15 (01B3)	PID Sleep Function Start Level	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the output level that triggers the PID Sleep function.	Determined by A1-02 (0.0 - 590.0)	630
b5-16 (01B4)	PID Sleep Delay Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets a delay time to activate or deactivate the PID Sleep function.	0.0 s (0.0 - 25.5 s)	630
b5-17 (01B5)	PID Accel/Decel Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Raises or lowers the PID setpoint using the acceleration and deceleration times set to the drive. This is a soft-starter for the PID setpoint.	0.0 s (0.0 - 6000.0 s)	630
b5-18 (01DC)	PID Setpoint Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Enables or disables <i>b5-19 [PID Setpoint Value]</i>. 0 : Disabled 1 : Enabled	0 (0, 1)	630
b5-19 (01DD) RUN	PID Setpoint Value	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV This setting is the PID setpoint when <i>b5-18 = 1 [PID Setpoint Selection = Enabled]</i>.	0.00% (0.00 - 100.00%)	631
b5-20 (01E2)	PID Setpoint Scaling	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Determines the units used to set and display <i>b5-19 [PID Setpoint Value]</i>.	1 (0 - 3)	631
b5-34 (019F) RUN	PID Output Lower Limit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the output lower limit for the PID control as a percentage of <i>E1-04 [Maximum Output Frequency]</i>.	0.0% (-100.0 - +100.0%)	631
b5-35 (01A0) RUN	PID Input Limit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the input upper limit for the PID control as a percentage of <i>E1-04 [Maximum Output Frequency]</i>.	1000.0% (0.0 - 1000.0%)	631
b5-36 (01A1)	PID Feedback High Detection Lvl	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the level that triggers <i>Excessive PID Feedback [FbH]</i> as a percentage of <i>E1-04 [Maximum Output Frequency]</i>.	100% (0 - 100%)	632
b5-37 (01A2)	PID Feedback High Detection Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the time that triggers <i>Excessive PID Feedback [FbH]</i> when the feedback signal rises above the level set in <i>b5-36 [PID Feedback High Detection Lvl]</i>.	1.0 s (0.0 - 25.5 s)	632
b5-38 (01FE)	PID Setpoint User Display	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Determines the value for setting and displaying <i>U5-01, U5-04</i> when outputting the maximum output frequency.	Determined by b5-20 (1 - 60000)	632
b5-39 (01FF)	PID Setpoint Display Digits	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the number of digits for setting and displaying the PID setpoint. 0 : No decimal places 1 : One decimal place 2 : Two decimal places 3 : Three decimal places	Determined by b5-20 (0 - 3)	632

No. (Hex.)	Name	Description	Default (Range)	Ref.
b5-40 (017F)	Frequency Reference Monitor @PID	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the contents displayed in monitor <i>U1-01 [Freq Reference]</i> when using PID control. 0 : with PID 1 : without PID	0 (0, 1)	632
b5-47 (017D)	PID Output Reverse Selection 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets whether or not the motor should rotate in reverse when the PID control output is negative. 0 : Disabled: 0 lower limit 1 : Enabled: Negative lower limit	1 (0, 1)	633
b5-53 (0B8F) RUN	Integrator Ramp Limit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Adjusts the responsiveness of PID control when the PID feedback changes rapidly.	0.0 Hz (0.0 - 10.0 Hz)	633
b5-54 (0BB7)	PID softstarter cancel selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the relationship between the soft starter and PID input/output. 0 : None 1 : Softstarter is canceled	0 (0, 1)	633
b5-55 (0BE1)	PID feedback monitor selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the monitor (<i>Ux-xx</i>) used as the PID Feedback.	000 (000 - 999)	633
b5-56 (0BE2)	PID feedback monitor gain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the gain for the monitor selected with <i>b5-55 [PID feedback monitor selection]</i> .	1.00 (0.00 - 10.00)	634
b5-57 (11DD)	PID Feedback Monitor Bias	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the bias for the monitor selected with <i>b5-55 [PID feedback monitor selection]</i> .	0.00 (-10.00 - +10.00)	634
b5-58 through b5- 60 (1182 - 1184) RUN	PID Setpoints 2 through 4	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the PID setpoint selected with <i>H1-xx = 3E or 3F [Terminal Sx Function Select = PID Setpoint Selection 1/2]</i> . This parameter is configured on the basis of the maximum output frequency being the 100% value.	0.00% (0.00 - 100.00%)	634
b5-61 (119A)	PID Trim lower limit selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the function used to adjust the PID output in proportion with the frequency reference. 0 : Disabled 1 : Enabled	0 (0, 1)	634
b5-62 (119B)	PID Trim lower limit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the lower limit of the PID frequency reference trim on the basis of the maximum output frequency as the 100% value.	0.00% (0.00 - 100.00%)	635
b5-63 (119C)	Differential PID Feedback Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the monitor (<i>Ux-xx</i>) used as the PID Differential Feedback.	000 (000 - 999)	635
b5-64 (119D)	Differential PID Feedback Gain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the gain of the monitor configured with <i>b5-63 [Differential PID Feedback Select]</i> .	1.00 (0.00 - 10.00)	635
b5-65 (119F)	PID Diff Fdbk Monitor Bias	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the bias for the monitor selected with <i>b5-63 [Differential PID Feedback Select]</i> .	0.00 (-10.00 - +10.00)	635
b5-66 (11DE)	PID feedback monitor Lvl	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the signal level for the monitor selected with <i>b5-55 [PID feedback monitor selection]</i> . 0 : Absolute 1 : +-	0 (0, 1)	635
b5-67 (11DF)	PID Diff Fdbk monitor Lvl	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the signal level for the monitor selected with <i>b5-63 [Differential PID Feedback Select]</i> . 0 : Absolute 1 : +-	0 (0, 1)	636

No. (Hex.)	Name	Description	Default (Range)	Ref.
b5-89 (0B89) RUN	Sleep Method Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the sleep and start operations modes when using the PID function. 0 : Standard 1 : EZ Sleep/Wake-up	0 (0, 1)	636
b5-90 (0B90)	EZ Sleep Unit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the unit of measure for b5-91 [EZ Minimum Speed] and b5-92 [EZ Sleep Level]. 0 : 0.1Hz units 1 : rev/min	0 (0, 1)	636
b5-91 (0B91) RUN	EZ Minimum Speed	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the minimum speed for the EZ Sleep/Wake-up function. The larger value between b5-91, b5-34 [PID Output Lower Limit], and d2-02 [Frequency Reference Lower Limit] is used internally to configure this parameter.	0.0 Hz or 0 min ⁻¹ (r/min) (0.0 to 590.0 Hz or 0 to 35400 min ⁻¹ (r/min))	636
b5-92 (0B92) RUN	EZ Sleep Level	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The drive enters the sleep state when the output frequency or motor speed drops below the value of b5-92 for a time longer than the setting value of b5-93 [EZ Sleep Time].	0.0 Hz or 0 min ⁻¹ (r/min) (0.0 to 590.0 Hz or 0 to 35400 min ⁻¹ (r/min))	636
b5-93 (0B93) RUN	EZ Sleep Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The drive enters the sleep state when the output frequency or motor speed drops below the value of b5-92 [EZ Sleep Level] for a time longer than the setting value of b5-93.	5.0 s (0.0 - 1000.0 s)	637
b5-94 (0B94) RUN	EZ Wake-up Level	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the level at which the drive resumes operation from the Sleep mode.	0.00% (0.00 - 600.00%)	637
b5-95 (0B95)	EZ Wake-up Mode	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the wake-up mode used when resuming operation from the Sleep mode. 0 : Absolute 1 : Setpoint Delta	0 (0, 1)	637
b5-96 (0B96)	EZ Wake-up Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the EZ Wake-up time.	1.0 s (0.0 - 1000.0 s)	637

◆ b6: Dwell Function

No. (Hex.)	Name	Description	Default (Range)	Ref.
b6-01 (01B6)	Dwell Reference at Start	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the output frequency to briefly maintain when starting the motor.	0.0 (Determined by A1-02)	638
b6-02 (01B7)	Dwell Time at Start	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the time for the drive to hold the output frequency when starting the motor.	0.0 s (0.0 - 10.0 s)	638
b6-03 (01B8)	Dwell Reference at Stop	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the output frequency to briefly maintain when ramping to stop.	0.0 (Determined by A1-02)	638
b6-04 (01B9)	Dwell Time at Stop	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the time for the drive to hold the output frequency when ramping to stop.	0.0 s (0.0 - 10.0 s)	638

◆ b7: Droop Control

No. (Hex.)	Name	Description	Default (Range)	Ref.
b7-01 (01CA) RUN	Droop Control Gain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the amount of deceleration when the torque reference is at 100% as a percentage of Maximum Output Frequency.	0.0% (0.0 - 100.0%)	639
b7-02 (01CB) RUN	Droop Control Delay Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Adjusts the responsiveness of Droop control. Lower this setting when drive response is slow. Raise this setting when hunting or oscillation occur.	0.05 s (0.03 - 2.00 s)	639
b7-03 (017E)	Droop Control Limit Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Enables or disables the limit when using Droop control. 0 : Disabled 1 : Enabled	1 (0, 1)	639

◆ b8: Energy Saving

No. (Hex.)	Name	Description	Default (Range)	Ref.
b8-01 (01CC)	Energy Saving Control Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Enables or disables Energy-saving control. 0 : Disabled 1 : Enabled 2 : Search Enabled	0 (Determined by A1-02)	640
b8-02 (01CD) RUN	Energy Saving Gain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the gain for energy-saving control.	Determined by A1-02 (0.0 - 10.0)	640
b8-03 (01CE) RUN	Energy Saving Filter Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the responsiveness for energy-saving control.	Determined by A1-02, C6-01, and o2-04 (0.00 - 10.00 s)	640
b8-04 (01CF)	Energy Saving Coefficient Value	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the energy-saving coefficient. The energy-saving coefficient is used to maintain maximum motor efficiency. The default setting is the Yaskawa motor value.	Determined by C6-01, E2-11, and o2-04. (0.00 - 655.00)	640
b8-05 (01D0)	Power Detection Filter Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the time constant used by the drive to measure output power.	20 ms (0 - 2000 ms)	641
b8-06 (01D1)	Search Operation Voltage Limit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV On the basis that the motor rated voltage is the 100% value, this parameter configures the limit value of the voltage range defined for Search Operations as a percentage.	0% (0 - 100%)	641
b8-16 (01F8)	PM E-Save Coefficient Ki	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV This coefficient maintains torque linearity. Enter the Ki value written on the motor nameplate. Normally there is no need to change this setting.	1.00 (0.00 - 3.00)	641
b8-17 (01F9)	PM E-Save Coefficient Kt	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV This coefficient maintains torque linearity. Enter the Kt value written on the motor nameplate. Normally there is no need to change this setting.	1.00 (0.00 - 3.00)	641
b8-18 (01FA)	E-Save d-axis Current FilterTime	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the d-axis current reference filter time constant.	0.100 s (0.000 - 5.000 s)	642
b8-19 (0B40)	E-Save Search Injection Freq	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the energy-saving control search operation frequency. Normally there is no need to change this setting.	Determined by A1-02 (20 - 300 Hz)	642
b8-20 (0B41)	PM E-Save Search Width	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the amplitude of energy-saving control search operations.	1.0 degrees (0.1 to 5.0 degrees)	642

No. (Hex.)	Name	Description	Default (Range)	Ref.
b8-21 (0B42)	PM E-Save Search Gain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the gain of search operations.	0.3Hz (0.1 - 20.0 Hz)	642
b8-22 (0B43)	PM E-Save Search LPF Cutoff Freq	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the frequency of the filter used to extract the high-efficiency phase from search operations. Normally there is no need to change this setting.	10.0 Hz (1.0 - 30.0 Hz)	642
b8-23 (0B44)	PM E-Save Search Limit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the limit value of search operation output. Normally there is no need to change this setting.	15.0 degrees (0.0 to 30.0 degrees)	642
b8-24 (0B45)	PM E-Save High Freq ACR Gain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the gain for high-frequency current control.	200.0 Hz (100.0 - 1000.0 Hz)	643
b8-25 (0B46)	PM E-Save Search Start level	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the level at which search operations start.	10.0% (0.0 - 100.0%)	643
b8-26 (0B47)	PM E-Save Power Setpoint	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Adjust this parameter when torque accuracy cannot be ensured.	0.0% (-10.0 - +10.0%)	643
b8-28 (0B8B)	Save Energy Priority Function	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Enables and disables operation toward an overexcitation state. 0 : Priority:Followingness 1 : Priority:Save Energy	0 (0, 1)	643
b8-29 (0B8C)	Save Energy Priority Function	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Switches between prioritizing the capability to respond to load changes or the energy-saving control. 0 : Priority:Followingness 1 : Priority:Save Energy	0 (0, 1)	643
b8-50 (0B0D)	Standby mode selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Enables and disables standby mode. 0 : Disabled 1 : Enabled	0 (0, 1)	643
b8-51 (0B01)	Standby mode waiting time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the delay time until the input side electromagnetic contactor shuts off after the drive stops.	600 s (0 - 6000 s)	644

◆ b9: Zero Servo

No. (Hex.)	Name	Description	Default (Range)	Ref.
b9-01 (01DA)	Zero Servo Gain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the responsiveness for the Zero Servo function.	5 (0 - 100)	645
b9-02 (01DB)	Zero Servo Completion Width	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the output width that triggers the Zero Servo complete command. Set the allowable position displacement (deviation) from Zero Servo start position.	10 (0 - 16383)	645

10.6 C: Tuning

◆ C1: Accel and Decel Times

No. (Hex.)	Name	Description	Default (Range)	Ref.
C1-01 (0200) RUN	Acceleration Time 1	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the time it takes to accelerate from zero to the maximum output frequency.	10.0 s (0.0 - 6000.0 s)	648
C1-02 (0201) RUN	Deceleration Time 1	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the time it takes to decelerate from the maximum output frequency to zero.	10.0 s (0.0 - 6000.0 s)	648
C1-03 (0202) RUN	Acceleration Time 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the time to accelerate from 0 to maximum frequency.	10.0 s (0.0 - 6000.0 s)	648
C1-04 (0203) RUN	Deceleration Time 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the time to decelerate from maximum frequency to 0.	10.0 s (0.0 - 6000.0 s)	648
C1-05 (0204) RUN	Acceleration Time 3	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the time to accelerate from 0 to maximum frequency.	10.0 s (0.0 - 6000.0 s)	648
C1-06 (0205) RUN	Deceleration Time 3	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the time to decelerate from maximum frequency to 0.	10.0 s (0.0 - 6000.0 s)	648
C1-07 (0206) RUN	Acceleration Time 4	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the time to accelerate from 0 to maximum frequency.	10.0 s (0.0 - 6000.0 s)	649
C1-08 (0207) RUN	Deceleration Time 4	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the time to decelerate from maximum frequency to 0.	10.0 s (0.0 - 6000.0 s)	649
C1-09 (0208)	Fast Stop Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the deceleration time for the drive to trigger the Fast Stop. Note: Rapid deceleration can trigger <i>ov</i> [Overvoltage]. The drive output shuts off when <i>ov</i> is detected and the motor coasts. Set an appropriate Fast Stop time to C1-09 to avoid motor coasting and to ensure that the motor stops quickly and safely.	10.0 s (0.0 - 6000.0 s)	649
C1-10 (0209)	Accel/Decel Time Setting Units	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the setting units for C1-01 to C1-08 [Acceleration Time 1 through Deceleration Time 4], C1-09 [Fast Stop Time], L2-06 [KEB Deceleration Time], and L2-07 [KEB Acceleration Time]. 0 : 0.01 s units 1 : 0.1 s units	1 (0, 1)	649
C1-11 (020A)	Accel/Decel Time Switchover Freq	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the frequency at which acceleration and deceleration times are automatically changed.	Determined by A1-02 (0.0 - 590.0 Hz)	650

◆ C2: S-Curve Characteristics

No. (Hex.)	Name	Description	Default (Range)	Ref.
C2-01 (020B)	S-Curve Time @ Start of Accel	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the S-curve time to the time to start acceleration.	Determined by A1-02 (0.00 to 10.00 s)	651
C2-02 (020C)	S-Curve Time @ End of Accel	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the S-curve time to the time to complete acceleration.	0.20 s (0.00 - 10.00 s)	651

No. (Hex.)	Name	Description	Default (Range)	Ref.
C2-03 (020D)	S-Curve Time @ Start of Decel	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the S-curve time to the time to start deceleration.	0.20 s (0.00 - 10.00 s)	651
C2-04 (020E)	S-Curve Time @ End of Decel	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the S-curve time to the time to complete deceleration.	0.00 s (0.00 - 10.00 s)	651

◆ C3: Slip Compensation

No. (Hex.)	Name	Description	Default (Range)	Ref.
C3-01 (020F) RUN	Slip Compensation Gain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the gain for the slip compensation function. Normally there is no need to change this setting. Note: Confirm that the following parameters are correctly set before changing the slip compensation gain. If <i>A1-02 = 2 [Control Method Selection = Open Loop Vector Control]</i> , <i>E2-02 [Motor Rated Slip]</i> can be set by Auto-Tuning. • <i>E2-01 [Motor Rated Current (FLA)]</i> • <i>E2-02 [Motor Rated Slip]</i> • <i>E2-03 [Motor No-Load Current]</i>	Determined by A1-02 (0.0 - 2.5)	651
C3-02 (0210) RUN	Slip Compensation Delay Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Adjusts the slip compensation delay time when speed is unstable or when the slip compensation response is too slow. There is normally no need to change this parameter from the default value.	Determined by A1-02 (0 - 10000 ms)	652
C3-03 (0211)	Slip Compensation Limit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the upper limit for the slip compensation function as a percentage of the motor rated slip.	200% (0 - 250%)	652
C3-04 (0212) RUN	Slip Compensation @ Regen Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Enables or disables slip compensation during regenerative operation. 0 : Disabled 1 : Enabled (6 Hz and above) 2 : Enabled (compensation provided wherever possible)	0 (0 - 2)	652
C3-05 (0213)	Output Voltage limit Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects whether to automatically reduce the motor magnetic flux when the output voltage is saturated. 0 : Disabled 1 : Enabled	0 (0, 1)	653
C3-16 (0261)	Vout Modulation Limit Start Lvl	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the level (modulation factor) used to start the output voltage limit operation when <i>C3-05 = 1 [Output Voltage limit Selection = Enabled]</i> .	90.0% (70.0 - 90.0%)	653
C3-17 (0262)	Maximum Output Voltage Limit Level	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the level (modulation factor) used with <i>C3-18 [Output Voltage Limit Level]</i> for the output voltage limit operation when <i>C3-05 = 1 [Output Voltage limit Selection = Enabled]</i> .	100.0% (85.0 - 100.0%)	653
C3-18 (0263)	Output Voltage Limit Level	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the maximum drop width of the voltage reference when <i>C3-05 = 1 [Output Voltage limit Selection = Enabled]</i> .	90.0% (50.0 - 100.0%)	653
C3-21 (033E) RUN	Motor 2 Slip Compensation Gain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the gain for the motor 2 slip compensation function. There is normally no need to change this parameter from the default value. Note: Confirm that the following parameters are correctly set before changing the slip compensation gain. If <i>E3-01 = 2 [Motor 2 Control Mode Selection = Open Loop Vector Control]</i> , <i>E4-02 [Motor 2 Rated Slip]</i> can be set by Auto-Tuning. • <i>E4-01 [Motor 2 Rated Current]</i> • <i>E4-02 [Motor 2 Rated Slip]</i> • <i>E4-03 [Motor 2 Rated No-Load Current]</i>	Determined by E3-01 (0.0 - 2.5)	653

No. (Hex.)	Name	Description	Default (Range)	Ref.
C3-22 (0241) RUN	Motor 2 Slip Comp DelayTime	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Adjusts the slip compensation delay time for motor 2 when speed is unstable or when the slip compensation response is too slow. Normally there is no need to change this setting.	Determined by E3-01 (0 - 10000 ms)	654
C3-23 (0242) RUN	Motor 2 Slip Compensation Limit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the upper limit for the slip compensation function as a percentage of the motor 2 rated slip.	200% (0 - 250%)	654
C3-24 (0243) RUN	Motor 2 Slip Comp @ Regen Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Enables or disables slip compensation for motor 2 during regenerative operation. 0 : Disabled 1 : Enabled (6 Hz and above) 2 : Enabled (compensation provided wherever possible)	0 (0 - 2)	654
C3-28 (1B5B)	Adaptive Slip Contrl Mode Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the mode for the slip compensation function. 0 : Normal 1 : Advance	0 (0, 1)	655

◆ C4: Torque Compensation

No. (Hex.)	Name	Description	Default (Range)	Ref.
C4-01 (0215) RUN	Torque Compensation Gain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the gain for the torque compensation function. Sets the gain for motor 1 when multiple motors are driven.	Determined by A1-02 (0.00 - 2.50)	655
C4-02 (0216) RUN	Torque Compensation Delay Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the torque compensation delay time in milliseconds. Normally there is no need to change this setting.	Determined by A1-02 (0 - 60000 ms)	656
C4-03 (0217)	Torque Compensation @ FWD Start	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the amount of torque reference at start in the forward direction as a percentage of the motor rated torque.	0.0% (0.0 - 200.0%)	656
C4-04 (0218)	Torque Compensation @ REV Start	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the amount of torque reference at start in the reverse direction as a percentage of the motor rated torque.	0.0% (-200.0 - 0.0%)	656
C4-05 (0219)	Torque Compensation Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the starting torque constant used with C4-03 and C4-04 [Torque Compensation @ REV Start].	10 ms (0 - 200 ms)	656
C4-06 (021A)	Motor 2 Torque Comp Delay Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the value if <i>ov</i> [Overvoltage] occurs with sudden changes in the load, at the end of acceleration, or at the start of deceleration.	150 ms (0 - 10000 ms)	656
C4-07 (0341) RUN	Motor 2 Torque Compensation Gain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the motor 2 gain for the torque compensation function when using the Motor Switch function.	1.00 (0.00 - 2.50)	657
C4-19 (0B8D)	T-ripple suppress freq	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Increase the setting in increments of approximately 1.0 when current ripples and torque ripples occur during low-speed operation. If this still does not improve the situation, set C4-19 = 0 to disable this function. Normally there is no need to change this setting.	0.1 Hz (0.0 - 10.0 Hz)	657
C4-20 (0BCB)	vol compensation adjust 1	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV This parameter compensates the voltage accuracy. Normally there is no need to change this setting.	120 (0 - 200)	657
C4-21 (0BCC)	vol compensation adjust 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV This parameter compensates the voltage accuracy. Normally there is no need to change this setting.	5 (0 - 10)	657

◆ C5: Automatic Speed Regulator Automatic Speed Regulator)

No. (Hex.)	Name	Description	Default (Range)	Ref.
C5-01 (021B) RUN	ASR Proportional Gain 1	 Sets the gain to adjust ASR response.	Determined by A1-02 (0.00 - 300.00)	661
C5-02 (021C) RUN	ASR Integral Time 1	 Sets the ASR integral time.	Determined by A1-02 (0.000 - 10.000 s)	661
C5-03 (021D) RUN	ASR Proportional Gain 2	 Sets the gain to adjust ASR response.	Determined by A1-02 (0.00 - 300.00)	662
C5-04 (021E) RUN	ASR Integral Time 2	 Sets the ASR integral time.	Determined by A1-02 (0.000 - 10.000 s)	662
C5-05 (021F) RUN	ASR Limit	 Sets the ASR output limit as a percentage of E1-04 [Maximum Output Frequency].	5.0% (0.0 - 20.0%)	662
C5-06 (0220) RUN	ASR Delay Time	 Sets the filter time constant for the time from the speed loop to the torque command output. There is normally no need to change this parameter from the default value.	Determined by A1-02 (0.000 - 0.500 s)	662
C5-07 (0221)	ASR Gain Switchover Frequency	 Sets the frequency where the drive should switch between C5-01 [ASR Proportional Gain 1] and C5-03 [ASR Proportional Gain 2] as well as between integral time 1 and 2 (C5-02, C5-04). Sets the frequency where the drive should switch between C5-02 [ASR Integral Time 1] and C5-04 [ASR Integral Time 2] as well as between C5-01 and C5-03.	Determined by A1-02 (Determined by A1-02)	662
C5-08 (0222)	ASR Integral Limit	 Sets the upper limit for ASR as a percentage of the rated load.	400% (0 - 400%)	663
C5-12 (0386)	Integral Operation @ Accel/Decel	 Enables or disables integral operation during acceleration and deceleration. 0 : Disabled 1 : Enabled	0 (0, 1)	663
C5-17 (0276)	Motor inertia	 Sets the motor inertia.	Determined by o2-04, C6-01, and E5-01 (0.0001 - 600.00 kgm ²)	663
C5-18 (0277)	Load Inertia Ratio	 Sets the load inertia ratio for the motor inertia.	1.0 (0.0 - 6000.0)	663
C5-21 (0356) RUN	Motor 2 ASR Proportional Gain 1	 Sets the gain to adjust ASR response.	Determined by E3-01 (0.00 - 300.00)	663
C5-22 (0357) RUN	Motor 2 ASR Integral Time 1	 Sets the ASR integral time.	Determined by E3-01 (0.000 - 10.000 s)	664
C5-23 (0358) RUN	Motor 2 ASR Proportional Gain 2	 Sets the gain to adjust ASR response.	Determined by E3-01 (0.00 - 300.00)	664
C5-24 (0359) RUN	Motor 2 ASR Integral Time 2	 Sets the ASR integral time.	Determined by E3-01 (0.000 - 10.000 s)	664

No. (Hex.)	Name	Description	Default (Range)	Ref.
C5-25 (035A)	Motor 2 ASR Limit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the ASR output limit as a percentage of E1-04 [Maximum Output Frequency].	5.0% (0.0 - 20.0%)	664
C5-26 (035B)	Motor 2 ASR Delay Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the filter time constant for the time from the speed loop to the torque command output. There is normally no need to change this parameter from the default value.	Determined by E3-01 (0.000 - 0.500 s)	665
C5-27 (035C)	Motor 2 ASR Gain Switchover Freq	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the frequency where the drive should switch between C5-21 [Motor 2 ASR Proportional Gain 1] and C5-23 [Motor 2 ASR Proportional Gain 2]. Sets the frequency where the drive should switch between C5-22 [Motor 2 ASR Integral Time 1] and C5-24 [Motor 2 ASR Integral Time 2] as well as between C5-21 and C5-23.	0.0 (Determined by A1-02)	665
C5-28 (035D)	Motor 2 ASR Integral Limit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the upper limit for ASR in units of %. Sets the percentage of the rated load.	400% (0 - 400%)	665
C5-29 (0B18)	Speed Control Response	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the responsiveness of speed control. Normally there is no need to change this setting. 0 : Standard 1 : High speed	0 (0, 1)	665
C5-32 (0361)	Motor 2 Integral Oper @ Acc/Dec	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Enables or disables integral operation during acceleration and deceleration. 0 : Disabled 1 : Enabled	0 (0, 1)	665
C5-37 (0278)	Motor 2 Inertia	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the motor inertia.	Determined by o2-04 and C6-01 (0.0001 - 6.0000 kgm ²)	666
C5-38 (0279)	Motor 2 Load Inertia Ratio	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the load inertia ratio for the motor inertia.	1.0 (0.0 - 6000.0)	666
C5-39 (030D)	ASR Primary Delay Time Const 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the filter time constant used for the time from the speed loop to the torque command output. There is normally no need to change this parameter from the default value.	0.000 s (0.000 - 0.500 s)	666
C5-50 (0B14)	Notch filter frequency	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the machine resonance frequency in increments of 1 Hz. Note: Setting C5-50 to 0 Hz disables the notch filter.	0 Hz (0, or 2 to 100 Hz)	666
C5-51 (0B15)	Notch Filter Bandwidth	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the notch width of the notch filter. Note: Setting this parameter to 0 Hz disables the function.	1.0 (0.5 - 5.0)	666

◆ C6: Carrier Frequency

No. (Hex.)	Name	Description	Default (Range)	Ref.
C6-01 (0223)	Normal / Heavy Duty Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the drive duty rating. 0 : Constant Torque Application Heavy Duty Rating 1 (HD1 / HD2) 1 : Variable Torque Application Normal Duty Rating 1 (ND1 / ND2)	0 (0, 1)	667
C6-02 (0224)	Carrier Frequency Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the switching frequency (carrier frequency) for the transistors inside the drive. 1 : 2.0 kHz 2 : 5.0 kHz (4.0 kHz) 3 : 8.0 kHz (6.0 kHz) 4 : 10.0 kHz (8.0 kHz) 5 : 12.5 kHz (10.0 kHz) 6 : 15.0 kHz (12.0 kHz) 7 : Swing PWM1 (Audible Sound 1) 8 : Swing PWM2 (Audible Sound 2) 9 : Swing PWM3 (Audible Sound 3) A : Swing PWM4 (Audible Sound 4) F : User Defined (C6-03 to C6-05) Note: •The carrier frequency for Swing PWM 1 to 4 is equivalent to 2.0 kHz. •The value in parenthesis indicates the carrier frequency when <i>A1-02 = 6 [Control Method Selection = PM Advanced Open Loop Vector]</i>.	Determined by A1-02, C6-01, and o2-04 (Determined by A1-02)	668
C6-03 (0225)	Carrier Frequency Upper Limit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the upper limit of the carrier frequency. This parameter can only be set when <i>C6-02 = F [Carrier Frequency Selection = User Defined (C6-03 to C6-05)]</i>.	Determined by C6-02 (1.0 - 15.0 kHz)	669
C6-04 (0226)	Carrier Frequency Lower Limit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the lower limit of the carrier frequency. This parameter can only be set when <i>C6-02 = F [Carrier Frequency Selection = User Defined (C6-03 to C6-05)]</i>.	Determined by C6-02 (1.0 - 15.0 kHz)	669
C6-05 (0227)	Carrier Freq Proportional Gain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the proportional gain for the carrier frequency. This parameter can only be set when <i>C6-02 = F [Carrier Frequency Selection = User Defined (C6-03 to C6-05)]</i>.	Determined by C6-02 (0 - 99)	670
C6-09 (022B)	Carrier Frequency @ RotateTuning	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the carrier frequency while performing Auto-Tuning. There is normally no need to change this parameter from the default value. 0 : 5 kHz 1 : use C6-03	0 (0, 1)	670

10.7 d: Reference Settings

◆ d1: Frequency reference

No. (Hex.)	Name	Description	Default (Range)	Ref.
d1-01 (0280) RUN	Reference 1	V/f CL-V/f OLV CLV AOLV OLVP/AOLVP CLVP/EZOLV Sets the frequency reference in the unit set in <i>o1-03 [Keypad Display Selection]</i>. Note: The value set to <i>o1-03 [Keypad Display Selection]</i> is changed to 1 [0.01% (100% = E1-04)] when A1-02 = 6, 7 [Control Method Selection = AOLV/PM, CLV/PM].	0.00 Hz (0.00 - 590.00 Hz)	673
d1-02 (0281) RUN	Reference 2	V/f CL-V/f OLV CLV AOLV OLVP/AOLVP CLVP/EZOLV Sets the frequency reference in the unit set in <i>o1-03 [Keypad Display Selection]</i>. Note: The value set to <i>o1-03 [Keypad Display Selection]</i> is changed to 1 [0.01% (100% = E1-04)] when A1-02 = 6, 7 [Control Method Selection = AOLV/PM, CLV/PM].	0.00 Hz (0.00 - 590.00 Hz)	674
d1-03 (0282) RUN	Reference 3	V/f CL-V/f OLV CLV AOLV OLVP/AOLVP CLVP/EZOLV Sets the frequency reference in the unit set in <i>o1-03 [Keypad Display Selection]</i>. Note: The value set to <i>o1-03 [Keypad Display Selection]</i> is changed to 1 [0.01% (100% = E1-04)] when A1-02 = 6, 7 [AOLV/PM, CLV/PM].	0.00 Hz (0.00 - 590.00 Hz)	674
d1-04 (0283) RUN	Reference 4	V/f CL-V/f OLV CLV AOLV OLVP/AOLVP CLVP/EZOLV Sets the frequency reference in the unit set in <i>o1-03 [Keypad Display Selection]</i>. Note: The value set to <i>o1-03 [Keypad Display Selection]</i> is changed to 1 [0.01% (100% = E1-04)] when A1-02 = 6, 7 [AOLV/PM, CLV/PM].	0.00 Hz (0.00 - 590.00 Hz)	674
d1-05 (0284) RUN	Reference 5	V/f CL-V/f OLV CLV AOLV OLVP/AOLVP CLVP/EZOLV Sets the frequency reference in the unit set in <i>o1-03 [Keypad Display Selection]</i>. Note: The value set to <i>o1-03 [Keypad Display Selection]</i> is changed to 1 [0.01% (100% = E1-04)] when A1-02 = 6, 7 [Control Method Selection = AOLV/PM, CLV/PM].	0.00 Hz (0.00 - 590.00 Hz)	674
d1-06 (0285) RUN	Reference 6	V/f CL-V/f OLV CLV AOLV OLVP/AOLVP CLVP/EZOLV Sets the frequency reference in the unit set in <i>o1-03 [Keypad Display Selection]</i>. Note: The value set to <i>o1-03 [Keypad Display Selection]</i> is changed to 1 [0.01% (100% = E1-04)] when A1-02 = 6, 7 [Control Method Selection = PM Advanced Open Loop Vector, PM Closed Loop Vector Control].	0.00 Hz (0.00 - 590.00 Hz)	675
d1-07 (0286) RUN	Reference 7	V/f CL-V/f OLV CLV AOLV OLVP/AOLVP CLVP/EZOLV Sets the frequency reference in the unit set in <i>o1-03 [Keypad Display Selection]</i>. Note: The value set to <i>o1-03 [Keypad Display Selection]</i> is changed to 1 [0.01% (100% = E1-04)] when A1-02 = 6, 7 [Control Method Selection = AOLV/PM, CLV/PM].	0.00 Hz (0.00 - 590.00 Hz)	675
d1-08 (0287) RUN	Reference 8	V/f CL-V/f OLV CLV AOLV OLVP/AOLVP CLVP/EZOLV Sets the frequency reference in the unit set in <i>o1-03 [Keypad Display Selection]</i>. Note: The value set to <i>o1-03 [Keypad Display Selection]</i> is changed to 1 [0.01% (100% = E1-04)] when A1-02 = 6, 7 [Control Method Selection = AOLV/PM, CLV/PM].	0.00 Hz (0.00 - 590.00 Hz)	675

10.7 d: Reference Settings

No. (Hex.)	Name	Description	Default (Range)	Ref.
d1-09 (0288) RUN	Reference 9	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the frequency reference in the unit set in <i>o1-03 [Keypad Display Selection]</i>. Note: The value set to <i>o1-03 [Keypad Display Selection]</i> is changed to 1 [0.01% (100% = E1-04)] when A1-02 = 6, 7 [Control Method Selection = AOLV/PM, CLV/PM].	0.00 Hz (0.00 - 590.00 Hz)	675
d1-10 (028B) RUN	Reference 10	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the frequency reference in the unit set in <i>o1-03 [Keypad Display Selection]</i>. Note: The value set to <i>o1-03 [Keypad Display Selection]</i> is changed to 1 [0.01% (100% = E1-04)] when A1-02 = 6, 7 [Control Method Selection = AOLV/PM, CLV/PM].	0.00 Hz (0.00 - 590.00 Hz)	675
d1-11 (028C) RUN	Reference 11	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the frequency reference in the unit set in <i>o1-03 [Keypad Display Selection]</i>. Note: The value set to <i>o1-03 [Keypad Display Selection]</i> is changed to 1 [0.01% (100% = E1-04)] when A1-02 = 6, 7 [Control Method Selection = AOLV/PM, CLV/PM].	0.00 Hz (0.00 - 590.00 Hz)	676
d1-12 (028D) RUN	Reference 12	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the frequency reference in the unit set in <i>o1-03 [Keypad Display Selection]</i>. Note: The value set to <i>o1-03 [Keypad Display Selection]</i> is changed to 1 [0.01% (100% = E1-04)] when A1-02 = 6, 7 [Control Method Selection = AOLV/PM, CLV/PM].	0.00 Hz (0.00 - 590.00 Hz)	676
d1-13 (028E) RUN	Reference 13	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the frequency reference in the unit set in <i>o1-03 [Keypad Display Selection]</i>. Note: The value set to <i>o1-03 [Keypad Display Selection]</i> is changed to 1 [0.01% (100% = E1-04)] when A1-02 = 6, 7 [Control Method Selection = AOLV/PM, CLV/PM].	0.00 Hz (0.00 - 590.00 Hz)	676
d1-14 (028F) RUN	Reference 14	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the frequency reference in the unit set in <i>o1-03 [Keypad Display Selection]</i>. Note: The value set to <i>o1-03 [Keypad Display Selection]</i> is changed to 1 [0.01% (100% = E1-04)] when A1-02 = 6, 7 [Control Method Selection = AOLV/PM, CLV/PM].	0.00 Hz (0.00 - 590.00 Hz)	676
d1-15 (0290) RUN	Reference 15	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the frequency reference in the unit set in <i>o1-03 [Keypad Display Selection]</i>. Note: The value set to <i>o1-03 [Keypad Display Selection]</i> is changed to 1 [0.01% (100% = E1-04)] when A1-02 = 6, 7 [Control Method Selection = AOLV/PM, CLV/PM].	0.00 Hz (0.00 - 590.00 Hz)	677

No. (Hex.)	Name	Description	Default (Range)	Ref.
d1-16 (0291) RUN	Reference 16	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the frequency reference in the unit set in <i>o1-03 [Keypad Display Selection]</i>. Note: The value set to <i>o1-03 [Keypad Display Selection]</i> is changed to 1 [0.01% (100% = <i>E1-04</i>)] when <i>A1-02</i> = 6, 7 [Control Method Selection = AOLV/PM, CLV/PM].	0.00 Hz (0.00 - 590.00 Hz)	677
d1-17 (0292) RUN	Jog Reference	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the Jog frequency reference according to the units set in <i>o1-03 [Frequency Display Unit Selection]</i>. Set <i>H1-xx</i> = 6 [Terminal Sx Function Selection = Jog Reference Selection] to use the Jog frequency reference. Note: The value set to <i>o1-03 [Keypad Display Selection]</i> is changed to 1 [0.01% (100% = <i>E1-04</i>)] when <i>A1-02</i> = 6, 7 [Control Method Selection = AOLV/PM, CLV/PM].	6.00 Hz (0.00 - 590.00 Hz)	677

◆ d2: Reference Limits

No. (Hex.)	Name	Description	Default (Range)	Ref.
d2-01 (0289)	Frequency Reference Upper Limit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the maximum frequency reference as a percentage of <i>E1-04 [Maximum Output Frequency]</i>. This limit applies to all frequency references.	100.0% (0.0 - 110.0%)	678
d2-02 (028A)	Frequency Reference Lower Limit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the minimum frequency reference as a percentage of <i>E1-04 [Maximum Output Frequency]</i>. This limit applies to all frequency references.	0.0% (0.0 - 110.0%)	678
d2-03 (0293)	Analog Speed Reference Low Limit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the lower limit of the master frequency reference (Multi-Step Speed 1) as a percentage of <i>E1-04 [Maximum Output Frequency]</i>.	0.0% (0.0 - 110.0%)	678

◆ d3: Jump Frequency

No. (Hex.)	Name	Description	Default (Range)	Ref.
d3-01 (0294)	Jump Frequency 1	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the median value of the specific frequency band that needs to be jumped.	0.0 Hz (Determined by A1-02)	679
d3-02 (0295)	Jump Frequency 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the median value of the specific frequency band that needs to be jumped.	0.0 Hz (Determined by A1-02)	679
d3-03 (0296)	Jump Frequency 3	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the median value of the specific frequency band that needs to be jumped.	0.0 Hz (Determined by A1-02)	679
d3-04 (0297)	Jump Frequency Width	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the width of a specific frequency band that needs to be jumped.	1.0 Hz (Determined by A1-02)	679

◆ d4: Frequency Reference Hold and Up/Down 2 Function

No. (Hex.)	Name	Description	Default (Range)	Ref.
d4-01 (0298)	Freq Reference Retention Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Determines whether the frequency reference or the frequency bias (Up/Down 2) value is saved when the Stop command is entered or the power supply is shut down. This parameter is effective when <i>H1-xx</i> [MFDI Function Select] has been set to one of the following. <i>H1-xx</i> = A [Accel/Decel Ramp Hold] <i>H1-xx</i> = 10/11 [Up/Down Command] <i>H1-xx</i> = 75/76 [Up/Down 2 Command] The Frequency Reference Hold function depends on which function it is combined with. 0 : Disabled 1 : Enabled	0 (0, 1)	680
d4-03 (02AA) RUN	Up/Down 2 Bias Level	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the bias added to or subtracted from the frequency reference by the Up/Down 2 function.	0.00 Hz (0.00 - 99.99 Hz)	682
d4-04 (02AB) RUN	Up/Down 2 Ramp Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the acceleration and deceleration times to use when adding or subtracting the bias to/from the frequency reference when using the Up/Down 2 function. 0 : Use selected Accel/Decel time 1 : Use Accel/Decel Time 4	0 (0, 1)	682
d4-05 (02AC) RUN	Up/Down 2 Bias Mode Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Determines if the bias value is saved to the drive when <i>Up/Down 2 Command</i> [<i>H1-xx</i> = 75, 76] are both released or both enabled. This parameter is effective only when <i>d4-03</i> [Up/Down 2 Bias Level] = 0.00. 0 : Hold bias @ no Up/Down selected 1 : 0 bias @ neither/both selected	0 (0, 1)	683
d4-06 (02AD)	Frequency Ref Bias (Up/Down 2)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Saves the bias value from the Up/Down 2 Command assuming that <i>E1-04</i> [Maximum Output Frequency] is 100%.	0.0% (-99.9 - +100.0%)	683
d4-07 (02AE) RUN	Analog Freq Ref Fluctuate Limit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV If the frequency reference changes for more than the level set to this parameter, then the bias value will be held. Parameter <i>E1-04</i> [Maximum Output Frequency] is 100%.	1.0% (0.1 - 100.0%)	683
d4-08 (02AF) RUN	Up/Down 2 Bias Upper Limit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the upper limit of the Up/Down 2 bias as a percentage of <i>E1-04</i> [Maximum Output Frequency].	100.0% (0.0 - 100.0%)	684
d4-09 (02B0) RUN	Up/Down 2 Bias Lower limit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the lower limit of the Up/Down 2 bias as a percentage of <i>E1-04</i> [Maximum Output Frequency].	0.0% (-99.9 - 0.0%)	684
d4-10 (02B6)	Up/Down Freq Lower Limit Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects how the lower frequency limit is set when using the Up/Down function. 0 : d2-02 or Analog (larger level) 1 : d2-02	0 (0, 1)	684
d4-11 (02B7)	Bi-Directional Output Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects whether to change the frequency reference to a Bi-Directional internal frequency reference. 0 : Disabled 1 : Enabled	0 (0, 1)	684
d4-12 (02B8)	Stop Position Gain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the gain to adjust the stopping accuracy. Set this parameter when <i>b1-03</i> = 9 [Stopping Method Selection = Stop in Position].	1.00 (0.50 - 2.55)	686

◆ d5: Torque Control

No. (Hex.)	Name	Description	Default (Range)	Ref.
d5-01 (029A)	Torque Control Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Enables/disables Torque Control. 0 : Speed Control 1 : Torque Control	0 (0, 1)	689
d5-02 (029B)	Torque Reference Delay Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the primary delay time constant for the torque reference filter.	Determined by A1-02 (0 - 1000 ms)	689
d5-03 (029C)	Speed Limit Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the speed limit method associated with torque control. 1 : Active frequency reference 2 : d5-04 setting	1 (1, 2)	690
d5-04 (029D)	Speed Limit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the speed limit during Torque Control as a percentage of the Maximum Output Frequency. This parameter is effective when $d5-03 = 2$ [<i>Speed Limit Selection = d5-04 setting</i>].	0% (-120 - +120%)	690
d5-05 (029E)	Speed Limit Bias	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets a bias to the speed limit value as a percentage of E1-04 [Maximum Output Frequency].	10% (0 - 120%)	690
d5-06 (029F)	Speed/Torque Changeover Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the delay time for switching between Speed Control and Torque Control using the multi-function digital input terminal. This parameter is effective when $H1-xx = 71$ [<i>MFDI Function Select = Speed/Torque Control Switch</i>] has been set.	0 ms (0 - 1000 ms)	690
d5-08 (02B5)	Unidirectional Speed Limit Bias	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the direction of the speed limit reference to which <i>Speed Limit Bias</i> [d5-05] applies. 0 : Disabled 1 : Enabled	1 (0, 1)	690

◆ d6: Field Weak & Field Force

No. (Hex.)	Name	Description	Default (Range)	Ref.
d6-01 (02A0)	Field Weakening Level	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the output voltage of the drive when the <i>Field weakening</i> [$H1-xx = 63$] is input as a percentage of the maximum output voltage.	80% (0 - 100%)	691
d6-02 (02A1)	Field Weakening Frequency Limit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the minimum output frequency at which field weakening can be activated.	0.0 Hz (0.0 - 590.0 Hz)	691
d6-03 (02A2)	Field Forcing Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Enables or disables the field forcing function. 0 : Disabled 1 : Enabled	0 (0, 1)	691
d6-06 (02A5)	Field Forcing Limit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the maximum level at which the Field Forcing function can boost the excitation current reference as a percentage of the motor no load current. Normally there is no need to change this setting.	400% (100 - 400%)	691

◆ d7: Offset Frequency

No. (Hex.)	Name	Description	Default (Range)	Ref.
d7-01 (02B2) RUN	Offset Frequency 1	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Adds or subtracts the set frequency to/from the frequency reference using $H1-xx = 44$ [MFDI Function Select = Offset frequency 1] as a percentage of the maximum output frequency.	0.0% (-100.0 - +100.0%)	692
d7-02 (02B3) RUN	Offset Frequency 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Adds or subtracts the set frequency to/from the frequency reference using $H1-xx = 45$ [MFDI Function Select = Offset frequency 2] as a percentage of the maximum output frequency.	0.0% (-100.0 - +100.0%)	692
d7-03 (02B4) RUN	Offset Frequency 3	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Adds or subtracts the set frequency to/from the frequency reference using $H1-xx = 46$ [MFDI Function Select = Offset frequency 3] as a percentage of the maximum output frequency.	0.0% (-100.0 - +100.0%)	692

10.8 E: Motor Parameters

◆ E1: V/f Pattern for Motor 1

No. (Hex.)	Name	Description	Default (Range)	Ref.
E1-01 (0300)	Input AC Supply Voltage	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the drive input voltage. Set this parameter to the nominal voltage of the AC power supply. NOTICE: Set parameter E1-01 [Input AC Supply Voltage] to match the input voltage of the drive. Drive input voltage (not motor voltage) must be set in E1-01 for the protective features to function properly. Failure to set the correct drive input voltage will result in improper drive operation.	200 V Class: 230 V, 400 V: 400 V (200 V Class: 155 to 255 V, 400 V Class: 310 to 510 V)	694
E1-03 (0302)	V/f Pattern Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the V/f pattern for the drive and motor from 15 predefined patterns (settings: 0 through E) or creates a custom V/f pattern (setting: F). 0 : Constant Trq_50Hz base_50Hz max 1 : Constant Trq_60Hz base_60Hz max 2 : Constant Trq_50Hz base_60Hz max 3 : Constant Trq_60Hz base_72Hz max 4 : Variable Trq_50Hz base_35% mid V 5 : Variable Trq_50Hz base_50% mid V 6 : Variable Trq_60Hz base_35% mid V 7 : Variable Trq_60Hz base_50% mid V 8 : High Start Trq_50Hz base_125% V 9 : High Start Trq_50Hz base_165% V A : High Start Trq_60Hz base_125% V B : High Start Trq_60Hz base_165% V C : High Freq_60Hz base_90Hz max D : High Freq_60Hz base_120Hz max E : High Freq_60Hz base_180Hz max F : Custom Note: - Setting 0 through E cannot be selected when A1-02 = 2 [Control Method Selection = Open Loop Vector Control]. - Select the appropriate V/f pattern in accordance with the application and usage environment. Setting an improper V/f pattern may result in low motor torque or increased current due to overexcitation.	F (Determined by A1-02)	694
E1-04 (0303)	Maximum Output Frequency	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the maximum output frequency for the V/f pattern.	Determined by A1-02 and E5-01 (Determined by A1-02 and E5-01)	699
E1-05 (0304)	Maximum Output Voltage	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the maximum voltage for the V/f pattern.	Determined by A1-02 (200 V Class: 0.0 - 255.0 V, 400 V Class: 0.0 - 510.0 V)	700
E1-06 (0305)	Base Frequency	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the base frequency for the V/f pattern.	Determined by A1-02 and E5-01 (0.0 to E1-04)	700
E1-07 (0306)	Mid Point A Frequency	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the middle output frequency.	Determined by A1-02 (0.0 to E-04)	700
E1-08 (0307)	Mid Point A Voltage	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the middle output frequency.	Determined by A1-02 (200 V Class: 0.0 to 255.0 V, 400 V Class: 0.0 to 510.0 V)	700

10.8 E: Motor Parameters

No. (Hex.)	Name	Description	Default (Range)	Ref.
E1-09 (0308)	Minimum Output Frequency	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the minimum output frequency for the V/f pattern.	Determined by A1-02 and E5-01 (Determined by A1-02, E1-04, and E5-01)	700
E1-10 (0309)	Minimum Output Voltage	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the minimum output voltage.	Determined by A1-02 (200 V Class: 0.0 - 255.0 V, 400 V Class: 0.0 - 510.0 V)	700
E1-11 (030A)	Mid Point B Frequency	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the mid point B frequency.	0.0 Hz (0.0 to E-04)	700
E1-12 (030B)	Mid Point B Voltage	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the mid point B voltage.	0.0 V (200 V Class: 0.0 - 255.0 V, 400 V Class: 0.0 - 510.0 V)	701
E1-13 (030C)	Base Voltage	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the base voltage.	0.0 V (200 V Class: 0.0 - 255.0 V, 400 V Class: 0.0 - 510.0 V)	701

◆ E2: Motor Parameters

No. (Hex.)	Name	Description	Default Setting (Range)	Ref.
E2-01 (030E)	Motor Rated Current	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the motor rated current in amps.	Determined by o2-04 and C6-01 (10% to 200% of the drive rated current)	220
E2-02 (030F) RUN	Motor Rated Slip	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the amount of motor rated slip.	Determined by o2-04 and C6-01 (0.000 - 20.000 Hz)	702
E2-03 (0310)	Motor No-Load Current	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Set the no-load current for the motor in amperes when operating at the rated frequency and the no-load voltage.	Determined by o2-04 and C6-01 (0 to E2-01)	702
E2-04 (0311)	Motor Pole Count	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the number of motor poles.	4 (2 - 48)	702
E2-05 (0312)	Motor Line-to-Line Resistance	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the line-to-line resistance for motor stator windings.	Determined by o2-04 and C6-01 (0.000 - 65.000 Ω)	702
E2-06 (0313)	Motor Leakage Inductance	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the voltage drop due to motor leakage inductance as a percentage of motor rated voltage while the motor is operating at the rated frequency and rated current.	Determined by o2-04 and C6-01 (0.0 - 40.0%)	703
E2-07 (0314)	Motor Saturation Coefficient 1	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the motor iron-core saturation coefficient at 50% of the magnetic flux.	0.50 (0.00 - 0.50)	703
E2-08 (0315)	Motor Saturation Coefficient 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the motor iron-core saturation coefficient at 75% of the magnetic flux.	0.75 (E2-07 to 0.75)	703
E2-09 (0316)	Motor Mechanical Loss	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the mechanical loss of the motor. The motor rated power (kW) is 100%. Normally there is no need to change this setting.	0.0% (0.0 - 10.0%)	703

No. (Hex.)	Name	Description	Default Setting (Range)	Ref.
E2-10 (0317)	Motor Iron Loss	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the motor iron loss in watts.	Determined by o2-04 and C6-01 (0 - 65535 W)	703
E2-11 (0318)	Motor Rated Power (kW)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the motor rated power in 0.01 kW units. (1 HP = 0.746 kW)	Determined by o2-04 and C6-01 (0.00 - 650.00 kW)	703

◆ E3: V/f Pattern for Motor 2

No. (Hex.)	Name	Description	Default (Range)	Ref.
E3-01 (0319)	Motor 2 Control Mode Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the control method for motor 2. Note: Changing the motor 2 control mode selection changes the settings value of parameters dependent on <i>E3-01</i> to the default settings. 0 : V/f Control 1 : Closed Loop V/f Control 2 : Open Loop Vector Control 3 : Closed Loop Vector Control	0 (0 - 3)	704
E3-04 (031A)	Motor 2 Maximum Output Frequency	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the maximum output frequency used for motor 2.	Determined by E3-01 (40.0 - 590.0 Hz)	704
E3-05 (031B)	Motor 2 Maximum Output Voltage	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the maximum output voltage used for motor 2.	Determined by E3-01 (200 V Class: 0.0 to 255.0 V, 400 V Class: 0.0 to 510.0 V)	705
E3-06 (031C)	Motor 2 Base Frequency	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the base frequency used for motor 2.	Determined by E3-01 (0.0 to E3-04)	705
E3-07 (031D)	Motor 2 Mid Point A Frequency	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the middle output frequency used for motor 2.	Determined by E3-01 (0.0 to E3-04)	705
E3-08 (031E)	Motor 2 Mid Point A Voltage	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the middle output frequency voltage used for motor 2.	Determined by E3-01 (200 V Class: 0.0 to 255.0 V, 400 V Class: 0.0 to 510.0 V)	705
E3-09 (031F)	Motor 2 Minimum Output Frequency	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the minimum output frequency used for motor 2.	Determined by E3-01 (0.0 to E3-04)	705
E3-10 (0320)	Motor 2 Minimum Output Voltage	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the minimum output voltage used for motor 2.	Determined by E3-01 (200 V Class: 0.0 to 255.0 V, 400 V Class: 0.0 to 510.0 V)	705
E3-11 (0345)	Motor 2 Mid Point B Frequency	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the mid point B frequency used for motor 2. Configure this parameter only when the V/f pattern for the constant output range needs to be adjusted. Normally there is no need to configure this setting.	0.0 Hz (0.0 to E3-04)	705

No. (Hex.)	Name	Description	Default (Range)	Ref.
E3-12 (0346)	Motor 2 Mid Point B Voltage	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the mid point B voltage used for motor 2. Configure this parameter only when the V/f pattern for the constant output range needs to be adjusted. Normally there is no need to configure this setting.	0.0 V (200 V Class: 0.0 - 255.0 V, 400 V Class: 0.0 - 510.0 V)	706
E3-13 (0347)	Motor 2 Base Voltage	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the base voltage used for motor 2. Configure this parameter only when the V/f pattern for the constant output range needs to be adjusted. Normally there is no need to configure this setting.	0.0 V (200 V Class: 0.0 - 255.0 V, 400 V Class: 0.0 - 510.0 V)	706

◆ E4: Motor 2 Parameters

No. (Hex.)	Name	Description	Default (Range)	Ref.
E4-01 (0321)	Motor 2 Rated Current	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the motor rated current for motor 2 in amperes.	Determined by o2-04 and C6-01 (10% to 200% of the drive rated current)	706
E4-02 (0322) RUN	Motor 2 Rated Slip	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the amount of motor 2 rated slip.	Determined by o2-04 and C6-01 (0.000 - 20.000 Hz)	706
E4-03 (0323)	Motor 2 Rated No-Load Current	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the no-load current for motor 2 in amperes when operating at the rated frequency and the no-load voltage.	Determined by o2-04 and C6-01 (0 to E4-01)	707
E4-04 (0324)	Motor 2 Motor Poles	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the number of poles for motor 2.	4 (2 - 48)	707
E4-05 (0325)	Motor 2 Line-to-Line Resistance	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the line-to-line resistance for motor 2 stator windings.	Determined by o2-04 and C6-01 (0.000 - 65.000 Ω)	707
E4-06 (0326)	Motor 2 Leakage Inductance	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the voltage drop due to motor 2 leakage inductance as a percentage of motor 2 rated voltage while the motor 2 is operating at the rated frequency and rated current.	Determined by o2-04 and C6-01 (0.0 - 40.0%)	707
E4-07 (0343)	Motor2 Saturation Coefficient 1	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the motor 2 iron-core saturation coefficient at 50% of the magnetic flux.	0.50 (0.00 - 0.50)	708
E4-08 (0344)	Motor 2 Saturation Coefficient 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the motor 2 iron-core saturation coefficient at 75% of the magnetic flux.	0.75 (E4-07 to 0.75)	708
E4-09 (033F)	Motor 2 Mechanical Loss	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the mechanical loss of motor 2. The motor rated power (kW) is 100%. Normally there is no need to change this setting.	0.0% (0.0 - 10.0%)	708
E4-10 (0340)	Motor 2 Iron Loss	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the motor iron loss for motor 2 in watts.	Determined by o2-04 and C6-01 (0 - 65535 W)	708
E4-11 (0327)	Motor 2 Rated Power	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the motor 2 rated power in 0.01 kW. (1 HP = 0.746 kW)	Determined by o2-04 and C6-01 (0.00 - 650.00 kW)	708

◆ E5: PM Motor Settings

No. (Hex.)	Name	Description	Default (Range)	Ref.
E5-01 (0329)	PM Motor Code Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV When using Yaskawa motors, set the motor code for the PM motor being used. The drive automatically sets several parameters to appropriate values depending on the motor code.	Determined by A1-02, o2-04, and C6-01 (0000 - FFFF)	709
E5-02 (032A)	Motor Rated Capacity	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the rated power of PM motors.	Determined by E5-01 (0.10 - 650.00 kW)	709
E5-03 (032B)	PM Motor Rated Current (FLA)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the motor rated current (FLA) for PM motors.	Determined by E5-01 (10 to 200% of the drive rated current)	220
E5-04 (032C)	PM Motor Pole Count	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the number of PM motor poles.	Determined by E5-01 (2 - 48)	710
E5-05 (032D)	PM Motor Resistance (ohms/phase)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the resistance per phase of the PM motors. Do not enter the line-to-line resistance into E5-05 when measuring the resistance manually.	Determined by E5-01 (0.000 - 65.000 Ω)	710
E5-06 (032E)	PM d-axis Inductance (mH/phase)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the PM motor d-Axis inductance.	Determined by E5-01 (0.00 - 300.00 mH)	710
E5-07 (032F)	PM q-axis Inductance (mH/phase)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the PM motor q-Axis inductance.	Determined by E5-01 (0.00 - 600.00 mH)	710
E5-09 (0331)	PM Back-EMF V _{peak} (mV/(rad/s))	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the peak value of PM motor induced voltage in units of electrical angles.	Determined by E5-01 (0.0 - 2000.0 mV/(rad/s))	711
E5-11 (0333)	Encoder Z-Pulse Offset	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the encoder Z-pulse offset.	0.0 degrees (-180.0 - +180.0 degrees)	711
E5-24 (0353)	PM Back-EMF L-L Vrms (mV/rpm)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the rms value for PM motor line voltage in units of mechanical angles.	Determined by E5-01 (0.0 - 6500.0 mV/min ⁻¹)	711
E5-25 (035E)	Polarity Estimation Reversal	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Changes the polarity estimate used when estimating initial polarity. Normally there is no need to change this setting. 0 : Disabled 1 : Enabled	0 (0, 1)	711

◆ E9: Motor Setting

No. (Hex.)	Name	Description	Default (Range)	Ref.
E9-01 (11E4)	Motor Type Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the motor type. 0 : IM 1 : PM 2 : SynRM	0 (0 to 2)	712
E9-02 (11E5)	Motor Max Revolutions	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the max revolutions of the motor.	Determined by E9-01 (40.0 - 120.0 Hz)	712
E9-03 (11E6)	Rated Speed	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the rated rotation speed of the motor.	Determined by E9-01 (100 - 7200 min ⁻¹)	712

10.8 E: Motor Parameters

No. (Hex.)	Name	Description	Default (Range)	Ref.
E9-04 (11E7)	Base Frequency	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the rated frequency of the motor.	Determined by E9-01 (40.0 - 120.0 Hz)	712
E9-05 (11E8)	Base Voltage	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the rated voltage of the motor.	Determined by E9-01 (200 V Class: 0.0 to 255.0 V, 400 V Class: 0.0 to 510.0 V)	713
E9-06 (11E9)	Motor Rated Current	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the motor rated current in amperes.	Determined by E9-01 and o2-04 (10% to 200% of the drive rated current)	221
E9-07 (11EA)	Motor Rated Power (kW)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the motor rated power in 0.01 kW. (1 HP = 0.746 kW)	Determined by E9-02 and o2-04 (0.00 - 650.00 kW)	713
E9-08 (11EB)	Motor Pole Count	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the number of motor poles.	4 (2 to 48)	713
E9-09 (11EC)	Motor Rated Slip	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the motor rated slip.	0.0 Hz (0.0 - 20.0 Hz)	713
E9-10 (11ED)	Motor Line-to-Line Resistance	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the line-to-line resistance for motor stator windings.	Determined by o2-04 (0.000 - 65.000 Ω)	714

10.9 F: Options

◆ F1: PG Speed Control Card (Encoder)

No. (Hex.)	Name	Description	Default (Range)	Ref.
F1-01 (0380)	PG 1 Pulses Per Revolution	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the number of output pulses per revolution of the encoder.	1024 ppr (1 - 60000 ppr)	716
F1-02 (0381)	PG Feedback Loss Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the motor operation when <i>PGo</i> [<i>PG Disconnect</i>] is detected. 0 : Ramp to stop 1 : Coast to stop 2 : Fast Stop (use C1-09) 3 : Alarm only 4 : No alarm display	1 (0 - 4)	716
F1-03 (0382)	Operation Select at Overspeed	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the motor operation when <i>oS</i> [<i>Overspeed</i>] is detected. 0 : Ramp to stop 1 : Coast to stop 2 : Fast Stop (use C1-09) 3 : Alarm only 4 : No alarm display	1 (0 - 3)	716
F1-04 (0383)	Operation Selection at Deviation	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the motor operation when <i>dEv</i> [<i>Speed Deviation</i>] is detected. 0 : Ramp to stop 1 : Coast to stop 2 : Fast Stop (use C1-09) 3 : Alarm only	3 (0 - 3)	717
F1-05 (0384)	PG 1 Rotation Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the output sequence for phases A and B of the pulses output from the encoder, given that the motor is running forward. 0 : Pulse A leads 1 : Pulse B leads	Determined by A1-02 (0, 1)	717
F1-06 (0385)	PG1 Division Rate for Pulse Mon	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the dividing ratio for monitor signals output from the encoder option card.	001 (001 - 032, 102 - 132 (1 - 1/32))	717
F1-08 (0387)	Overspeed Detection Level	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the detection level of <i>oS</i> [<i>Overspeed</i>] as a percentage when the maximum output frequency is 100%.	115% (0 - 120%)	718
F1-09 (0388)	Overspeed Detection Delay Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the detection time for <i>oS</i> [<i>Overspeed</i>].	Determined by A1-02 (0.0 - 2.0 s)	718
F1-10 (0389)	Speed Deviation Detection Level	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the detection level of <i>dEv</i> [<i>Speed Deviation</i>] as a percentage when the maximum output frequency is 100%.	10% (0 - 50%)	718
F1-11 (038A)	Speed Deviation Detect Delay Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the detection time for <i>dEv</i> [<i>Speed Deviation</i>].	0.5 s (0.0 - 10.0 s)	718
F1-12 (038B)	PG 1 Gear Teeth 1	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the number of gear teeth (gear ratio) between the motor and encoder, in combination with <i>F1-13</i> [<i>PG 1 Gear Teeth 2</i>]. <i>F1-12</i> is set with the number of gear teeth for the motor side.	0 (0 - 1000)	718
F1-13 (038C)	PG 1 Gear Teeth 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the number of gear teeth (gear ratio) between the motor and encoder, in combination with <i>F1-12</i> [<i>PG 1 Gear Teeth 1</i>]. Parameter <i>F1-13</i> is set with the number of gear teeth for the load side.	0 (0 - 1000)	719

No. (Hex.)	Name	Description	Default (Range)	Ref.
F1-14 (038D)	PG Open-Circuit Detection Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the detection time for <i>PGo</i> [<i>PG Disconnect</i>]. Note: Faults such as <i>ov</i> [<i>DB Bus Overvoltage</i>] and <i>oC</i> [<i>Overcurrent</i>] may occur depending on the motor speed and load conditions.	2.0 s (0.0 - 10.0 s)	719
F1-18 (03AD)	Deviation 3 Detection Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the number of rotations to detect scenarios in which the torque reference and rate of acceleration are inverted, which function as the detection conditions for <i>dv3</i> [<i>Inversion Detection</i>].	10 (0 - 10)	719
F1-19 (03AE)	Deviation 4 Detection Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the number of pulses used to detect <i>dv4</i> [<i>Inversion Prevention Detection</i>].	128 (0 - 5000)	719
F1-20 (03B4)	PG Option PCB Disconnect Detect1	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects whether to enable or disable the disconnection detection function for the encoder connection cable regarding the PG-X3 and PG-F3. Detects <i>PGoH</i> [<i>PG Hardware Fault</i>] when <i>F1-20</i> = 1. 0 : Disabled 1 : Enabled	1 (0, 1)	720
F1-21 (03BC)	PG 1 Signal Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the type of pulse signal (channel) used for the encoder option card. 0 : A pulse detection 1 : AB pulse detection	0 (0, 1)	720
F1-30 (03AA)	Motor 2 PG Option Port Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the connector used when the motor 2 encoder option card is mounted in the drive. 0 : CN5-C 1 : CN5-B	1 (0, 1)	720
F1-31 (03B0)	PG 2 Pulses Per Revolution	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the number of output pulses per revolution of the encoder. This parameter is for motor 2.	1024 ppr (1 - 60000 ppr)	720
F1-32 (03B1)	PG 2 Rotation Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the output sequence for phases A and B of the pulses output from the encoder, given that the motor is running forward. This parameter is for motor 2. 0 : Pulse A leads 1 : Pulse B leads	0 (0, 1)	720
F1-33 (03B2)	PG 2 Gear Teeth 1	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Set the number of gear teeth (gear ratio) between the motor and encoder, in combination with <i>F1-34</i> [<i>PG 2 Gear Teeth 2</i>]. <i>F1-33</i> is set with the number of gear teeth for the motor side. This parameter is for motor 2.	0 (0 - 1000)	721
F1-34 (03B3)	PG 2 Gear Teeth 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the number of gear teeth (gear ratio) between the motor and encoder, in combination with <i>F1-33</i> [<i>PG 2 Gear Teeth 1</i>]. <i>F1-34</i> is set with the number of gear teeth for the load side. This parameter is for motor 2.	0 (0 - 1000)	721
F1-35 (03BE)	PG2 Division Rate for Pulse Mon	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the dividing ratio for monitor signals output from the encoder option card. This parameter is for motor 2.	001 (001 - 032, 102 - 132 (1 - 1/32))	721
F1-36 (03B5)	PG Option PCB Disconnect Detect2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects whether to enable or disable the disconnection detection function for the encoder connection cable regarding the PG-X3. <i>PGoH</i> [<i>PG Hardware Fault</i>] is detected when this parameter is enabled. This parameter is for motor 2. 0 : Disabled 1 : Enabled	1 (0, 1)	721

No. (Hex.)	Name	Description	Default (Range)	Ref.
F1-37 (03BD)	PG 2 Signal Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the type of pulse signal (channel) used for the encoder option card. This parameter is for motor 2. 0 : A pulse detection 1 : AB pulse detection	0 (0, 1)	721
F1-50 (03D2)	Encoder Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the encoder connected to PG-F3. 0 : EnDat Sin/Cos 1 : EnDat Serial Only 2 : Hiperface	0 (0 - 2)	722
F1-51 (03D3)	PGoH Detection Level	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV $\sqrt{\sin^2\theta + \cos^2\theta}$ Sets the detection level for <i>PGoH</i> [<i>PG Hardware Fault</i>] of PG-F3 as a percentage when XXX is 100%. Note: This function is enabled when <i>F1-20 = 1</i> [<i>PG Hardware Disconnection Detection Selection = Enabled</i>].	80% (1 - 100%)	722
F1-52 (03D4)	Serial Encoder Communication bps	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the speed of communication between the PG-F3 and serial encoder. 0 : 1M/9600bps 1 : 500k/19200bps 2 : 1M/38400bps	0 (0 - 2)	722

◆ F2: Analog Input Option

No. (Hex.)	Name	Description	Default (Range)	Ref.
F2-01 (038F)	Analog Input Function Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the input method for the analog reference used with AI-A3. 0 : 3 channel individual 1 : 3 channel addition	0 (0, 1)	722
F2-02 (0368) RUN	Analog Input Option Card Gain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the analog reference gain as a percentage when the maximum output frequency is 100%. Note: This parameter is only enabled when <i>F2-01 = 1</i> [<i>Analog Input Function Selection = 3 channel addition</i>].	100.0% (-999.9 - +999.9%)	724
F2-03 (0369) RUN	Analog Input Option Card Bias	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the analog reference bias as a percentage when the maximum output frequency is 100%. Note: This parameter is only enabled when <i>F2-01 = 1</i> [<i>Analog Input Function Selection = 3 channel addition</i>].	0.0% (-999.9 - +999.9%)	724

◆ F3: Digital Input Option

No. (Hex.)	Name	Description	Default (Range)	Ref.
F3-01 (0390)	Digital Input Function Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the data format of digital input signals. This parameter is enabled when $o1-03 = 0$ or 1 [Keypad Display Selection = 0.01 Hz or 0.01% (100% = E1-04)]. Note: The DI-A3 input method is set to the BCD input method regardless of the setting of F3-01 when $o1-03 = 2$ or 3 [r/min or User-selected units]. In this scenario, the value set in $o1-03$ is used as the setting unit. 0 : BCD, 1% units 1 : BCD, 0.1% units 2 : BCD, 0.01% units 3 : BCD, 1 Hz units 4 : BCD, 0.1 Hz units 5 : BCD, 0.01 Hz units 6 : BCD (5-digit), 0.01 Hz 7 : Binary input 8 : Multi-function Digital input	0 (0 - 8)	725
F3-03 (03B9)	DI Data Length Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the number of bits used to set the frequency reference with DI-A3. 0 : 8bit 1 : 12bit 2 : 16bit	2 (0 to 2)	725
F3-10 (0BE3)	Terminal D0 Function Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the function for terminal D0 of the DI-A3 when $F3-01 = 8$ [Digital Input Function Selection = Multi-function Digital input].	F (1 - 19F)	726
F3-11 (0BE4)	Terminal D1 Function Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the function for terminal D1 of the DI-A3 when $F3-01 = 8$ [Digital Input Function Selection = Multi-function Digital input].	F (1 - 19F)	727
F3-12 (0BE5)	Terminal D2 Function Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the function for terminal D2 of the DI-A3 when $F3-01 = 8$ [Digital Input Function Selection = Multi-function Digital input].	F (1 - 19F)	727
F3-13 (0BE6)	Terminal D3 Function Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the function for terminal D3 of the DI-A3 when $F3-01 = 8$ [Digital Input Function Selection = Multi-function Digital input].	F (1 - 19F)	727
F3-14 (0BE7)	Terminal D4 Function Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the function for terminal D4 of the DI-A3 when $F3-01 = 8$ [Digital Input Function Selection = Multi-function Digital input].	F (1 - 19F)	727
F3-15 (0BE8)	Terminal D5 Function Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the function for terminal D5 of the DI-A3 when $F3-01 = 8$ [Digital Input Function Selection = Multi-function Digital input].	F (1 - 19F)	727
F3-16 (0BE9)	Terminal D6 Function Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the function for terminal D6 of the DI-A3 when $F3-01 = 8$ [Digital Input Function Selection = Multi-function Digital input].	F (1 - 19F)	727
F3-17 (0BEA)	Terminal D7 Function Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the function for terminal D7 of the DI-A3 when $F3-01 = 8$ [Digital Input Function Selection = Multi-function Digital input].	F (1 - 19F)	727
F3-18 (0BEB)	Terminal D8 Function Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the function for terminal D8 of the DI-A3 when $F3-01 = 8$ [Digital Input Function Selection = Multi-function Digital input].	F (1 - 19F)	727
F3-19 (0BEC)	Terminal D9 Function Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the function for terminal D9 of the DI-A3 when $F3-01 = 8$ [Digital Input Function Selection = Multi-function Digital input].	F (1 - 19F)	728
F3-20 (0BED)	Terminal DA Function Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the function for terminal DA of the DI-A3 when $F3-01 = 8$ [Digital Input Function Selection = Multi-function Digital input].	F (1 - 19F)	728

No. (Hex.)	Name	Description	Default (Range)	Ref.
F3-21 (0BEE)	Terminal DB Function Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the function for terminal DB of the DI-A3 when $F3-01 = 8$ [Digital Input Function Selection = Multi-function Digital input].	F (1 - 19F)	728
F3-22 (0BEF)	Terminal DC Function Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the function for terminal DC of the DI-A3 by when $F3-01 = 8$ [Digital Input Function Selection = Multi-function Digital input].	F (1 - 19F)	728
F3-23 (0BF0)	Terminal DD Function Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the function for terminal DD of the DI-A3 when $F3-01 = 8$ [Digital Input Function Selection = Multi-function Digital input].	F (1 - 19F)	728
F3-24 (0BF1)	Terminal DE Function Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the function for terminal DE of the DI-A3 when $F3-01 = 8$ [Digital Input Function Selection = Multi-function Digital input].	F (1 - 19F)	728
F3-25 (0BF2)	Terminal DF Function Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the function for terminal DF of the DI-A3 when $F3-01 = 8$ [Digital Input Function Selection = Multi-function Digital input].	F (1 - 19F)	728

◆ F4: Analog Monitor Option

No. (Hex.)	Name	Description	Default (Range)	Ref.
F4-01 (0391)	Terminal V1 Monitor Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the number for monitor item of output from terminal V1.	102 (000 - 999)	729
F4-02 (0392) RUN	Terminal V1 Monitor Gain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the gain of monitor signal output from terminal V1 as a percentage. Sets the voltage level output from terminal V1 to a 100% value of 10 V when a monitoring item is at 100% while an output of 0% for monitoring items is 0 V.	100.0% (-999.9 - +999.9%)	729
F4-03 (0393)	Terminal V2 Monitor Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the number for monitor item of output from terminal V2.	103 (000 - 999)	730
F4-04 (0394) RUN	Terminal V2 Monitor Gain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the gain of monitor signal output from terminal V2 as a percentage. Sets the voltage level output from terminal V2 to a 100% value of 10 V when a monitoring item is at 100% while an output of 0% for monitoring items is 0 V.	50.0% (-999.9 - +999.9%)	730
F4-05 (0395) RUN	Terminal V1 Monitor Bias	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the bias of monitor signal output from terminal V1 as a percentage. Sets the voltage level output from terminal V1 to a 100% value of 10 V when the output for monitoring items is 0%.	0.0% (-999.9 - +999.9%)	730
F4-06 (0396) RUN	Terminal V2 Monitor Bias	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the bias of monitor signal output from terminal V2 as a percentage. Sets the voltage level output from terminal V2 to a 100% value of 10 V when the output for monitoring items is 0%.	0.0% (-999.9 - +999.9%)	730
F4-07 (0397)	Terminal V1 Signal Level	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the level of signals output from the MFAO terminal V1. 0 : 0 to 10 V 1 : -10 to 10 V	0 (0, 1)	731
F4-08 (0398)	Terminal V2 Signal Level	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the level of signals output from the MFAO terminal V2. 0 : 0 to 10 V 1 : -10 to 10 V	0 (0, 1)	731

◆ F5: Digital Output Option

No. (Hex.)	Name	Description	Default (Range)	Ref.
F5-01 (0399)	Terminal P1-PC Output Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The function output from terminal P1-PC on the DO-A3 card is selected by the MFDO setting value. This parameter is enabled when $F5-09 = 2$ [DO-A3 Output Mode Selection = 8 channel select (F5-01 to F5-08)].	0 (0 - 1A7)	733
F5-02 (039A)	Terminal P2-PC Output Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The function output from terminal P2-PC on the DO-A3 card is selected by the MFDO setting value. This parameter is enabled when $F5-09 = 2$ [DO-A3 Output Mode Selection = 8 channel select (F5-01 to F5-08)].	1 (0 - 1A7)	733
F5-03 (039B)	Terminal P3-PC Output Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The function output from terminal P3-PC on the DO-A3 card is selected by the MFDO setting value. This parameter is enabled when $F5-09 = 2$ [DO-A3 Output Mode Selection = 8 channel select (F5-01 to F5-08)].	2 (0 - 1A7)	733
F5-04 (039C)	Terminal P4-PC Output Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The function output from terminal P4-PC on the DO-A3 card is selected by the MFDO setting value. This parameter is enabled when $F5-09 = 2$ [DO-A3 Output Mode Selection = 8 channel select (F5-01 to F5-08)].	4 (0 - 1A7)	733
F5-05 (039D)	Terminal P5-PC Output Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The function output from terminal P5-PC on the DO-A3 card is selected by the MFDO setting value. This parameter is enabled when $F5-09 = 2$ [DO-A3 Output Mode Selection = 8 channel select (F5-01 to F5-08)].	6 (0 - 1A7)	734
F5-06 (039E)	Terminal P6-PC Output Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The function output from terminal P6-PC on the DO-A3 card is selected by the MFDO setting value. This parameter is enabled when $F5-09 = 2$ [DO-A3 Output Mode Selection = 8 channel select (F5-01 to F5-08)].	37 (0 - 1A7)	734
F5-07 (039F)	Terminal M1-M2 Output Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the function output from terminal M3-M2 on the DO-A3 card by the setting value for the multi-function digital output. This parameter is enabled when $F5-09 = 2$ [DO-A3 Output Mode Selection = 8 channel select (F5-01 to F5-08)].	F (0 - 1A7)	734
F5-08 (03A0)	Terminal M3-M4 Output Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the function output from terminal M3-M4 on the DO-A3 card by the setting value for the multi-function digital output. This parameter is enabled when $F5-09 = 2$ [DO-A3 Output Mode Selection = 8 channel select (F5-01 to F5-08)].	F (0 - 1A7)	734
F5-09 (03A1)	DO-A3 Output Mode Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the output mode of signals output from the DO-A3 card. 0 : 8 channel individual 1 : Binary code output 2 : 8 channel select (F5-01 to F5-08)	0 (0 - 2)	734

◆ F6: Communication Option and Ethernet Option

No. (Hex.)	Name	Description	Default (Range)	Ref.
F6-01 (03A2)	Communication Error Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the operation of the drive when <i>bUS [Option Communication Error]</i> is detected. 0 : Ramp to stop 1 : Coast to stop 2 : Fast Stop (use C1-09) 3 : Alarm only 4 : Alarm - run at <i>d1-04</i> 5 : Alarm - Ramp Stop	1 (0 - 5)	736
F6-02 (03A3)	Comm External Fault (EF0) Detect	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the conditions at which <i>EF0 [Option Card External Fault]</i> is detected. 0 : Always detected 1 : Detection during run only	0 (0, 1)	736
F6-03 (03A4)	Comm External Fault (EF0) Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the operation of the drive when <i>EF0 [Option Card External Fault]</i> is detected. 0 : Ramp to stop 1 : Coast to stop 2 : Fast Stop (use C1-09) 3 : Alarm only	1 (0 - 3)	736
F6-04 (03A5)	bUS Error Detection Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the delay time until <i>bUS [Option Communication Error]</i> issues are detected. Note: The setting value changes to 0.0 s when the option card is mounted in the drive.	2.0 s (0.0 - 5.0 s)	737
F6-06 (03A7)	Torque Reference/Limit by Comm	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects whether to enable or disable the torque reference and torque limit received from the communication option card. 0 : Disabled 1 : Enabled	0 (0, 1)	737
F6-07 (03A8)	MultiStep Ref Priority Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects whether to enable/disable the multi-step speed reference when NetRef (communication option card) or ComRef (MEMOBUS/Modbus communications) is selected as the frequency reference source. 0 : MultiStep References Disabled 1 : MultiStep References Enabled	0 (0, 1)	737
F6-08 (036A)	Comm Parameter Reset @Initialize	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects whether or not to initialize <i>communication parameters [F6-xx and F7-xx]</i> when the drive is initialized by <i>A1-03 [Initialize Parameters]</i>. 0 : No Reset - parameters retained 1 : Reset - back to factory default	0 (0, 1)	737
F6-10 (03B6)	CC-Link Node Address	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the node address for CC-Link communication. The drive must be restarted when the setting is changed. Note: Set a node address that does overlap with other nodes. Do not set this parameter to a value of 0. If the parameter is set incorrectly, the L.ERR LED on the option is lit, and the drive will detect the <i>AEr [Node Address Setting Error]</i> error.	0 (0 - 64)	737

No. (Hex.)	Name	Description	Default (Range)	Ref.
F6-11 (03B7)	CC-Link Communication Speed	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the communication speed for CC-Link communication. The drive must be restarted when the setting is changed. 0 : 156 kbps 1 : 625 kbps 2 : 2.5 Mbps 3 : 5 Mbps 4 : 10 Mbps	0 (0 - 4)	738
F6-14 (03BB)	CC-Link bUS Error Auto Reset	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects whether to enable or disable the automatic reset of <i>bUS</i> [Option Communication Error] issues. 0 : Disabled 1 : Enabled	0 (0, 1)	738
F6-16 (0B8A)	Gateway Mode	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the operation mode of the gateway mode and the number of connected slave drives. 0 : Uses the drive as a slave drive or disables the gateway function. 1 : Uses the drive as a master drive (slave drives: 1). 2 : Uses the drive as a master drive (slave drives: 2). 3 : Uses the drive as a master drive (slave drives: 3). 4 : Uses the drive as a master drive (slave drives: 4).	0 (0 to 4)	437
F6-20 (036B)	MECHATROLINK Station Address	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the station address for MECHATROLINK communication. The drive must be restarted when the setting is changed. Note: - The setting range varies depending on the type of MECHATROLINK communication. –MECHATROLINK-II (SI-T3): 20 - 3F –MECHATROLINK-III (SI-ET3): 03 - EF - Set an address that does overlap with other nodes. If the parameter is set incorrectly, the ERR on the option card flashes, and the drive will detect the <i>AEr</i> [Node Address Setting Error] error. <i>AEr</i> issues are detected when the station address is set to either 20 or 3F.	0021h (MECHATRO LINK-II : 0020h - 003Fh , MECHATRO LINK-III : 0003h - 00EFh)	742
F6-21 (036C)	MECHATROLINK Frame Size	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the frame size for MECHATROLINK communication. The drive must be restarted when the setting is changed. 0 : 32-byte 1 : 17-byte	0 (0, 1)	742
F6-22 (036D)	MECHATROLINK Link Speed	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the communications speed for MECHATROLINK-II. The drive must be restarted when the setting is changed. Note: This parameter can only be used when the MECHATROLINK-II option is connected. 0 : 10 Mbps 1 : 4 Mbps	0 (0, 1)	742
F6-23 (036E)	MECHATROLINK Monitor Select (E)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the MEMOBUS register used for the monitor functions of INV_CTL (drive operation control command) and INV_I/O (drive I/O control command). The drive must be restarted when the setting is changed.	0000h (0000h - FFFFh)	743
F6-24 (036F)	MECHATROLINK Monitor Select (F)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the MEMOBUS register used for the monitor functions of INV_CTL (drive operation control command) and INV_I/O (drive I/O control command). The drive must be restarted when the setting is changed.	0000h (0000h - FFFFh)	743

No. (Hex.)	Name	Description	Default (Range)	Ref.
F6-25 (03C9)	MECHATROLINK Watchdog Error Sel	V/f CL-V/f OLV CLV AOLV OLVP/PM AOLV/PM CLV/PM EZOLV Selects the motor operation when <i>E5 [MECHATROLINK Watchdog Timer Err]</i> is detected. 0 : Ramp to stop 1 : Coast to stop 2 : Fast stop (use C1-09) 3 : Alarm Only	1 (0 - 3)	743
F6-26 (03CA)	MECHATROLINK bUS Errors Detected	V/f CL-V/f OLV CLV AOLV OLVP/PM AOLV/PM CLV/PM EZOLV <i>bUS [Option Communication Error]</i> is detected when the option card detects the <i>bUS</i> alarm for a number of times that exceeds the number set in <i>F6-26</i>.	2 times (2 to 10 times)	743
F6-30 (03CB)	PROFIBUS-DP Node Address	V/f CL-V/f OLV CLV AOLV OLVP/PM AOLV/PM CLV/PM EZOLV Sets the node address for PROFIBUS-DP communication. The drive must be restarted when the setting is changed.	0 (0 - 125)	743
F6-31 (03CC)	PROFIBUS-DP Clear Mode selection	V/f CL-V/f OLV CLV AOLV OLVP/PM AOLV/PM CLV/PM EZOLV Selects the operation of the drive after the Clear mode command is received. 0 : Reset 1 : Hold previous state	0 (0, 1)	744
F6-32 (03CD)	PROFIBUS-DP Data Format Select	V/f CL-V/f OLV CLV AOLV OLVP/PM AOLV/PM CLV/PM EZOLV Selects the data format of PROFIBUS-DP communication. The drive must be restarted when the setting is changed. 0 : PPO Type 1 : Conventional 2 : PPO (bit0) 3 : PPO (Enter) 4 : Conv (Enter) 5 : PPO (bit0,Enter)	0 (0 - 5)	744
F6-35 (03D0)	CANopen Node ID Selection	V/f CL-V/f OLV CLV AOLV OLVP/PM AOLV/PM CLV/PM EZOLV Sets the node address for CANopen communication. The drive must be restarted when the setting is changed. Note: Select an address that does not overlap with other nodes. Do not set this parameter to a value of 0. If the parameter is set incorrectly, the ERR on the option card flashes, and the drive will detect the <i>AEr [Node Address Setting Error]</i> error.	0 (0 - 126)	744
F6-36 (03D1)	CANopen Communication Speed	V/f CL-V/f OLV CLV AOLV OLVP/PM AOLV/PM CLV/PM EZOLV Selects the communications speed for CANopen communication. The drive must be restarted when the setting is changed. 0 : Auto-detection 1 : 10 kbps 2 : 20 kbps 3 : 50 kbps 4 : 125 kbps 5 : 250 kbps 6 : 500 kbps 7 : 800 kbps 8 : 1 Mbps	0 (0 - 8)	744
F6-45 (02FB)	BACnet Node Address	V/f CL-V/f OLV CLV AOLV OLVP/PM AOLV/PM CLV/PM EZOLV Sets the node address for BACnet communication.	1 (0 - 127)	745

No. (Hex.)	Name	Description	Default (Range)	Ref.
F6-46 (02FC)	BACnet Baud Rate	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the BACnet communications speed. 0 : 1200 bps 1 : 2400 bps 2 : 4800 bps 3 : 9600 bps 4 : 19.2 kbps 5 : 38.4 kbps 6 : 57.6 kbps 7 : 76.8 kbps 8 : 115.2 kbps	3 (0 - 8)	745
F6-47 (02FD)	Rx to Tx Wait Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the wait time for reception and transmission of BACnet communication.	5 ms (5 - 65 ms)	745
F6-48 (02FE)	BACnet Device Object Identifier0	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the last word of addresses for BACnet communication.	0 (0 - FFFF)	745
F6-49 (02FF)	BACnet Device Object Identifier1	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the last word of addresses for BACnet communication.	0 (0 - 3F)	745
F6-50 (03C1)	DN MAC Address	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the MAC address for DeviceNet communication. The drive must be restarted when the setting is changed. Note: Select a MAC address that does not overlap with other nodes. Do not set this parameter to a value of 0. If the parameter is set incorrectly, the ERR on the option card flashes, and the drive will detect the <i>AEr [Node Address Setting Error]</i> error.	0 (0 - 64)	746
F6-51 (03C2)	DN Baud Rate	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the communications speed for DeviceNet communication. The drive must be restarted when the setting is changed. 0 : 125 kbps 1 : 250 kbps 2 : 500 kbps 3 : Adjustable from network 4 : Detect automatically	0 (0 - 4)	746
F6-52 (03C3)	DeviceNet PCA Setting	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the format of data sent from the DeviceNet communication master to the drive.	21 (0 - 255)	746
F6-53 (03C4)	DeviceNet PPA Setting	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the format of data sent from the drive to the DeviceNet communication master.	71 (0 - 255)	746
F6-54 (03C5)	DN Idle Flt Det	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects whether or not to detect issues of <i>EF0 [Option Card External Fault]</i> when data is not received from the DeviceNet master. 0 : Enabled 1 : Disabled, no fault detection	0 (0 - 4)	746
F6-55 (03C6)	DN BAUD RATE MEM	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV This parameter is used to enable confirmation of the currently valid communications speed for DeviceNet communication via the keypad. This parameter is used for monitoring only. 0 : 125 kbps 1 : 250 kbps 2 : 500 kbps	0 (0 - 2)	747
F6-56 (03D7)	DeviceNet Speed Scaling	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the speed scale for DeviceNet communication.	0 (-15 - +15)	747
F6-57 (03D8)	DeviceNet Current Scaling	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the current scale of the DeviceNet communication master.	0 (-15 - +15)	747

No. (Hex.)	Name	Description	Default (Range)	Ref.
F6-58 (03D9)	DeviceNet Torque Scaling	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the torque scale of the DeviceNet communication master.	0 (-15 - +15)	747
F6-59 (03DA)	DeviceNet Power Scaling	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the power scale of the DeviceNet communication master.	0 (-15 - +15)	747
F6-60 (03DB)	DeviceNet Voltage Scaling	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the voltage scale of the DeviceNet communication master.	0 (-15 - +15)	747
F6-61 (03DC)	DeviceNet Time Scaling	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the time scale of the DeviceNet communication master.	0 (-15 - +15)	747
F6-62 (03DD)	DeviceNet Heartbeat Interval	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the Heart Beat for DeviceNet communication. A setting of 0 disables the Heart Beat function.	0 (0 - 10)	748
F6-63 (03DE)	DeviceNet Network MAC ID	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV This parameter is used to enable confirmation of the currently valid MAC address for DeviceNet communication via the keypad. This parameter is used for monitoring only.	0 (0 - 63)	748
F6-64 through F6-67 (03DF to 03E2)	Dynamic Output Assembly 109 Programmable Output 1 through 4	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the Configurable Output 1 through 4 written to the MEMOBUS register.	0000h (0000h - FFFFh)	748
F6-68 through F6-71 (03E3, 03E4, 03C7, and 03C8)	Dynamic Input Assembly 159 Programmable Input 1 to 4	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the Configurable Input 1 through 4 loaded from the MEMOBUS register.	0000h (0000h - FFFFh)	748
F6-72 (081B)	PowerLink Node Address	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the node ID for PowerLink communication.	0 (0 - 255)	748

◆ F7: Communication Option and Ethernet Option

No. (Hex.)	Name	Description	Default (Range)	Ref.
F7-01 (03E5)	IP Address 1	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the IP Address of the device used to connect to the network. Sets the first octet. The drive must be restarted when the setting is changed. Note: • Set the IP Address using F7-01 through F7-04 [IP Address 4] when F7-13 = 0 [Address Mode at Startup = Static]. Set IP Addresses so that they do not overlap on the same network. • Set F7-01 through F7-12 when F7-13 = 0.	192 (0 to 255)	748
F7-02 (03E6)	IP Address 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the IP Address of the device used to connect to the network. Sets the second octet. The drive must be restarted when the setting is changed. Note: • Set the IP Address using F7-01 through F7-04 [IP Address 1 through IP Address 4] when F7-13 = 0 [Address Mode at Startup = Static]. Set the IP Addresses so that they do not overlap on the same network. • Set F7-01 through F7-12 when F7-13 = 0.	168 (0 to 255)	748

No. (Hex.)	Name	Description	Default (Range)	Ref.
F7-03 (03E7)	IP Address 3	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the IP Address of the device used to connect to the network. Sets the third octet. The drive must be restarted when the setting is changed. Note: - Set the IP Address using <i>F7-01 through F7-04 [IP Address 1 through IP Address 4]</i> when <i>F7-13 = 0 [Address Mode at Startup = Static]</i>. Set the IP Addresses so that they do not overlap on the same network. - Set <i>F7-01 through F7-12</i> when <i>F7-13 = 0</i>.	1 (0 to 255)	749
F7-04 (03E8)	IP Address 4	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the IP Address of the device used to connect to the network. Sets the fourth octet. The drive must be restarted when the setting is changed. Note: - Set the IP Address using <i>F7-01 through F7-04 [IP Address 1 through IP Address 4]</i> when <i>F7-13 = 0 [Address Mode at Startup = Static]</i>. Set the IP Addresses so that they do not overlap on the same network. - Set <i>F7-01 through F7-12</i> when <i>F7-13 = 0</i>.	20 (0 to 255)	749
F7-05 (03E9)	Subnet Mask 1	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the subnet mask of the connected network. Sets the first octet. Note: Set this parameter when <i>F7-13 = 0 [Address Mode at Startup = Static]</i>.	255 (0 to 255)	749
F7-06 (03EA)	Subnet Mask 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the subnet mask of the connected network. Sets the second octet. Note: Set this parameter when <i>F7-13 = 0 [Address Mode at Startup = Static]</i>.	255 (0 to 255)	749
F7-07 (03EB)	Subnet Mask 3	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the subnet mask of the connected network. Sets the third octet. Note: Set this parameter when <i>F7-13 = 0 [Address Mode at Startup = Static]</i>.	255 (0 to 255)	749
F7-08 (03EC)	Subnet Mask 4	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the subnet mask of the connected network. Sets the fourth octet. Note: Set this parameter when <i>F7-13 = 0 [Address Mode at Startup = Static]</i>.	0 (0 to 255)	750
F7-09 (03ED)	Gateway Address 1	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the Gateway address for the connected network. Sets the first octet. Note: Set this parameter when <i>F7-13 = 0 [Address Mode at Startup = Static]</i>.	192 (0 to 255)	750
F7-10 (03EE)	Gateway Address 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the Gateway address for the connected network. Sets the second octet. Note: Set this parameter when <i>F7-13 = 0 [Address Mode at Startup = Static]</i>.	168 (0 to 255)	750
F7-11 (03EF)	Gateway Address 3	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the Gateway address for the connected network. Sets the third octet. Note: Set this parameter when <i>F7-13 = 0 [Address Mode at Startup = Static]</i>.	1 (0 to 255)	750

No. (Hex.)	Name	Description	Default (Range)	Ref.
F7-12 (03F0)	Gateway Address 4	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the Gateway address for the connected network. Sets the fourth octet. Note: Set this parameter when $F7-13 = 0$ [Address Mode at Startup = Static].	1 (0 to 255)	750
F7-13 (03F1)	Address Mode at Startup	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the method to set addresses for option cards. 0 : Static 1 : BOOTP 2 : DHCP Note: - The following setting values are available when using the PROFINET communication option card (SI-EP3). 0: Static 2: DCP - Set $F7-01$ through $F7-12$ [IP Address 1 = Gateway Address 4] when $F7-13 = 0$. Set the IP Addresses so that they do not overlap on the same network.	2 (0 - 2)	750
F7-14 (03F2)	Duplex Mode Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the setting method for the duplex mode. 0 : Auto/Auto 1 : Half/Half 2 : Full/Full	1 (0 - 8)	751
F7-15 (03F3)	Communication Speed Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the communications speed. 10 : 10/10 Mbps 102 : 100/100 Mbps	10 (10, 102)	751
F7-16 (03F4)	Timeout Value	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the detection time of the timeout value for communications in increments of 0.1 s. Note: Setting this parameter to 0 disables the connection timeout function.	0.0 s (0.0 to 30.0 s)	751
F7-17 (03F5)	EtherNet/IP Speed Scaling Factor	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the scaling factor for the speed monitor for the EtherNet/IP object with the Class ID 2AH.	0 (-15 to 15)	751
F7-18 (03F6)	EtherNet/IP Current Scale Factor	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the scaling factor for the output current monitor for the EtherNet/IP object with the Class ID 2AH.	0 (-15 to 15)	751
F7-19 (03F7)	EtherNet/IP Torque Scale Factor	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the scaling factor for the torque monitor for the EtherNet/IP object with the Class ID 2AH.	0 (-15 to 15)	752
F7-20 (03F8)	EtherNet/IP Power Scaling Factor	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the scaling factor for the power monitor for the EtherNet/IP object with the Class ID 2AH.	0 (-15 to 15)	752
F7-21 (03F9)	EtherNet/IP Voltage Scale Factor	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the scaling factor for the voltage monitor for the EtherNet/IP object with the Class ID 2AH.	0 (-15 to 15)	752
F7-22 (03FA)	EtherNet/IP Time Scaling	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the scaling factor for the time monitor for the EtherNet/IP object with the Class ID 2AH.	0 (-15 to 15)	752

No. (Hex.)	Name	Description	Default (Range)	Ref.
F7-23 through F7-27 (03FB - 03FF) F7-28 through F7-32 (0370 - 0374)	Dynamic Out Assembly 115 Param 1 through 10	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Used for setting the Output Assembly 116. The values received from the Output Assembly 116 are written to the MEMOBUS/Modbus address register stored for each parameter. When the MEMOBUS/Modbus address is 0, the values received from the Output Assembly 116 are not written to the registers.	0	752
F7-33 through F7-42 (0375 - 037E)	Dynamic In Assembly 165 Param 1 through 10	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Used for setting the input assembly 166. The values sent to the input assembly 166 are loaded from the MEMOBUS/Modbus address register stored for each parameter. When the MEMOBUS/Modbus address is 0, the value sent to the input assembly 166 is not defined, and so the default register value for the option card is returned.	0	752
F7-60 (0780)	PZD1 Write	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the MEMOBUS/Modbus address for PZD1 (PPO output). PZD1 (PPO output) functions as the STW when $F7-60 = 0, 1, \text{ or } 2$.	0	752
F7-61 (0781)	PZD2 Write	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the MEMOBUS/Modbus address for PZD2 (PPO output). PZD2 (PPO output) functions as the HSW when $F7-61 = 0, 1, \text{ or } 2$.	0	753
F7-62 (0782)	PZD3 Write	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the MEMOBUS/Modbus address for PZD3 (PPO output). When $F7-62 = 0, 1, \text{ or } 2$, the write operation to the MEMOBUS/Modbus register performed by the PZD3 (PPO output) is disabled.	0	753
F7-63 (0783)	PZD4 Write	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the MEMOBUS/Modbus address for PZD4 (PPO output). When $F7-63 = 0, 1, \text{ or } 2$, the write operation to the MEMOBUS/Modbus register performed by the PZD4 (PPO output) is disabled.	0	753
F7-64 (0784)	PZD5 Write	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the MEMOBUS/Modbus address for PZD5 (PPO output). When $F7-64 = 0, 1, \text{ or } 2$, the write operation to the MEMOBUS/Modbus register performed by the PZD5 (PPO output) is disabled.	0	753
F7-65 (0785)	PZD6 Write	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the MEMOBUS/Modbus address for PZD6 (PPO output). When $F7-65 = 0, 1, \text{ or } 2$, the write operation to the MEMOBUS/Modbus register performed by the PZD6 (PPO output) is disabled.	0	753
F7-66 (0786)	PZD7 Write	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the MEMOBUS/Modbus address for PZD7 (PPO output). When $F7-66 = 0, 1, \text{ or } 2$, the write operation to the MEMOBUS register performed by the PZD7 (PPO output) is disabled.	0	753
F7-67 (0787)	PZD8 Write	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the MEMOBUS/Modbus address for PZD8 (PPO output). Setting $F7-67 = 0, 1, \text{ or } 2$ disables the PZD8 Write.	0	753
F7-68 (0788)	PZD9 Write	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the MEMOBUS/Modbus address for PZD9 (PPO output). When $F7-68 = 0, 1, \text{ or } 2$, the write operation to the MEMOBUS/Modbus register performed by the PZD9 (PPO output) is disabled.	0	754
F7-69 (0789)	PZD10 Write	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the MEMOBUS/Modbus address for PZD10 (PPO output). When $F7-69 = 0, 1, \text{ or } 2$, the write operation to the MEMOBUS/Modbus register performed by the PZD10 (PPO output) is disabled.	0	754
F7-70 (078A)	PZD1 Read	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the MEMOBUS/Modbus address for PZD1 (PPO Read). PZD1 (PPO input) functions as the ZSW when $F7-70 = 0$.	0	754
F7-71 (078B)	PZD2 Read	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the MEMOBUS/Modbus address for PZD2 (PPO Read). PZD2 (PPO input) functions as the HIW when $F7-71 = 0$.	0	754
F7-72 (078C)	PZD3 Read	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the MEMOBUS/Modbus address for PZD3 (PPO Read). When $F7-72 = 0$, the load operation from the MEMOBUS/Modbus register performed by the PZD3 (PPO input) is disabled.	0	754

No. (Hex.)	Name	Description	Default (Range)	Ref.
F7-73 (078D)	PZD4 Read	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the MEMOBUS/Modbus address for PZD4 (PPO Read). When $F7-73 = 0$, the load operation from the MEMOBUS register performed by the PZD4 (PPO input) is disabled.	0	754
F7-74 (078E)	PZD5 Read	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the MEMOBUS/Modbus address for PZD5 (PPO Read). When $F7-74 = 0$, the load operation from the MEMOBUS/Modbus register performed by the PZD5 (PPO input) is disabled.	0	754
F7-75 (078F)	PZD6 Read	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the MEMOBUS/Modbus address for PZD6 (PPO Read). When $F7-75 = 0$, the load operation from the MEMOBUS/Modbus register performed by the PZD6 (PPO input) is disabled.	0	755
F7-76 (0790)	PZD7 Read	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the MEMOBUS/Modbus address for PZD7 (PPO Read). When $F7-76 = 0$, the load operation from the MEMOBUS register performed by the PZD7 (PPO input) is disabled.	0	755
F7-77 (0791)	PZD8 Read	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the MEMOBUS/Modbus address for PZD8 (PPO Read). When $F7-77 = 0$, the load operation from the MEMOBUS/Modbus register performed by the PZD8 (PPO input) is disabled.	0	755
F7-78 (0792)	PZD9 Read	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the MEMOBUS/Modbus address for PZD9 (PPO Read). When $F7-78 = 0$, the load operation from the MEMOBUS/Modbus register performed by the PZD9 (PPO input) is disabled.	0	755
F7-79 (0793)	PZD10 Read	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the MEMOBUS/Modbus address for PZD10 (PPO Read). When $F7-79 = 0$, the load operation from the MEMOBUS/Modbus register performed by the PZD10 (PPO input) is disabled.	0	755

10.10 H: Terminal Functions

◆ H1: Multi-Function Digital Inputs

No. (Hex.)	Name	Description	Default (Range)	Ref.
H1-01 (0438)	Terminal S1 Function Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects a function assigned to the MFDI terminal S1. Note: When <i>Initialization</i> [A1-03 = 3330] has been performed for a 3-wire sequence, the default setting is <i>F</i> .	40 (1-19F)	757
H1-02 (0439)	Terminal S2 Function Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects a function assigned to the MFDI terminal S2. Note: When <i>Initialization</i> [A1-03 = 3330] has been performed for a 3-wire sequence, the default setting is <i>F</i> .	41 (1 - 19F)	757
H1-03 (0400)	Terminal S3 Function Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects a function assigned to the MFDI terminal S3.	24 (0 - 19F)	757
H1-04 (0401)	Terminal S4 Function Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects a function assigned to the MFDI terminal S4.	14 (0 - 19F)	757
H1-05 (0402)	Terminal S5 Function Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects a function assigned to the MFDI terminal S5. Note: When <i>Initialization</i> [A1-03 = 3330] has been performed for a 3-wire sequence, the default setting is <i>0</i> .	3 (0 - 19F)	758
H1-06 (0403)	Terminal S6 Function Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects a function assigned to the MFDI terminal S6. Note: When <i>Initialization</i> [A1-03 = 3330] has been performed for a 3-wire sequence, the default setting is <i>3</i> .	4 (0 - 19F)	758
H1-07 (0404)	Terminal S7 Function Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects a function assigned to the MFDI terminal S7. Note: When <i>Initialization</i> [A1-03 = 3330] has been performed for a 3-wire sequence, the default setting is <i>4</i> .	6 (0 - 19F)	758
H1-08 (0405)	Terminal S8 Function Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects a function assigned to the MFDI terminal S8.	8 (0 - 19F)	758
H1-21 (0B70)	Terminal S1 Function Select 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the 2nd function for the MFDI terminal S1.	F (1 - 19F)	758
H1-22 (0B71)	Terminal S2 Function Select 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the 2nd function for the MFDI terminal S2.	F (1 - 19F)	758
H1-23 (0B72)	Terminal S3 Function Select 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the 2nd function for the MFDI terminal S3.	F (1 - 19F)	759
H1-24 (0B73)	Terminal S4 Function Select 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the 2nd function for the MFDI terminal S4.	F (1 - 19F)	759
H1-25 (0B74)	Terminal S5 Function Select 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the 2nd function for the MFDI terminal S5.	F (1 - 19F)	759
H1-26 (0B75)	Terminal S6 Function Select 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the 2nd function for the MFDI terminal S6.	F (1 - 19F)	759
H1-27 (0B76)	Terminal S7 Function Select 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the 2nd function for the MFDI terminal S7.	F (1 - 19F)	759
H1-28 (0B77)	Terminal S8 Function Select 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the 2nd function for the MFDI terminal S8.	F (1 - 19F)	759
H1-40 (0B54)	Extend MFDI1 Function Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects MFDI function assigned to <i>bit 0</i> of the MEMOBUS register 15C0(Hex.).	F (1 - 19F)	760

No. (Hex.)	Name	Description	Default (Range)	Ref.
H1-41 (0B55)	Extend MFDI2 Function Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects MFDI function assigned to <i>bit 1</i> of the MEMOBUS register <i>15C0(Hex.)</i> .	F (1 - 19F)	760
H1-42 (0B56)	Extend MFDI3 Function Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects MFDI function assigned to <i>bit 2</i> of the MEMOBUS register <i>15C0(Hex.)</i> .	F (1 - 19F)	760

■ H1-xx: Multi-Function Digital Input Setting Values

Setting	Function	Description	Ref.
0	3-Wire Sequence	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the direction of motor rotation for 3-wire sequence.	760
1	LOCAL/REMOTE Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Switches drive control between the keypad (LOCAL) and an external source (REMOTE). ON : LOCAL OFF : REMOTE	761
2	External Reference 1/2 Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Switches between the Run command source 1/2 and Reference command source 1/2 when in REMOTE mode. ON : <i>b1-15</i> = [Frequency Reference Selection 2], <i>b1-16</i> [Run Command Selection 2] OFF : <i>b1-01</i> = [Frequency Reference Selection 1], <i>b1-02</i> [Run Command Selection 1]	761
3	Multi-Step Speed Reference 1	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Switches <i>d1-01</i> to <i>d1-08</i> [Multi-Step Speed Reference] using a combination of multi-step speed references 1, 2 and 3.	762
4	Multi-Step Speed Reference 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Switches <i>d1-01</i> to <i>d1-08</i> [Multi-Step Speed Reference] using a combination of multi-step speed references 1, 2 and 3.	762
5	Multi-Step Speed Reference 3	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Switches <i>d1-01</i> to <i>d1-08</i> [Multi-Step Speed Reference] using a combination of multi-step speed references 1, 2 and 3.	762
6	Jog Reference Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Enables the Jog Reference (JOG command) that was set <i>ind1-17</i> . The Jog Reference (JOG command) overrides even References 1 to 16 (<i>d1-01</i> to <i>d1-16</i>).	762
7	Accel/decel Time Selection 1	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Switches between <i>C1-01</i> , <i>C1-02</i> [Acceleration/Deceleration Time 1] and <i>C1-03</i> , <i>C1-04</i> [Acceleration/Deceleration Time 2].	762
8	Baseblock Command (N.O.)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV If a baseblock command (N.O.) is input, the drive output will stop and the motor will coast to stop. ON : Baseblock (drive output stop) OFF : Normal operation	762
9	Baseblock Command (N.C.)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV If a baseblock command (N.C.) is input (turned OFF), the drive output will stop and the motor will coast to stop. ON : Normal operation OFF : Baseblock (drive output stop)	763
A	Accel/Decel Ramp Hold	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Momentarily pauses motor acceleration and deceleration when the terminal is turned ON, retains the output frequency that was stored in the drive at the time of the pause, and restarts motor operation.	763
B	Drive Overheat Alarm (oH2)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV If the terminal is turned ON, the keypad flashes an <i>oH2</i> [Drive Overheat Warning] minor fault message. The fault does not affect drive operation.	763
C	Analog Terminal Input Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Enables and disables the terminal selected with the <i>H3-14</i> [Analog Input Term Enable Select] function. ON : Input to the terminal selected with <i>H3-14</i> is enabled OFF : Input to the terminal selected with <i>H3-14</i> is disabled	763

Setting	Function	Description	Ref.
D	PG Encoder Disable	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Disregards feedback control from the encoder and runs V/f Control run if the terminal is turned ON. Controls the motor speed using feedback from the encoder if the terminal is turned OFF. ON : Speed feedback control disable (V/f Control) OFF : Speed feedback control enable (Closed Loop V/f Control)	764
E	ASR Integral Reset	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Resets the integral value and switches the speed control loop between PI control and P control. ON : P control OFF : PI control	764
F	Through Mode	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Set when a terminal is not used or when using a terminal in through mode.	764
10	Up Command	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV It is paired with setting value 11 (Down command). When using the Up command and Down command, the user can raise and lower the frequency reference of the drive using two push buttons. ON : Raises the frequency reference. OFF : Holds the current frequency reference.	764
11	Down Command	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV It is paired with setting value 10 (Up command). When using the Up command and Down command, the user can raise and lower the frequency reference of the drive using two push buttons. ON : Lowers the frequency reference. OFF : Holds the current frequency reference.	766
12	Forward Jog	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Inputting the Forward JOG command runs the motor in the forward direction at the jog frequency set in <i>d1-17 [Jog Reference]</i>.	766
13	Reverse Jog	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Inputting the Reverse JOG command reverses the motor at the jog frequency set in <i>d1-17 [Jog Reference]</i>.	766
14	Fault Reset	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV When the terminal is ON while the Run command is inactive, the fault currently detected by the drive will be reset. Note: The fault reset signal is disregarded when the Run command is enabled. Remove the Run command before attempting to clear a fault situation.	767
15	Fast Stop (N.O.)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV When Fast Stop (N.O.) is input while the drive is running, the drive performs ramp to stop in a deceleration time configured with <i>C1-09 [Fast Stop Time]</i>.	767
16	Motor 2 Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Switches between motors 1 and 2. Switch between motors when they are stopped. ON : Selects motor 2 OFF : Selects motor 1	767
17	Fast Stop (N.C.)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV When Fast Stop (N.C.) is input while the drive is running, the drive performs ramp to stop in a deceleration time configured with <i>C1-09 [Fast Stop Time]</i>.	768
18	Timer Function Input	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Used as the input terminal for the timer function. It is paired with <i>Timer Output [H2-xx = 12]</i>.	768
19	PID Disable	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Disables PID control using an external input when <i>b5-01 = 1 to 8 [PID Function Setting = Enabled]</i>. ON : PID control disabled OFF : PID control enabled	769
1A	Accel/Decel Time Selection 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV It is used in combination with the <i>Accel/dec el Time Selection 1 [H1-xx = 7]</i>. Switches between <i>C1-01 to C1-08 [Acceleration and Deceleration Times 1 to 4]</i>.	769

Setting	Function	Description	Ref.
1B	Program Lockout	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The following parameter setting values can be changed when the terminal set for program lockout is ON. When the terminal is OFF, the setting values of parameters cannot be changed. ON : Program Lockout OFF : Parameter Write Prohibit	769
1E	Reference sample hold	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The analog frequency reference input via terminal A1, A2 or A3 will be sampled and operation will continue at that frequency.	769
20 to 2F	External fault	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the drive operation that was active at the time the failure or fault was detected in the external device connected to the drive from any of the patterns between 20 to 2F.	770
30	PID integral reset	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Resets the value of the PID control integral to 0 while the terminal is ON, and holds the value.	770
31	PID integral hold	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV This function force holds the integral value of the PID control as long as the terminal is ON.	771
32	Multi-Step Speed Reference 4	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Switches <i>d1-09</i> to <i>d1-16</i> [Reference 9 to 16] using a combination of multi-step speed references 1, 2 and 3.	771
34	PID soft starter cancel	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Enables or disables the PID soft starter. ON : Disabled OFF : Enabled	771
35	PID input level selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Switches the PID input level (polarity) by turning the terminal on and off.	771
3E	PID Setpoint Selection 1	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV It is used in combination with <i>PID Setpoint Selection 2</i> [<i>H1-xx</i> = 3F]. Switches the PID setpoint to <i>b5-58</i> to <i>b5-60</i> [<i>PID setpoint2</i> to 4].	771
3F	PID Setpoint Selection 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV It is used in combination with <i>PID Setpoint Selection 1</i> [<i>H1-xx</i> = 3E]. Switches the PID setpoint to <i>b5-58</i> to <i>b5-60</i> [<i>PID setpoint2</i> to 4].	771
40	Forward Run Command (2-Wire Seq)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Inputs the Forward run command for 2-wire sequence 1. Use it paired with the <i>Reverse Run Command (2-Wire Seq)</i> [<i>H1-xx</i> = 41]. ON : Forward Run OFF : Run Stop Note: - When the both Forward run command and Reverse run command terminals have been turned ON, the drive detects <i>EF</i> [<i>FWD/REV Run Command Input Error</i>] (minor fault), and the motor ramps to stop. - The Forward run/Reverse run command is set to terminals S1 and S2 when the drive is initialized using a 2-wire sequence. - Simultaneous use with <i>H1-xx</i> = 42, 43 [<i>Run Command/FWD/REV Command (2-Wire Seq 2)</i>] is not possible.	772
41	Reverse Run Command (2-Wire Seq)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Inputs the Reverse run command for 2-wire sequence 1. Use it paired with the <i>Forward Run Command (2-Wire Seq)</i> [<i>H1-xx</i> = 40]. ON : Reverse Run OFF : Run Stop Note: - When the both Forward run command and Reverse run command terminals have been turned ON, the drive detects <i>EF</i> [<i>FWD/REV Run Command Input Error</i>] (minor fault), and the motor ramps to stop. - The Reverse run command is set to terminal S2 when the drive is initialized using a 2-wire sequence. - Simultaneous use with <i>H1-xx</i> = 42, 43 [<i>Run Command/FWD/REV Command (2-Wire Seq 2)</i>] is not possible.	772

Setting	Function	Description	Ref.
42	Run Command (2-Wire Sequence 2)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Inputs the Run command for 2-wire sequence 2. Use it paired with the <i>FWD/REV Command (2-Wire Seq 2)</i> [H1-xx = 43]. ON : Run OFF : Stop Note: Run Command (2-Wire Sequence 2) cannot be used at the same time as <i>Forward/Reverse Run Command (2-Wire Seq)</i> [H1-xx = 40, 41].	772
43	FWD/REV Command (2-Wire Seq 2)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the direction of motor rotation for 2-wire sequence 2. Use it paired with the <i>Run Command (2-Wire Sequence 2)</i> [H1-xx = 42]. ON : Reverse OFF : Forward Note: FWD/REV Command (2-Wire Seq 2) cannot be used at the same time as <i>Forward/Reverse Run Command (2-Wire Seq)</i> [H1-xx = 40, 41].	772
44	Offset frequency 1	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Adds the offset frequency set in d7-01 to the frequency reference. when the terminal is turned ON.	773
45	Offset frequency 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Adds the offset frequency set in d7-02 to the frequency reference. when the terminal is turned ON.	773
46	Offset frequency 3	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Adds the offset frequency set in d7-03 to the frequency reference. when the terminal is turned ON.	773
47	Node Setup	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV When the CANopen communication option is utilized, the Node Setup function (a function for setting the drive node address from the host controller) is enabled.	773
60	DC Injection Braking command	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV If the DC Injection Braking command is input when the drive is performing stopping operation, DC Injection Braking is applied to stop the motor. Note: This function enables only when the induction motor is used for A1-02 = 8 [Control Method Selection = EZ Open Loop Vector Control].	773
61	External Speed Search command 1	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Executes speed search using an external reference even when b3-01 = 0 [Speed Search Selection at Start = Disabled]. Note: If both H1-xx = 61 and 62 are set simultaneously, oPE03 [Multi-Function Input Setting Err] is detected. Set the external speed search command for only 1 or 2.	774
62	External Speed Search command 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Executes speed search using an external reference even when b3-01 = 0 [Speed Search Selection at Start = Disabled]. Note: If both H1-xx = 61 and 62 are set simultaneously, oPE03 [Multi-Function Input Setting Err] is detected. Set the external speed search command for only 1 or 2.	774
63	Field weakening	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV This function issues the commands of Field Weakening Level and Field Weakening Frequency Limit set in d6-01 and d6-02 when the input terminal is turned ON	774
65	KEB Ride-Thru 1 (N.C.)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Switches the KEB1 function between enable and disable via the KEB Ride-Thru 1 (N.C.). ON : Normal operation OFF : Deceleration during momentary power loss	774
66	KEB Ride-Thru 1 (N.O.)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Switches the KEB1 function between enable and disable via the KEB Ride-Thru 1 (N.O.). ON : Deceleration during momentary power loss OFF : Normal operation	774
67	Communications test mode	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Performs self-diagnosis on the RS-485 serial communications operation.	775

Setting	Function	Description	Ref.
68	High Slip Braking (HSB)	 Stops the motor using high-slip braking.	775
6A	Drive Enable	 The keypad displays <i>dnE [Drive Enabled]</i> when the terminal is turned OFF and the Run command will not be accepted.	775
71	Speed/Torque Control Switch	 Switches between torque and speed control. ON : Torque control OFF : Speed control	775
72	Zero Servo	 Holds the motor when it is stopped.	776
75	Up 2 Command	 Accelerates the motor by increasing the frequency reference bias value when the terminal is turned ON. Always set the Up2 command and Down 2 command as a pair. Note: When the Up2/Down2 function is used, set the optimal bias limit value using <i>d4-08</i> and <i>d4-09 [Up/Down 2 Bias Upper Limit/Lower Limit (Up/Down 2)]</i> .	776
76	Down 2 Command	 Decelerates the motor by reducing the frequency reference bias value when the terminal is turned ON. Always set the Up2 command and Down 2 command as a pair. Note: When the Up2/Down2 function is used, set the optimal bias limit value using <i>d4-08</i> and <i>d4-09 [Up/Down 2 Bias Upper Limit/Lower Limit (Up/Down 2)]</i> .	778
77	ASR Gain Switch	 Switches the ASR proportional gain set in <i>C5-01 [ASR Proportional Gain 1]</i> and <i>C5-03 [ASR Proportional Gain 1/2]</i> . ON : C5-03 OFF : C5-01	778
78	Ex.Torque Ref Polarity Inversion	 Switches the rotation direction of the external torque reference. ON : External torque reference reverse direction OFF : External torque reference forward direction	778
7A	KEB Ride-Thru 2 (N.C.)	 Switches the KEB 2 Ride-Thru function between enable and disable via the KEB Ride-Thru 2 (N.C.). ON : Normal operation OFF : Deceleration during momentary power loss	779
7B	KEB Ride-Thru 2 (N.O.)	 Switches the KEB 2 Ride-Thru function between enable and disable via the KEB Ride-Thru 2 (N.O.). ON : Deceleration during momentary power loss OFF : Normal operation	779
7C	Short Circuit Braking (N.O.)	 Enables and disables Short Circuit Braking. (N.O.) ON : Short Circuit Braking is enabled. OFF : Normal operation Note: This function enables only when the PM motor is used for <i>A1-02 = 8 [Control Method Selection = EZ Open Loop Vector Control]</i> .	779
7D	Short Circuit Braking (N.C.)	 Enables and disables Short Circuit Braking. (N.C.) ON : Normal operation OFF : Short Circuit Braking is enabled. Note: This function enables only when the PM motor is used for <i>A1-02 = 8 [Control Method Selection = EZ Open Loop Vector Control]</i> .	779

Setting	Function	Description	Ref.
7E	FWD/REV Detect (V/f w/ simplePG)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Allows the rotation direction of the motor to be set when $F1-21, F1-37 = 0$ [Encoder Option Function Selection = A pulse detection] for Simple Closed Loop V/f Control method and Closed Loop V/f Control method. ON : Reverse OFF : Forward	780
7F	PID Bi-Directional Enable	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Switches between PID Bi-Directional enable and disable. ON : Enabled OFF : Disabled	780
90 to 97	DriveWorksEZ Digital Inputs 1 to 8	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV A setting parameter for digital inputs used with DriveWorksEZ. Refer to the DriveWorksEZ Online Manual for more details.	780
9F	DriveWorksEZ Disable	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Enables and disables the DriveWorksEZ program that is saved in the drive. ON : Disabled OFF : Enabled Note: This function can only be used when $A1-07 = 2$ [DriveWorksEZ Function Selection = Digital input].	780

◆ H2: Multi-function digital output

No. (Hex.)	Name	Description	Default (Range)	Ref.
H2-01 (040B)	Term M1-M2 Function Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the function assigned to MFDO terminal M1-M2. Note: Set this parameter to <i>F</i> when not using the terminal or to use the terminal in through mode.	0 (0 - 1A7)	783
H2-02 (040C)	Term M3-M4 Function Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the function assigned to MFDO terminal M3-M4. Note: Set this parameter to <i>F</i> when not using the terminal or to use the terminal in through mode.	1 (0 - 1A7)	783
H2-03 (040D)	Term M5-M6 Function Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the function assigned to MFDO terminal M5-M6. Note: Set this parameter to <i>F</i> when not using the terminal or to use the terminal in through mode.	2 (0 - 1A7)	783
H2-06 (0437)	Watt Hour Output Unit Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the output signal unit when $H2-01$ through $H2-03 = 39$ [MFDO Function Select = Watt Hour Pulse Output] is selected. 0 : 0.1 kWh units 1 : 1 kWh units 2 : 10 kWh units 3 : 100 kWh units 4 : 1000 kWh units	0 (0 - 4)	783
H2-07 (0B3A)	MEMOBUS Register1 Address Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the address of the MEMOBUS/Modbus register output to the MFDO terminal.	0001 (0001 - 1FFF)	784
H2-08 (0B3B)	MEMOBUS Register 1 Bit Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the bit of the MEMOBUS/Modbus register output to the MFDO terminal.	0000 (0000 - FFFF)	784
H2-09 (0B3C)	MEMOBUS Register2 Address Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the address of the MEMOBUS/Modbus register output to the MFDO terminal.	0001 (0001 - 1FFF)	784

No. (Hex.)	Name	Description	Default (Range)	Ref.
H2-10 (0B3D)	MEMOBUS Register 2 Bit Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the bit of the MEMOBUS/Modbus register output to the MFDO terminal.	0000 (0000 - FFFF)	784
H2-20 (1540)	Comparator 1 Monitor Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the monitor number for comparator 1. Configure the x-xx portion of <i>Ux-xx</i> [Monitor]. For example, set x-xx to 102 to monitor <i>U1-02</i> [Output Frequency].	102 (000 - 999)	784
H2-21 (1541)	Comparator 1 Lower Limit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the lower limit detection level for comparator 1 on the basis that the full scale analog output for the monitor number selected with <i>H2-20</i> [Comparator 1 Monitor Selection] is the 100% value.	0.0% (0.0 - 300.0%)	785
H2-22 (1542)	Comparator 1 Upper Limit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the upper limit detection level for comparator 1 on the basis that the full scale analog output for the monitor number selected with <i>H2-20</i> [Comparator 1 Monitor Selection] is the 100% value.	0.0% (0.0 - 300.0%)	785
H2-23 (1543)	Comparator 1 Hysteresis	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the hysteresis level for comparator 1 on the basis that the full scale analog output for the monitor number selected with <i>H2-20</i> [Comparator 1 Monitor Selection] is the 100% value.	0.0% (0.0 - 10.0%)	785
H2-24 (1544)	Comparator 1 On- Delay Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the comparator 1 on delay time.	0.0 s (0.0 - 600.0 s)	785
H2-25 (1545)	Comparator 1 Off- Delay Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the comparator 1 off delay time.	0.0 s (0.0 - 600.0 s)	785
H2-26 (1546)	Comparator 2 Monitor Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the monitor number for comparator 2. Configure the x-xx portion of <i>Ux-xx</i> [Monitor]. For example, to monitor <i>U1-03</i> [Output Current], set a value of 103.	103 (000 - 999)	786
H2-27 (1547)	Comparator 2 Lower Limit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the lower limit detection level for comparator 2 on the basis that the full scale analog output for the monitor number selected with <i>H2-26</i> [Comparator 2 Monitor Selection] is the 100% value.	0.0% (0.0 - 300.0%)	786
H2-28 (1548)	Comparator 2 Upper Limit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the upper limit detection level for comparator 2 on the basis that the full scale analog output for the monitor number selected with <i>H2-26</i> [Comparator 2 Monitor Selection] is the 100% value.	0.0% (0.0 - 300.0%)	786
H2-29 (1549)	Comparator 2 Hysteresis	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the hysteresis level for comparator 2 on the basis that the full scale analog output for the monitor number selected with <i>H2-26</i> [Comparator 2 Monitor Selection] is the 100% value.	0.0% (0.0 - 10.0%)	786
H2-30 (154A)	Comparator 2 On- Delay Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the time for the monitor number configured with <i>H2-26</i> [Comparator 2 Monitor Selection].	0.0 s (0.0 - 6000.0 s)	786
H2-31 (154B)	Comparator 2 Off- Delay Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the time for the monitor number configured with <i>H2-26</i> [Comparator 2 Monitor Selection].	0.0 s (0.0 - 600.0 s)	787
H2-40 (0B58)	Extend MFDI1 Function Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the MFDO assigned to bit 0 of MEMOBUS register 15E0 (Hex.).	F (0 - 1A7)	787
H2-41 (0B59)	Extend MFDI2 Function Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the MFDO assigned to bit 1 of MEMOBUS register 15E0 (Hex.).	F (0 - 1A7)	787
H2-42 (0B5A)	Extend MFDI3 Function Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the MFDO assigned to bit 2 of MEMOBUS register 15E0 (Hex.).	F (0 - 1A7)	787

No. (Hex.)	Name	Description	Default (Range)	Ref.
H2-60 (1B46)	Terminal M1-M2 Function B Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the second function for terminal M1-M2. The logical calculation results of the terminals assigned to functions by <i>H2-01 [Terminal M1-M2 Function Select]</i> is output.	F (0 - A7)	787
H2-61 (1B47)	Terminal M1-M2 Logical Operation	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the logical operation for the two functions selected by <i>H2-01 [Terminal M1-M2 Function Select]</i> and <i>H2-60 [Terminal M1-M2 Function B Select]</i> .	0 (0 - 8)	787
H2-62 (1B48)	Terminal M1-M2 Delay Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the minimum on time used to output the logical calculation results from terminal M1-M2.	0.1 s (0.0 - 25.0 s)	787
H2-63 (1B49)	Terminal M3-M4 Function B Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the second function for terminal M3-M4. The logical calculation results of the terminals assigned to functions by <i>H2-02 [Terminal M3-M4 Function Select]</i> is output.	F (0 - A7)	788
H2-64 (1B4A)	Terminal M3-M4 Logical Operation	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the logical operation for the two functions selected by <i>H2-02 [Terminal M3-M4 Function Select]</i> and <i>H2-63 [Terminal M3-M4 Function B Select]</i> .	0 (0 - 8)	788
H2-65 (1B4B)	Terminal M3-M4 Delay Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the minimum on time used to output the logical calculation results from terminal M3-M4.	0.1 s (0.0 - 25.0 s)	788
H2-66 (1B4C)	Terminal M5-M6 Function B Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the second function for terminal M5-M6. The logical calculation results of the terminals assigned to functions by <i>H2-03 [Terminal M5-M6 Function Select]</i> is output.	F (0 - A7)	788
H2-67 (1B4D)	Terminal M5-M6 Logical Operation	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the logical operation for the two functions selected by <i>H2-03 [Terminal M5-M6 Function Select]</i> and <i>H2-66 [Terminal M5-M6 Function B Select]</i> .	0 (0 - 8)	788
H2-68 (1B4E)	Terminal M5-M6 Delay Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the minimum on time used to output the logical calculation results from terminal M5-M6.	0.1 s (0.0 - 25.0 s)	788

■ H2-xx: MFDO setting value

Setting	Function	Description	Ref.																														
0	During Run	V/fCL-V/fOLVCLVAOLVOLV/PM AOV/PMCLV/PM EZOLV The terminal turns on when the Run command is input or the drive is outputting voltage. ON : Drive is running OFF : Drive is stopping	789																														
1	Zero Speed	V/fCL-V/fOLVCLVAOLVOLV/PM AOV/PMCLV/PM EZOLV The terminal turns on when the output frequency drops below the value of <i>E1-09</i> [Minimum Output Frequency] or <i>b2-01</i> [DC Injection/Zero SpeedThreshold]. Note: The parameter used as the reference is determined by the setting of <i>A1-02</i> [Control Method Selection]. <table><tr><th>A1-02 Settings</th><th>Description</th><th>Parameter used as the reference</th></tr><tr><td>0</td><td>V/f Control</td><td><i>E1-09</i></td></tr><tr><td>1</td><td>Closed Loop V/f Control</td><td><i>E1-09</i></td></tr><tr><td>2</td><td>Open Loop Vector Control</td><td><i>b2-01</i></td></tr><tr><td>3</td><td>Closed Loop Vector Control</td><td><i>E1-09</i></td></tr><tr><td>4</td><td>Advanced OpenLoop Vector Control</td><td><i>E1-09</i></td></tr><tr><td>5</td><td>PM Open Loop Vector Control</td><td><i>E1-09</i></td></tr><tr><td>6</td><td>PM Advanced Open Loop Vector</td><td><i>E1-09</i></td></tr><tr><td>7</td><td>PM Closed Loop Vector Control</td><td><i>b2-01</i></td></tr><tr><td>8</td><td>EZ Open Loop Vector Control</td><td><i>E1-09</i></td></tr></table> ON : The output frequency is less than the value of <i>E1-09</i> or <i>b2-01</i>. OFF : The output frequency is the value of <i>E1-09</i> or more, or <i>b2-01</i> or more.	A1-02 Settings	Description	Parameter used as the reference	0	V/f Control	<i>E1-09</i>	1	Closed Loop V/f Control	<i>E1-09</i>	2	Open Loop Vector Control	<i>b2-01</i>	3	Closed Loop Vector Control	<i>E1-09</i>	4	Advanced OpenLoop Vector Control	<i>E1-09</i>	5	PM Open Loop Vector Control	<i>E1-09</i>	6	PM Advanced Open Loop Vector	<i>E1-09</i>	7	PM Closed Loop Vector Control	<i>b2-01</i>	8	EZ Open Loop Vector Control	<i>E1-09</i>	789
A1-02 Settings	Description	Parameter used as the reference																															
0	V/f Control	<i>E1-09</i>																															
1	Closed Loop V/f Control	<i>E1-09</i>																															
2	Open Loop Vector Control	<i>b2-01</i>																															
3	Closed Loop Vector Control	<i>E1-09</i>																															
4	Advanced OpenLoop Vector Control	<i>E1-09</i>																															
5	PM Open Loop Vector Control	<i>E1-09</i>																															
6	PM Advanced Open Loop Vector	<i>E1-09</i>																															
7	PM Closed Loop Vector Control	<i>b2-01</i>																															
8	EZ Open Loop Vector Control	<i>E1-09</i>																															
2	Speed Agree 1	V/fCL-V/fOLVCLVAOLVOLV/PM AOV/PMCLV/PM EZOLV The terminal turns on when the output frequency is within the range of the frequency reference $\pm L4-02$ [Speed Agree Detection Width]. Note: When using Closed Loop Vector Control, the motor speed is used as the reference. ON : The output frequency is within the range of "frequency reference $\pm L4-02$." OFF : The output frequency does not match the frequency reference even though the drive is running.	790																														
3	User-set Speed Agree 1	V/fCL-V/fOLVCLVAOLVOLV/PM AOV/PMCLV/PM EZOLV The terminal turns on when the output frequency is within the range of <i>L4-01</i> [Speed Agree Detection Level] $\pm L4-02$ [Speed Agree Detection Width] and within the range of the frequency reference $\pm L4-02$. Note: - The detection function operates regardless of the direction of motor rotation. The value of <i>L4-01</i> is used as the forward/reverse detection level. - When using Closed Loop Vector Control, this is the value of "Motor Speed $\pm L4-02$." ON : The output frequency is within the range of "<i>L4-01</i> $\pm L4-02$" and the range of frequency reference $\pm L4-02$. OFF : The output frequency is not within the range of "<i>L4-01</i> $\pm L4-02$" or the range of frequency reference $\pm L4-02$.	790																														
4	Frequency Detection 1	V/fCL-V/fOLVCLVAOLVOLV/PM AOV/PMCLV/PM EZOLV The terminal turns on when the output frequency is higher than the value of <i>L4-01</i> [Speed Agree Detection Level] + <i>L4-02</i> [Speed Agree Detection Width]. After the terminal turns off, the terminal continues to remain off until the output frequency reaches the level set with <i>L4-01</i>. Note: - The detection function operates regardless of the direction of motor rotation. The value of <i>L4-01</i> is used as the forward/reverse detection level. - When using Closed Loop Vector Control, the motor speed is used as the reference. ON : The output frequency is less than the value of <i>L4-01</i> or does not exceed the value of <i>L4-01</i> + <i>L4-02</i>. OFF : The output frequency exceeds the value of <i>L4-01</i> + <i>L4-02</i>.	791																														

Setting	Function	Description	Ref.
5	Frequency Detection 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on when the output frequency is higher than the setting value of <i>L4-01</i> [<i>Speed Agree Detection Level</i>]. After the terminal turns on, the terminal continues to stay on until the output frequency reaches the value of <i>L4-01</i> - <i>L4-02</i>. ON : The output frequency exceeds the value of <i>L4-01</i>. OFF : The output frequency is less than the value of "<i>L4-01</i> - <i>L4-02</i>," or it does not exceed the value of <i>L4-01</i>.	792
6	Drive Ready	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on when the drive is in the ready state and the drive is running.	792
7	DC Bus Undervoltage	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on when the DC bus voltage or control circuit power supply drops below the voltage set with <i>L2-05</i> [<i>Undervoltage Detect Level (Uv1)</i>]. The terminal also turns on when the DC bus voltage experiences a fault. ON : The DC bus voltage has dropped below the setting value of <i>L2-05</i>. OFF : The DC bus voltage exceeds the setting value of <i>L2-05</i>.	792
8	During Baseblock (N.O.)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on during baseblock. When the drive is in the baseblock state, the drive output transistor stops switching, and the DC bus voltage is not output. ON : During baseblock OFF : The drive is not in the baseblock state.	792
9	Frequency Reference Source	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the frequency reference source that is currently selected. ON : The keypad is the frequency reference source. OFF : Either <i>b1-01</i> or <i>b1-15</i> [<i>Frequency Reference Selection 1 or 2</i>] is the frequency reference source.	793
A	Run Command Source	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the Run command source that is currently selected. ON : The keypad is the Run command source. OFF : Either <i>b1-02</i> or <i>b1-16</i> [<i>Run Command Selection 1 or 2</i>] is the Run command source.	793
B	Torque Detection 1 (N.O.)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on when an overtorque/undertorque situation is detected. ON : The output current/torque exceeds the torque value set with <i>L6-02</i> [<i>Torque Detection Level 1</i>], or the level has dropped and remained in this state longer than the time set with <i>L6-03</i> [<i>Torque Detection Time 1</i>].	793
C	Frequency Reference Loss	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on when a loss of frequency reference is detected.	793
D	Braking Resistor Fault	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on when the mounting type braking resistor is overheating or the braking transistor is experiencing a fault.	793
E	Fault	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on when the drive experiences a fault. Note: Parameters <i>CPF00</i> and <i>CPF01</i> [<i>Control Circuit Error</i>] are excluded.	793
F	Through Mode	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Use this setting when terminals are not used or to use terminals in through mode. This can be used as the PLC contact output via MEMOBUS/Modbus or the communication option. This signal does not function as long as signals from the PLC are not configured.	794
10	Minor Fault	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on when the drive experiences a minor fault.	794
11	Fault Reset Command Active	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on when the drive receives the reset command from the control circuit terminal, serial communications, or the communication option.	794
12	Timer Output	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV This is configured when the timer function is used as an output terminal.	794

Setting	Function	Description	Ref.
13	Speed Agree 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on when the output frequency is within the range of the frequency reference $\pm L4-04$ [Speed Agree Detection Width (+/-)]. Note: The motor speed is used as the reference when using Closed Loop Vector Control or Closed Loop Vector Control for PM. ON : The output frequency is within the range of "frequency reference $\pm L4-04$." OFF : The output frequency is not within the range of "frequency reference $\pm L4-04$."	794
14	User-set Speed Agree 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on when the output frequency is within the range of $L4-03$ [Speed Agree Detect Level (+/-)] $\pm$ [L4-04 [Speed Agree Detect Width (+/-)]] and within the range of the frequency reference $\pm L4-04$. ON : The output frequency is within the range of "$L4-03 \pm L4-04$" and the range of the frequency reference $\pm L4-04$. OFF : The output frequency is not within the range of "$L4-03 \pm L4-04$" or the range of frequency reference $\pm L4-04$.	794
15	Frequency Detection 3	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns off when the output frequency is higher than the value of "$L4-03$ [Speed Agree Detect Level (+/-)] + $L4-04$ [Speed Agree Detect Width (+/-)]." After the terminal turns off, the terminal continues to remain off until the output frequency reaches the level set with $L4-03$. Note: - The detection level configured with $L4-03$ is a signed value. Detections only occur one specific orientation. - The motor speed is used as the reference when using Closed Loop Vector Control or Closed Loop Vector Control for PM. ON : The output frequency is less than the value of $L4-03$ or does not exceed the value of $L4-03 + L4-04$. OFF : The output frequency exceeds the value of $L4-03 + L4-04$.	795
16	Frequency Detection 4	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on when the output frequency is higher than the value of $L4-03$ [Speed Agree Detect Level (+/-)]. After the terminal turns on, the terminal continues to stay on until the output frequency reaches the value of $L4-03 - L4-04$. ON : The output frequency exceeds the value of $L4-03$. OFF : The output frequency is less than the value of "$L4-03 - L4-04$," or it does not exceed the value of $L4-03$.	796
17	Torque Detection 1 (N.C.)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns off when an overtorque/undertorque situation is detected. OFF : The output current/torque exceeds the torque value set with $L6-02$ [Torque Detection Level 1], or the level has dropped and remained in this state longer than the time set with $L6-03$ [Torque Detection Time 1].	796
18	Torque Detection 2 (N.O.)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on when an overtorque/undertorque situation is detected. ON : The output current/torque exceeds the torque value set with $L6-05$ [Torque Detection Level 2], or the level has dropped and remained in this state longer than the time set with $L6-06$ [Torque Detection Time 2].	796
19	Torque Detection 2 (N.C.)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns off when an overtorque/undertorque situation is detected. OFF : The output current/torque exceeds the torque value set with $L6-05$ [Torque Detection Level 2], or the level has dropped and remained in this state longer than the time set with $L6-06$ [Torque Detection Time 2].	796
1A	During Reverse	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on when the motor runs in reverse. ON : The motor is running in reverse. OFF : The motor is running forward or is stopped.	797
1B	During Baseblock (N.C.)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns off during baseblock. When the drive is in the baseblock state, the drive output transistor stops switching, and the DC bus voltage is not output. ON : The drive is not in the baseblock state. OFF : During baseblock	797

10.10 H: Terminal Functions

Setting	Function	Description	Ref.
1C	Motor 2 Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on when motor 2 is selected. ON : Motor 2 Selection OFF : Motor 1 Selection	797
1D	During Regeneration	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on when the motor is regenerating. ON : Motor is regenerating. OFF : Motor is operating or stopped.	797
1E	Restart Enabled	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on when a fault that can be restarted occurs and the Auto Restart function is attempting to operate.	798
1F	Motor Overload Alarm (oL1)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on when the electronic thermal protector value of the motor overload protective function reaches at least 90% of the detection level.	798
20	Drive Overheat Pre-Alarm (oH)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on when the drive heatsink temperature reaches the level set with <i>L8-02 [Overheat Alarm Level]</i>.	798
21	EDM	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on (safety stop state) when the safety circuit and safety diagnosis circuit is not experiencing a failure and when both terminals H1-HC and H2-HC are off (released). ON : Safety stop state OFF : Safety circuit fault or RUN/READY	798
22	Mechanical Weakening Detection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on when mechanical weakening is detected.	798
2F	Maintenance Period	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on when consumables reach the estimated maintenance period. Provides notification of the maintenance period for the following items. • IGBT • Cooling fan • Capacitor • Soft charge bypass relay	798
30	During Torque Limit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on when the torque reference reaches the torque limit configured with <i>L7 parameters, H3-02, H3-06, or H3-10 [Multi-Function Analog In]</i>.	799
31	During speed limit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on when the speed limit is active.	799
32	During Spd Limit inTorqueControl	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The motor accelerates in forward or reverse when torque control is enabled and the torque reference externally input is disproportionate to the load. The output terminal turns on when this speed is restricted to no higher than a constant speed and the motor speed is at the speed limit. Stopped operation is excluded.	799
33	Zero Servo Complete	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on when positioning within the range defined by <i>b9-02 [Zero Servo Completion Width]</i> completes after the input of the Zero-Servo command.	799
37	During Frequency Output	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on when the drive outputs frequency. ON : The drive outputs frequency. OFF : The drive does not output frequency.	799
38	Drive Enabled	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV This terminal turns on when the terminal allocated to <i>H1-xx = 6A [Drive Enable]</i> is turned on.	800
39	Watt Hour Pulse Output	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Outputs the pulse that represents the watt hours.	800

Setting	Function	Description	Ref.
3C	LOCAL/REMOTE Status	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on when the Run command source or frequency reference source is LOCAL. ON : LOCAL OFF : REMOTE	800
3D	During Speed Search	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on when speed search is executing.	800
3E	PID Feedback Low	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on when <i>FbL</i> [PID Feedback Loss] is detected.	800
3F	PID Feedback High	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on when <i>FbH</i> [Excessive PID Feedback] is detected.	801
4A	During KEB Ride-Thru	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on while the KEB Ride-Thru function is being executed.	801
4B	During Short Circuit Braking	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on during Short Circuit Braking. Note: This function is enabled only when using PM motors while <i>A1-02</i> = 8 [Control Method Selection = EZ Open Loop Vector Control].	801
4C	During Fast Stop	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on when the fast stop is active.	801
4D	oH Pre-Alarm Time Limit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on when <i>L8-03</i> = 4 [Overheat Pre-Alarm Ope Selection = Run@L8-19 Rate] and <i>oH</i> [Heatsink Overheat] does not clear even after the drive diminishes the frequency for 10 cycles.	801
4E	Braking Transistor Fault (rr)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on when the braking transistor integrated into the drive overheats and <i>rr</i> [Dynamic Braking Transistor] is detected.	801
4F	Braking Resistor Overheat (oH)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on when the braking resistor overheats and <i>rH</i> [Braking Resistor Overheat] is detected.	802
60	Internal Cooling Fan Alarm	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on when a failure is detected in the cooling fan inside the drive.	802
61	RotorPosition Detection Complete	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on when the Run command is input into the drive and the drive detects the motor magnetic pole position of the PM motor.	802
62	MEMOBUS Register 1 (H2-07&H2-08)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on when the bit specified by <i>H2-07</i> turns on regarding the MEMOBUS register address configured with <i>H2-08</i>.	802
63	MEMOBUS Register 2 (H2-09&H2-10)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on when the bit specified by <i>H2-10</i> turns on regarding the MEMOBUS register address configured with <i>H2-09</i>.	802
65	Standby output	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns off after the drive stops operating and the time set with <i>b8-51</i> [Standby Mode Wait Time] elapses. ON : The Run command turns on and the magnetic contactor on the input side turns off. OFF : The Run command turns off and the drive stops operating. Then, the magnetic contactor on the input side turns off after the time set with <i>b8-51</i> [Standby Mode Wait Time] elapses.	802
66	Comparator1	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The monitor value configured with <i>H2-20</i> is on while within range of the time configured with <i>H2-24</i> and the values of <i>H2-21</i> and <i>H2-22</i> are within range.	803
67	Comparator2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The monitor value configured with <i>H2-26</i> is on while within range of the time configured with <i>H2-30</i> and the values of <i>H2-27</i> and <i>H2-28</i> are within range.	803
69	External Power 24V Supply	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The terminal turns on when an external 24V power supply is provided between terminal PS-AC. ON : Power is supplied by an external 24V power supply. OFF : Power is not supplied by an external 24V power supply.	803

Setting	Function	Description	Ref.
90 - 92	DriveWorksEZ Digital Outputs 1 through 3	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the digital output used by DriveWorksEZ. Refer to the DriveWorksEZ online manual for more information.	804
100 through 1A7	Inverse output of 0 through A7	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Generates inverse output of the function for the selected MFDO. Selects the function for which to generate inverse output with the last two digits of 1xx.	804

◆ H3: Multi-Function Analog Inputs

No. (Hex.)	Name	Description	Default (Range)	Ref.
H3-01 (0410)	Terminal A1 Signal Level Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the input signal level for MFAI terminal A1. 0 : 0 to 10V (Lower Limit at 0) 1 : -10 to +10V (Bi-polar Reference) 2 : 4 to 20 mA 3 : 0 to 20 mA	0 (0 - 3)	806
H3-02 (0434)	Terminal A1 Function Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects a function assigned to the MFAI terminal A1.	0 (0 - 32)	806
H3-03 (0411) RUN	Terminal A1 Gain Setting	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the gain for the analog signal input to the MFAI terminal A1.	100.0% (-999.9 - +999.9%)	806
H3-04 (0412) RUN	Terminal A1 Bias Setting	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the bias of the analog signal to be input to the MFAI terminal A1.	0.0% (-999.9 - +999.9%)	807
H3-05 (0413)	Terminal A3 Signal Level Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the input signal level for MFAI terminal A3. 0 : 0 to 10V (Lower Limit at 0) 1 : -10 to +10V (Bi-polar Reference) 2 : 4 to 20 mA 3 : 0 to 20 mA	0 (0 - 3)	807
H3-06 (0414)	Terminal A3 Function Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the function to be set for the MFAI terminal A3.	2 (0 - 32)	807
H3-07 (0415) RUN	Terminal A3 Gain Setting	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the gain of the analog signal to be input to the MFAI terminal A3.	100.0% (-999.9 - +999.9%)	807
H3-08 (0416) RUN	Terminal A3 Bias Setting	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the bias of the analog signal to be input to the MFAI terminal A3.	0.0% (-999.9 - +999.9%)	808
H3-09 (0417)	Terminal A2 Signal Level Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the input signal level for MFAI terminal A2. 0 : 0 to 10V (Lower Limit at 0) 1 : -10 to +10V (Bi-polar Reference) 2 : 4 to 20 mA 3 : 0 to 20 mA	2 (0 - 3)	808
H3-10 (0418)	Terminal A2 Function Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects a function set to the MFAI terminal A2.	0 (0 - 32)	808
H3-11 (0419) RUN	Terminal A2 Gain Setting	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the gain of the analog signal to be input to the MFAI terminal A2.	100.0% (-999.9 - +999.9%)	808
H3-12 (041A) RUN	Terminal A2 Bias Setting	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the bias of the analog signal to be input to the MFAI terminal A2.	0.0% (-999.9 - +999.9%)	809

No. (Hex.)	Name	Description	Default (Range)	Ref.
H3-13 (041B)	Analog Input FilterTime Constant	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the time constant when applying a primary delay filter to the MFAI terminal.	0.03 s (0.00 - 2.00 s)	809
H3-14 (041C)	Analog Input Terminal Enable Sel	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the enabled terminal Sx when $H1-xx = C$ [Terminal Sx Function Selection = Analog Terminal Enable Selection] is ON. Also selects the disabled terminal Sx when the input is OFF. 1 : Terminal A1 only 2 : Terminal A2 only 3 : Terminals A1 and A2 4 : Terminal A3 only 5 : Terminals A1 and A3 6 : Terminals A2 and A3 7 : Terminals A1, A2, and A3	7 (1 - 7)	809
H3-16 (02F0)	Terminal A1 Offset	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the offset level for the analog signal input to terminal A1. Normally there is no need to change this setting.	0 (-500 - +500)	809
H3-17 (02F1)	Terminal A2 Offset	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the offset level for analog signals input to terminal A2. Normally there is no need to change this setting.	0 (-500 - +500)	810
H3-18 (02F2)	Terminal A3 Offset	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the offset level for analog signals input to terminal A3. Normally there is no need to change this setting.	0 (-500 - +500)	810
H3-40 (0B5C)	Extend MFAI1 Function Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the MEMOBUS AI1 function.	F (4 - 2F)	810
H3-41 (0B5F)	Extend MFAI2 Function Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the MEMOBUS AI2 function.	F (4 - 2F)	810
H3-42 (0B62)	Extend MFAI3 Function Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the MEMOBUS AI3 function.	F (4 - 2F)	810
H3-43 (117F)	Filter Time for MFAI	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the time constant when applying a primary delay filter to the MEMOBUS analog input terminal.	0.00 s (0.00 to 2.00 s)	810

■ H3-xx: Multi-Function Analog In Setting Value

Setting	Function	Description	Ref.
0	Frequency BiasMaster frequency reference	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The input value from the analog input terminal set with this function becomes the master frequency reference.	811
1	Frequency Gain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The input value from the analog input terminal set with this function is multiplied by the analog frequency reference.	811
2	Auxiliary Frequency Reference 1	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV When Reference 2 is selected via multi-step speed reference, the command reference (Auxiliary Frequency Reference 1) from the analog input terminal set with this setting is enabled. Set <i>E1-04 [Maximum Output Frequency]</i> as 100%.	811
3	Auxiliary Frequency Reference 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV When Reference 3 is selected via multi-step speed reference, the command reference (Auxiliary Frequency Reference 2) from the analog input terminal set with this setting is enabled. Set <i>E1-04 [Maximum Output Frequency]</i> as 100%.	811
4	Output Voltage Bias	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Set this parameter to input a bias signal that amplifies the output voltage.	811
5	Accel/Decel Time Gain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Inputs the signal that adjusts the gain used for <i>C1-01 through C1-08 [Accel & Decel Time 1 through 4]</i> assuming that the full scale analog signal (10 V or 20 mA) is 100%.	812
6	DC Injection Braking Current	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Inputs the signal that adjusts the current level used for DC Injection Braking assuming that the drive rated output current is 100%.	812

10.10 H: Terminal Functions

Setting	Function	Description	Ref.
7	Overtorque/ Undertorque DetectLvl	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Inputs the signal that adjusts the overtorque/undertorque detection level. Note: Use this function in conjunction with L6-01 [Torque Detection Selection 1]. This parameter functions in place of L6-02 [Torque Detection Level 1].	812
8	Stall Prevention Level DuringRun	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Inputs the signal that adjusts the stall prevention level during run assuming that the drive rated current is 100%.	813
9	Output Freq Lower Limit Level	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Inputs the signal that adjusts the output frequency lower limit level assuming that E1-04 [Maximum Output Frequency] is 100%.	813
B	PID Feedback	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Inputs the PID feedback value.	813
C	PID Setpoint	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Inputs the PID setpoint.	813
D	Frequency Bias	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Inputs the bias value added to the frequency reference assuming that E1-04 [Maximum Output Frequency] is 100%.	813
E	Motor Temperature (PTC input)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Uses the motor Positive Temperature Coefficient (PLC) thermistor to protect the motor from heat on the basis that the current value at the time the 10 V (or 20 mA) analog signal is input is 100%.	814
F	Through Mode	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Use this setting when terminals are not used or to use terminals in through mode.	814
10	Forward Torque Limit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Inputs the forward torque limit assuming that the motor rated torque is 100%.	814
11	Reverse Torque Limit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Inputs the load torque limit assuming that the motor rated torque is 100%.	815
12	Regenerative Torque Limit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Inputs the regenerative torque limit assuming that the motor rated torque is 100%.	815
13	Torque Reference / Torque Limit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Inputs the torque reference assuming that the motor rated torque is 100%. This parameter operates as the torque limit for speed control.	815
14	Torque Compensation	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Inputs the torque compensation value assuming that the motor rated torque is 100%.	816
15	General Torque Limit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Inputs the torque limit common to all quadrants for forward, reverse, and regenerative operation assuming that the motor rated torque is 100%.	816
16	Differential PID Feedback	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Inputs the PID differential feedback value assuming that the full scale analog signal (10 V or 20 mA) is 100%.	816
1F	Through Mode	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Use this setting when terminals are not used or to use terminals in through mode.	816
30	DriveWorksEZ analog input 1	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Used with DriveWorksEZ. Refer to the DriveWorksEZ Online Manual for more information.	816
31	DriveWorksEZ analog input 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Used with DriveWorksEZ. Refer to the DriveWorksEZ Online Manual for more information.	816
32	DriveWorksEZ analog input 3	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Used with DriveWorksEZ. Refer to the DriveWorksEZ Online Manual for more information.	816

◆ H4: Analog Outputs

No. (Hex.)	Name	Description	Default (Range)	Ref.
H4-01 (041D)	Terminal FM Analog Output Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the monitoring number to be output from multi-function analog output terminal FM. Set the x-xx portion of the monitoring parameter <i>Ux-xx</i>. For example, set x-xx to 102 to monitor <i>U1-02 [Output Frequency]</i>.	102 (000 - 999)	818
H4-02 (041E) RUN	Terminal FM Analog Output Gain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the gain of the monitoring signal that is output from the MFAO terminal FM as a percentage. Set the terminal FM output signal level when an output of 0% for monitoring items is 0 V (or 4 mA) and when an output for monitoring items is 100%, and sets 10 V (or 20 mA) as 100%.	100.0% (-999.9 - 999.9%)	818
H4-03 (041F) RUN	Terminal FM Analog Output Bias	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the bias of the monitoring signal that is output from the MFAO terminal FM as a percentage. Set the level of the analog signal output from the FM terminal at 10 V (or 20 mA) as 100% when an output for monitoring items is 0%.	0.0% (-999.9 - 999.9%)	818
H4-04 (0420)	Terminal AM Analog Output Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the monitoring number to be output from the MFAO terminal AM. Set the x-xx portion of the monitoring parameter <i>Ux-xx</i>. For example, set x-xx to 102 to monitor <i>U1-02 [Output Frequency]</i>.	103 (000 - 999)	818
H4-05 (0421) RUN	Terminal AM Analog Output Gain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the gain of the monitor signal that is output from the MFAO terminal AM as a percentage. Sets the level of the analog signal output from the AM terminal when an output of 0% for monitoring items is 0 V (or 4 mA) and when an output for monitoring items is 100%, and sets 10 V (or 20 mA) as 100%.	50.0% (-999.9 - 999.9%)	819
H4-06 (0422) RUN	Terminal AM Analog Output Bias	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the bias of the monitoring signal that is output from the MFAO terminal AM as a percentage. Set the level of the analog signal output from the AM terminal at 10 V (or 20 mA) as 100% when an output for monitoring items is 0%.	0.0% (-999.9 - 999.9%)	819
H4-07 (0423)	Terminal FM Signal Level Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the output signal level from the MFAO terminal FM. Note: Set jumper S5 on the terminal board accordingly when changing these parameters. 0 : 0 to 10 Vdc 1 : -10 to +10 Vdc 2 : 4 to 20 mA	0 (0 - 2)	819
H4-08 (0424)	Terminal AM Signal Level Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the signal level output from the MFAO terminal AM. Note: Set jumper S5 on the terminal board accordingly when changing these parameters. 0 : 0 to 10 Vdc 1 : -10 to +10 Vdc 2 : 4 to 20 mA	0 (0 - 2)	819
H4-20 (0B53)	Output power monitor level	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the level at 10 V when <i>U1-08 [Output Power]</i> executes analog output.	0.00 kW (0.00 - 650.00 kW)	820

◆ H5: MEMOBUS/Modbus Communication

No. (Hex.)	Name	Description	Default (Range)	Ref.
H5-01 (0425)	Drive Node Address	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the communication slave address for drives. Note: - Restart the drive to enable the settings. - Setting the parameter to 0 will cause the drive to stop responding to MEMOBUS/Modbus communications.	1FH (0 - FFH)	820
H5-02 (0426)	Communication Speed Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the communications speed for MEMOBUS/Modbus communications. Note: Restart the drive to enable the settings. 0 : 1200 bps 1 : 2400 bps 2 : 4800 bps 3 : 9600 bps 4 : 19.2 kbps 5 : 38.4 kbps 6 : 57.6 kbps 7 : 76.8 kbps 8 : 115.2 kbps	3 (0 - 8)	820
H5-03 (0427)	Communication Parity Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the communications parity used for MEMOBUS/Modbus communications. Note: Restart the drive to enable the settings. 0 : No parity 1 : Even parity 2 : Odd parity	0 (0 - 2)	821
H5-04 (0428)	Stopping Method after Com Error	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the motor Stopping Method when <i>CE [MEMOBUS/Modbus Communication Err]</i> issues are detected. 0 : Ramp to stop 1 : Coast to stop 2 : Fast Stop (use C1-09) 3 : Alarm only	3 (0 - 3)	821
H5-05 (0429)	Comm Fault Detection Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects whether to detect <i>CE [MEMOBUS/Modbus Communication Err]</i> issues during MEMOBUS/Modbus communications. 0 : Disabled 1 : Enabled	1 (0, 1)	821
H5-06 (042A)	Drive Transmit Wait Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the time to wait to send a response message after the drive receives a command message from the master. Note: Restart the drive to enable the settings.	5 ms (0 - 65 ms)	821
H5-09 (0435)	CE Detection Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the detection time for <i>CE [MEMOBUS/Modbus Communication Err]</i> issues when communication is disrupted.	2.0 s (0.0 - 10.0 s)	822
H5-10 (0436)	Unit Sel for MEMOBUS/Modbus 0025H	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the unit of measure used for the MEMOBUS/Modbus communications monitor register 0025H (output voltage reference monitor). 0 : 0.1 V units 1 : 1 V units	0 (0, 1)	822

No. (Hex.)	Name	Description	Default (Range)	Ref.
H5-11 (043C)	Communications ENTER Func Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects whether or not the Enter command is required to change parameters via MEMOBUS/Modbus communications. 0 : Enter Required 1 : No EnterRequired	0 (0, 1)	822
H5-12 (043D)	Run Command Method Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the input method for the Run command when <i>b1-02</i> or <i>b1-16</i> [<i>Run Command Selection</i>] are set to 2 [<i>MEMOBUS/Modbus Communications</i>]. 0 : FWD/Stop, REV/Stop 1 : Run/Stop, FWD/REV	0 (0, 1)	822
H5-17 (11A1)	Busy Enter Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the operation when the EEPROM write command is output without EEPROM write available. Normally there is no need to change this setting. 0 : Cannot write into EEPROM 1 : Write in RAM only	0 (0, 1)	823
H5-18 (11A2)	MtrSpd Monitor T	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the filter time constant used when monitoring the motor speed during MEMOBUS/Modbus communications or use of the communication option.	0 ms (0 - 100 ms)	823
H5-20 (0B57)	Comm. Parameters Activation Sel	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Updated MEMOBUS/Modbus communications parameters can take effect immediately after the update. 0 : Enabled when the drive is restarted. 1 : Enabled as soon as the setting value is changed.	0 (0, 1)	823

◆ H6: Pulse Train Input/Output

No. (Hex.)	Name	Description	Default (Range)	Ref.
H6-01 (042C)	PulseTrain InTerm RP Func Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects a function of the pulse train input terminal RP. 0 : Frequency reference 1 : PID feedback value 2 : PID setpoint value 3 : PG Feedback	0 (0 - 3)	824
H6-02 (042D) RUN	Pulse Train Input Scaling	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the frequency of the pulse train input signal used when the item selected with <i>H6-01</i> [<i>PulseTrain InTerm RP Func Select</i>] is input at 100%.	1440 Hz (100 - 32000 Hz)	825
H6-03 (042E) RUN	Pulse Train Input Gain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the gain used when the item configured with <i>H6-01</i> [<i>PulseTrain InTerm RP Func Select</i>] is input to terminal RP.	100.0% (0.0 - 1000.0%)	825
H6-04 (042F) RUN	Pulse Train Input Bias	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the bias used when the item configured with <i>H6-01</i> [<i>PulseTrain InTerm RP Func Select</i>] is input to terminal RP. Sets a value at the time when the pulse train is 0 Hz.	0.0% (-100.0 - 100.0%)	825
H6-05 (0430) RUN	Pulse Train Input Filter Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the primary delay filter time constant for the pulse train input.	0.10 s (0.00 – 2.00 s)	825
H6-06 (0431) RUN	Pulse Train Monitor Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects a function for the pulse train monitor output terminal MP. Inputs the “x-xx” portion of the <i>Ux-xx</i> parameter to be monitored.	102 (000, 031, 101, 102, 105, 116, 501, 502, and 801 through 809)	825

No. (Hex.)	Name	Description	Default (Range)	Ref.
H6-07 (0432) RUN	Pulse Train Monitor Scaling	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the frequency of the pulse train output signal used when the item selected with H6-06 [Pulse Train Monitor Selection] is output at 100%.	1440 Hz (0 - 32000 Hz)	826
H6-08 (043F)	Pulse Train Input Min Frequency	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the minimum frequency of the pulse train signal detectable by terminal RP in units of 0.1 Hz.	0.5 Hz (0.1 - 1000.0 Hz)	826

◆ H7: Virtual Multi-Function I/O

No. (Hex.)	Name	Description	Default (Range)	Ref.
H7-00 (116F)	Virtual MFIO selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Enables or disables the virtual I/O function. If enable is not set, the virtual I/O function will not operate. 0 : Disabled 1 : Enabled	0 (0, 1)	827
H7-01 (1185)	Virtual Multi-Function Input 1	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the function that enters the virtual input selected with the H7-10 [Virtual Multi-Function Output 1].	F (0 - 19F)	827
H7-02 (1186)	Virtual Multi-Function Input 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the function that enters the virtual input selected with the H7-12 [Virtual Multi-Function Output 2].	F (0 - 19F)	827
H7-03 (1187)	Virtual Multi-Function Input 3	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the function that enters the virtual input selected with H7-14 [Virtual Multi-Function Output 3].	F (0 - 19F)	828
H7-04 (1188)	Virtual Multi-Function Input 4	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the function that enters the virtual input selected with the H7-16 [Virtual Multi-Function Output 4].	F (0 - 19F)	828
H7-10 (11A4)	Virtual Multi-Function Output 1	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the function for virtual digital output 1.	F (0 - 1A7)	828
H7-11 (11A5)	Virtual Output 1 Delay Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the minimum ON time for virtual digital output 1.	0.1 s (0.0 - 25.0 s)	828
H7-12 (11A6)	Virtual Multi-Function Output 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the function for virtual digital output 2.	F (0 - 1A7)	828
H7-13 (11A7)	Virtual Output 2 Delay Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the minimum ON time for virtual digital output 2.	0.1 s (0.0 - 25.0 s)	828
H7-14 (11A8)	Virtual Multi-Function Output 3	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the function for virtual digital output 3.	F (0 - 1A7)	828
H7-15 (11A9)	Virtual Output 3 Delay Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the minimum ON time for virtual digital output 3.	0.1 s (0.0 - 25.0 s)	828
H7-16 (11AA)	Virtual Multi-Function Output 4	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the function for virtual digital output 4.	F (0 - 1A7)	829
H7-17 (11AB)	Virtual Output 4 Delay Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the minimum ON time for virtual digital output 4.	0.1 s (0.0 - 25.0 s)	829
H7-30 (1177)	Virtual Analog Input Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the virtual analog input function.	F (0 - 41)	829
H7-31 (1178) RUN	Virtual Analog Input Gain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the virtual analog input gain.	100.0% (-999.9 - 999.9%)	829
H7-32 (1179) RUN	Virtual Analog Input Bias	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the virtual analog input bias.	0.0% (-999.9 - 999.9%)	829

No. (Hex.)	Name	Description	Default (Range)	Ref.
H7-40 (1163)	Virtual Analog Out Signal Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the signal level of the virtual analog output. 0 : 0 ~ 100%(Absolute value) 1 : -10 +10 VDC 2 : 0-10 VDC	0 (0 - 2)	829
H7-41 (1164)	Virtual Analog Output Function	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the monitoring number to be output from the virtual analog output. Set the x-xx portion of the monitoring parameter <i>Ux-xx</i>. For example, set x-xx to 102 to monitor <i>U1-02 [Output Frequency]</i>.	102 (0 - 999)	829
H7-42 (1165)	Virtual Analog Output FilterTime	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the time constant for a primary filter of the virtual analog output.	0.00 s (0.00 to 2.00 s)	830

10.11 L: Protection Function

◆ L1: Motor Protection

No. (Hex.)	Name	Description	Default Setting (Range)	Ref.
L1-01 (0480)	Motor Overload (oL1) Protection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Enables or disables the motor overload protection using electronic thermal protectors. 0 : Disabled 1 : Variable Torque 2 : Constant Torque 10:1 Speed Range 3 : Constant Torque 100:1 SpeedRange 4 : PM Variable Torque 5 : PM Constant Torque 6 : Variable Torque (50Hz) Note: Set <i>L1-01 = 1 to 6 [Enabled]</i> when only one motor is connected to a drive. External thermal relays are not necessary in such cases.	Determined by A1-02 (0 - 6)	221
L1-02 (0481)	Motor Overload (oL1) Protection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the operation time for the electronic thermal protector of the drive to protect the motor. Normally there is no need to change this setting.	1.0 min (0.1 - 5.0 min)	224
L1-03 (0482)	Motor OH Alarm Operation Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the drive operation when the PTC input signal input into the drive reaches the detection level of <i>oH3 [Motor Overheat Alarm]</i>. 0 : Ramp to stop 1 : Coast to stop 2 : Fast Stop (use C1-09) 3 : Alarm only	3 (0 - 3)	224
L1-04 (0483)	Motor OH Fault Operation Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the drive operation when the PTC input signal input into the drive reaches the detection level of <i>oH4 [Motor Overheat Failure]</i>. 0 : Ramp to stop 1 : Coast to stop 2 : Fast Stop	1 (0 - 2)	225
L1-05 (0484)	Motor Temp Input Filter Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the primary delay time constant for the PTC input signal input to the drive. This parameter is used to prevent accidental detections of motor overheat fault.	0.20 s (0.00 - 10.00 s)	836
L1-08 (1103)	oL1 Current Level	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the reference current for motor thermal overload detection for motor 1 in amperes. Note: Cannot be set to a value smaller than 10% of drive rated current when the current level is set to a value greater than 0.0 A.	0.0 A (0.0 A or 10% to 150% of the drive rated current)	836
L1-09 (1104)	oL1 Current Level for Motor 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the current value used as the reference for detecting the motor overload state regarding the motor 2 electronic thermal protector. Note: Values greater than 0.0 A and less than 10% of the drive rated current cannot be set.	0.0 A (0.0 A or 10 to 150% of the drive rated current)	837
L1-13 (046D)	Cont Electrothermal Ope Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects whether to retain the current electronic thermal protector value when the power supply is interrupted. 0 : Disabled 1 : Enabled	1 (0, 1)	837

◆ L2: Momentary Power Loss Ride-Thru

No. (Hex.)	Name	Description	Default (Range)	Ref.
L2-01 (0485)	Momentary Power Loss Ope Select	V/f CL-V/f OLV CLV AOLV OLVP/AOLV/PM CLV/PM EZOLV Selects the drive operation performed when a momentary power loss occurs. 0 : Disabled 1 : Enbl with Timer 2 : Enbl whl CPU act 3 : KEB Mode 4 : KEB Stop Mode 5 : KEB Decel to Stp	0 (0 - 5)	842
L2-02 (0486)	Momentary Power Loss Ride-Thru Time	V/f CL-V/f OLV CLV AOLV OLVP/AOLV/PM CLV/PM EZOLV Sets the maximum time allowed to ride through a power loss until the drive restart is compensated.	Determined by o2-04 and C6-01 (0.0 - 25.5 s)	844
L2-03 (0487)	Momentary Power Loss Min BB Time	V/f CL-V/f OLV CLV AOLV OLVP/AOLV/PM CLV/PM EZOLV Sets the minimum baseblock time when power is restored following a momentary power loss.	Determined by o2-04 and C6-01 (0.1 - 5.0 s)	844
L2-04 (0488)	Momentary Power Loss Vol Recovery Ramp Time	V/f CL-V/f OLV CLV AOLV OLVP/AOLV/PM CLV/PM EZOLV Sets the time for the drive output voltage to return to normal voltage after completion of speed searches.	Determined by o2-04 and C6-01 (0.0 - 5.0 s)	844
L2-05 (0489)	Undervoltage Detect Level (Uv1)	V/f CL-V/f OLV CLV AOLV OLVP/AOLV/PM CLV/PM EZOLV Determines the voltage at which a <i>Uv1 [DC Bus Undervoltage]</i> fault is triggered or at which the KEB function is activated. Normally there is no need to change this setting. Note: Install an AC reactor option on the input side of the power supply when setting undervoltage detection level below the default value to prevent damage to drive circuitry.	Determined by E1-01 (Determined by E1-01)	844
L2-06 (048A)	KEB Deceleration Time	V/f CL-V/f OLV CLV AOLV OLVP/AOLV/PM CLV/PM EZOLV Sets the deceleration time during KEB operation used to reduce the maximum output frequency to 0.	0.0 s (0.0 to 6,000.0 s)	844
L2-07 (048B)	KEB Acceleration Time	V/f CL-V/f OLV CLV AOLV OLVP/AOLV/PM CLV/PM EZOLV Sets the acceleration time used for the frequency to return to the frequency reference in effect before a power loss after the KEB operation is canceled.	0.0 s (0.0 to 6000.0 s)	845
L2-08 (048C)	Frequency Gain at KEB Start	V/f CL-V/f OLV CLV AOLV OLVP/AOLV/PM CLV/PM EZOLV Sets the amount of output frequency reduction used when KEB operation starts, as a percentage of the motor rated slip before KEB operation startup.	100% (0 - 300%)	845
L2-09 (048D)	KEB Minimum Frequency Level	V/f CL-V/f OLV CLV AOLV OLVP/AOLV/PM CLV/PM EZOLV Sets the amount of output frequency reduction used when KEB operation starts, as a percentage of the motor rated slip.	20% (0 - 100%)	845
L2-10 (048E)	KEB Detect Time (Min KEB Time)	V/f CL-V/f OLV CLV AOLV OLVP/AOLV/PM CLV/PM EZOLV Sets the minimum duration to operate the KEB after a momentary power loss is detected.	50 ms (0 - 25500 ms)	846
L2-11 (0461)	DC Bus Vol Setpoint during KEB	V/f CL-V/f OLV CLV AOLV OLVP/AOLV/PM CLV/PM EZOLV Sets the target value used to control the DC bus voltage to a constant level in Single Drive KEB Ride-Thru 2. Sets the DC bus voltage level used to complete the KEB operation for all other KEB methods.	Determined by E1-01 (Determined by E1-01)	846
L2-29 (0475)	KEB Method Selection	V/f CL-V/f OLV CLV AOLV OLVP/AOLV/PM CLV/PM EZOLV Selects the operation mode of the KEB function. 0 : Single Drive KEB Ride-Thru 1 1 : Single Drive KEB Ride-Thru 2 2 : System KEB Ride-Thru 1 3 : System KEB Ride-Thru 2	0 (0 - 3)	846

No. (Hex.)	Name	Description	Default (Range)	Ref.
L2-30 (045E)	KEB Zero Speed Operation	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the operation when the output frequency drops below the zero level (DC braking injection starting frequency) during <i>KEB deceleration when L2-01 = 3 to 5 [Momentary Power Loss Operation Select = KEB Mode, KEB Stop Mode, or KEB Decel to Stop]</i>. 0 : Baseblock 1 : DC/SC Braking	0 (0, 1)	847
L2-31 (045D)	KEB Start Voltage Offset Level	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the KEB start voltage offset.	Determined by A1-02 (200 V Class: 0 - 100 V, 400 V Class: 0 - 200 V)	847

◆ L3: Stall Prevention

No. (Hex.)	Name	Description	Default (Range)	Ref.
L3-01 (048F)	Stall Prevent Select during Accel	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the method of the Stall Prevention During Acceleration function. 0 : Disabled 1 : General Purpose 2 : Automatic Decel Reduction 3 : ILim Mode	1 (0 to 3)	848
L3-02 (0490)	Stall Prevent Level during Accel	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the output current level used when the Stall Prevention function is enabled during acceleration, as a percentage of the drive rated output current. Note: The upper limit and default for this setting is determined by C6-01 [Normal / Heavy Duty Selection] and L8-38 [Carrier Frequency Reduction].	Determined by C6-01 and L8-38 (0 - 150%)	849
L3-03 (0491)	Stall Prevent Limit during Accel	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the lower limit for the stall prevention level during acceleration used for constant output ranges, as a percentage of the drive rated output current.	50% (0 - 100%)	850
L3-04 (0492)	Stall Prevention during Decel	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the type of Stall Prevention during deceleration. Note: Set L3-04 = 0, 3 [Disabled, General Purpose w/ DB resistor] when using a dynamic braking option. Using a dynamic braking option with another setting will give priority to the Stall Prevention Function and the dynamic braking option will not function. 0 : Disabled 1 : General Purpose 2 : Intelligent (Ignore Decel Ramp) 3 : General Purpose w/ DB resistor 4 : Overexcitation/High Flux 5 : Overexcitation/High Flux 2	1 (0 to Determined by A1-05)	850
L3-05 (0493)	Stall Prevent Select during Run	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Enables or disables the Stall Prevention During Run function. Note: The Stall Prevention during Run function is disabled regardless of the setting of L3-05 and L3-06 [Stall Prevent Level during Run] if the output frequency falls below 6 Hz. 0 : Disabled 1 : Decel Time 1 2 : Decel Time 2	Determined by A1-02 (0 to Determined by A1-02)	852

No. (Hex.)	Name	Description	Default (Range)	Ref.
L3-06 (0494)	Stall Prevent Level during Run	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the current level to trigger Stall Prevention during run. A setting of 100% is equal to the drive rated current. Note: - This parameter is valid if $L3-05 = 1, 2$ [Stall Prevent Select during Run = Decel time 1, Decel time 2]. - The upper limit and default for this setting is determined by C6-01 [Normal / Heavy Duty Selection] and L8-38 [Carrier Frequency Reduction].	Determined by C6-01 and L8-38 (30 - 150%)	852
L3-11 (04C7)	OV Suppression Function Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Enables or disables the overvoltage suppression function. 0 : Disabled 1 : Enabled	0 (0, 1)	852
L3-17 (0462)	DC Bus Reg Level	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Set the target value for the DC bus voltage used when the overvoltage suppression function and the Decel Stall Prevention function (Intelligent Stall Prevention) are running.	200 V Class: 375 V, 400 V Class: 750 V (200 V Class: 150 - 400 V, 400 V Class: 300 - 800 V)	853
L3-20 (0465)	DC Bus Voltage Adjustment Gain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the proportional gain used to control the DC bus voltage.	Determined by A1-02 (0.00 - 5.00)	853
L3-21 (0466)	Accel/Decel Rate Calculate Gain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the proportional gain used to calculate acceleration and deceleration rates.	Determined by A1-02 (0.10 - 10.00)	853
L3-22 (04F9)	DecTime at Stall Prevent during Acc	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the brief deceleration time used when stalling occurs while accelerating a PM motor. This function is valid when $L3-01 = 1$ [Stall Prevent Select during Accel = General Purpose].	0.0 s (0.0 - 6000.0 s)	854
L3-23 (04FD)	CHP Stall P Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects whether to automatically diminish the Stall Prevent Level during Run for constant output ranges. 0 : Level set in L3-06 1 : Automatic Reduction	0 (0, 1)	854
L3-24 (046E)	Motor Accel Time for Inertia Cal	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the motor acceleration time taken to reach the maximum frequency at the motor rated torque for single drive motors that are stopped.	Determined by o2-04, C6-01, E2-11, and E5-01 (0.001 - 10.000 s)	854
L3-25 (046F)	Load Inertia Ratio	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the ratio between motor inertia and machine inertia.	1.0 (1.0 - 1000.0)	855
L3-26 (0455)	Additional DC Bus Capacitors	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the capacity for external main circuit capacitors. Normally there is no need to change this setting. Set this parameter when using the KEB Ride-Thru function.	0 μF (0 to 65000 μF)	855
L3-27 (0456)	Stall Prevention Detection Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets a delay time from when the Stall Prevention level is reached and the actual Stall Prevention function is activated.	50 ms (0 - 5000 ms)	855
L3-34 (016F)	Torque Limit Delay Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the filter time constant in units of seconds used to return the torque limit to its original value while the KEB operation is executing under Single Drive KEB Ride-Thru mode.	Determined by A1-02 (0.000 - 1.000 s)	855
L3-35 (0747)	IntDecSp dAgrWdth	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the width for speed agreement when $L3-04 = 2$ [Decel Stall Prevention Selection = Automatic Decel Reduction]. Normally there is no need to change this setting.	0.00 Hz (0.00 - 1.00 Hz)	856
L3-36 (11D0)	VibraSuppression Gain during Accel	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the gain used to suppress current or motor speed hunting during operation when $L3-01 = 3$ [Stall Prevent Select during Accel = ILim Mode]. Normally there is no need to change this setting.	Determined by A1-02 (0.0 - 100.0)	856

No. (Hex.)	Name	Description	Default (Range)	Ref.
L3-37 (11D1)	Current Limit P Gain duringAccel	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Suppresses current hunting during acceleration. Normally there is no need to change this setting.	5 ms (0 - 100 ms)	856
L3-38 (11D2)	Current Limit I Time duringAccel	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Suppresses the hunting and overshooting of current that occurs when stalling occurs during acceleration. Normally there is no need to change this setting.	10.0 (0.0 - 100.0)	856
L3-39 (11D3)	CurlimIntegTime Con duringAcc/ Dec	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the time constant used to adjust the acceleration rate when <i>L3-01 = 3</i> [Stall Prevent Limit during Accel = ILim Mode]. Normally there is no need to change this setting.	100.0 ms (1.0 - 1000.0 ms)	856
L3-40 (11D4)	CurlimMaxScurve Sel duringAcc/Dec	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects whether to enable or disable the optimal S-curve characteristic used for current -limited acceleration. 0 : Disabled 1 : Enabled	0 (0, 1)	857

◆ L4: Speed Detection

No. (Hex.)	Name	Description	Default (Range)	Ref.
L4-01 (0499)	Speed Agree Detection Level	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the speed agree detection level or the motor speed detection level. Sets the speed detection level or motor speed detection level when <i>H2-01 to H2-03 = 2, 3, 4, 5</i> [MFDO Function Select = Speed Agree 1, User-set Speed Agree 1, Frequency Detection 1, Frequency Detection 2].	Determined by A1-02 (Determined by A1-02)	857
L4-02 (049A)	Speed Agree Detection Width	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the speed agree detection width or motor speed detection width. Sets the speed detection width or motor speed detection width when <i>H2-01 to H2-03 = 2, 3, 4, 5</i> [MFDO Function Select = Speed Agree 1, User-set Speed Agree 1, Frequency Detection 1, Frequency Detection 2].	Determined by A1-02 (Determined by A1-02)	857
L4-03 (049B)	Speed Agree Detect Level (+/-)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the speed agree detection level or the motor speed detection level. Sets the speed detection level or motor speed detection level when <i>H2-01 to H2-03 = 13, 14, 15, 16</i> [MFDO Function Select = Speed Agree 2, User-set Speed Agree 2, Frequency Detection 3, Frequency Detection 4].	Determined by A1-02 (Determined by A1-02)	857
L4-04 (049C)	Speed Agree Detect Width (+/-)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the speed agree detection width or motor speed detection width. Sets the speed detection width or motor speed detection width when <i>H2-01 to H2-03 = 13, 14, 15, 16</i> [MFDO Function Select = Speed Agree 2, User-set Speed Agree 2, Frequency Detection 3, Frequency Detection 4].	Determined by A1-02 (Determined by A1-02)	857
L4-05 (049D)	FreqReference Loss Detect Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the operation when a loss of the frequency reference is detected. 0 : Stop 1 : Run@L4-06PrevRef	0 (0, 1)	858
L4-06 (04C2)	FreqReference at Reference Loss	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the frequency reference as a percentage that is applicable when not stopping the drive so that it continues to operate after a loss of the frequency reference value is detected. The value is set as a percentage of the frequency reference before the loss was detected.	80.0% (0.0 - 100.0%)	858
L4-07 (0470)	Speed Agree Detection Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the condition for activating speed detection. 0 : No detection during baseblock 1 : Detection always enabled	0 (0, 1)	858

◆ L5: Fault Restart

No. (Hex.)	Name	Description	Default (Range)	Ref.
L5-01 (049E)	Number of Auto Restart Attempts	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the number of that can be automatically performed by the drive. Sets the number of Auto Restart operations that the drive may attempt to restart itself.	0 (0 - 10 times)	859
L5-02 (049F)	AutoRestartFaultOutputOpeSelect	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects whether to output signals to the MFDO terminal set for <i>Fault [H2-xx = E]</i> while the drive is executing Auto restart. 0 : Fault output not active 1 : Fault output active	0 (0, 1)	859
L5-04 (046C)	Auto Restart Interval Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the time interval between each Auto Restart process. This function is enabled when <i>L5-05 = 1</i> [<i>Auto Restart Operation Selection = Use L5-04 Time</i>].	10.0 s (0.5 - 600.0 s)	859
L5-05 (0467)	Auto Restart Operation Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the count method for the Auto Restart operation. 0 : Continuous 1 : Use L5-04 Time	0 (0, 1)	860

◆ L6: Torque Detection

No. (Hex.)	Name	Description	Default (Range)	Ref.
L6-01 (04A1)	Torque Detection Selection 1	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the speed range at which overtorque/undertorque is detected and the operation of drives (operation status) after detection. 0 : Disabled 1 : OL Alm at SpdAgr 2 : OL Alm dur RUN 3 : OL Flt at SpdAgr 4 : OL Flt dur RUN 5 : UL Alm at SpdAgr 6 : UL Alm dur RUN 7 : UL Flt at SpdAgr 8 : UL Flt dur RUN	0 (0 - 8)	862
L6-02 (04A2)	Torque Detection Level 1	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the detection level for Overtorque/Undertorque Detection 1. This parameter is set on the basis of the drive rated output current as the 100% value when using V/f Control. This parameter is set on the basis of the motor rated torque as the 100% value when using vector control.	150% (0 - 300%)	863
L6-03 (04A3)	Torque Detection Time 1	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the detection time for Overtorque/Undertorque Detection 1.	0.1 s (0.0 - 10.0 s)	863
L6-04 (04A4)	Torque Detection Selection 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the speed range at which overtorque/undertorque is detected and the operation of drives (operation status) after detection. 0 : Disabled 1 : OL Alm at SpdAgr 2 : OL Alm dur RUN 3 : OL Flt at SpdAgr 4 : OL Flt dur RUN 5 : UL Alm at SpdAgr 6 : UL Alm dur RUN 7 : UL Flt at SpdAgr 8 : UL Flt dur RUN	0 (0 - 8)	863

No. (Hex.)	Name	Description	Default (Range)	Ref.
L6-05 (04A5)	Torque Detection Level 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the detection level for Overtorque/Undertorque Detection 2. This parameter is set on the basis of the drive rated output current as the 100% value when using V/f Control. This parameter is set on the basis of the motor rated torque as the 100% value when using vector control.	150% (0 - 300%)	864
L6-06 (04A6)	Torque Detection Time 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the detection time for Overtorque/Undertorque Detection 2.	0.1 s (0.0 - 10.0 s)	864
L6-08 (0468)	Mechanical Weakening Detect Ope	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the speed range at which mechanical deterioration is detected and the operation of drives (operation status) after detection. 0 : Disabled 1 : Alm Spd>L6-09 2 : Alm [Spd]>L6-09 3 : Flt Spd>L6-09 4 : Flt [Spd]>L6-09 5 : Alm Spd<L6-09 6 : Alm [Spd]<L6-09 7 : Flt Spd<L6-09 8 : Flt [Spd]<L6-09	0 (0 - 8)	864
L6-09 (0469)	Mechanical Weakening Detect Spd Lvl	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV On the basis that <i>E1-04 [Maximum Output Frequency]</i> is the 100% value, this parameter sets the speed level at which the mechanical deterioration detection function operated as a percentage.	110.0% (-110.0 - 110.0%)	865
L6-10 (046A)	Mechanical Weakening Detect Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the time for mechanical deterioration detection.	0.1 s (0.0 - 10.0 s)	865
L6-11 (046B)	Mechanical Weakening Detect Srt Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the time at which mechanical deterioration detection is started using the cumulative operation time of the drive as a trigger.	0 h (0 - 65535 h)	865

◆ L7: Torque Limit

No. (Hex.)	Name	Description	Default (Range)	Ref.
L7-01 (04A7) RUN	Forward Torque Limit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV On the basis that the motor rated torque is the 100% value, this parameter sets the torque limit value for forward motoring as a percentage.	200% (0 - 300%)	866
L7-02 (04A8) RUN	Reverse Torque Limit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the torque limit value for reversed motoring as a percentage of the motor rated torque.	200% (0 - 300%)	866
L7-03 (04A9) RUN	Forward Regenerative Torque Limit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the torque limit value for forward regenerative states as a percentage of the motor rated torque.	200% (0 - 300%)	867
L7-04 (04AA) RUN	Reverse Regenerative Torque Limit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the torque limit value for reversed regenerative states as a percentage of the motor rated torque.	200% (0 - 300%)	867
L7-06 (04AC)	Torque Limit Integral Time Constant	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the integral time constant for the torque limit function.	200 ms (5 - 10000 ms)	867
L7-07 (04C9)	Torque Limit Control Method during Acc/Dec	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the function of torque limit during acceleration and deceleration. 0 : P-ctrl @ Acc/Dec 1 : I-ctrl @ Acc/Dec	0 (0, 1)	867

No. (Hex.)	Name	Description	Default (Range)	Ref.
L7-16 (044D)	Torque Limit Process at Start	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Assigns a time filter to allow the torque limit to build at start. 0 : Disabled 1 : Enabled	1 (0, 1)	868
L7-35 (1B57)	DeratingTrqLim ForLowFreq&Re gratn	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the torque limit used during low-speed regeneration. Normally there is no need to change this setting.	50.00% (0.00 - 200.00%)	868
L7-36 (1B58)	Ope Freq band for deratingTrqLim	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the frequency width at which L7-35 [Derating Trq Lim For Low Freq & Regeneration] operates.	6.00 Hz (0.00 - 30.00 Hz)	868

◆ L8: Hardware Protection

No. (Hex.)	Name	Description	Default (Range)	Ref.
L8-01 (04AD)	Internal DB Resistor Protect Sel	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Enabled braking resistor protection of the when using an optional heatsink mounted braking resistor (ERF type, 3% ED). 0 : Not Provided 1 : Provided	0 (0, 1)	869
L8-02 (04AE)	Overheat Alarm Level	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the <i>oH</i> detection level in temperature.	Determined by o2-04 and C6-01 (50 - 150 °C)	869
L8-03 (04AF)	Overheat Pre- Alarm Ope Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the operation of drives when an <i>oH</i> alarm is detected. 0 : Ramp to Stop 1 : Coast to Stop 2 : Fast-Stop 3 : Alarm Only 4 : Run@L8-19 Rate	3 (0 - 4)	869
L8-05 (04B1)	Input Phase Loss Protect Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Enables or disables input phase loss detection. 0 : Disabled 1 : Enabled	1 (0, 1)	870
L8-07 (04B3)	Output Phase Loss Protect Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Enables or disables the output phase loss detection. Output phase loss detection is triggered when the output current falls below 5% of the drive rated current. Note: Output phase loss detection can mistakenly be triggered in the following situations. Disable output phase loss protection. •The motor rated current is very small compared to the drive rating. •Operates PM motors with light loads. 0 : Disabled 1 : 1PH Loss Det 2 : 2/3PH Loss Det	0 (0 - 2)	870
L8-09 (04B5)	Output Ground Fault DetectSelect	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Enables or disables ground fault protection. 0 : Disabled 1 : Enabled	Determined by o2-04 (0, 1)	871
L8-10 (04B6)	Heatsink Cooling Fan Ope Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the heatsink cooling fan operation. 0 : Dur Run (OffDly) 1 : Always On 2 : Fan ON in heating of Drive	0 (0 - 2)	871

No. (Hex.)	Name	Description	Default (Range)	Ref.
L8-11 (04B7)	HeatsinkCooling Fan Off DelayTime	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the delay time that occurs before the cooling fan is stopped after the run command is canceled when $L8-10 = 0$ [<i>Heatsink Cooling Fan Ope Select = Dur Run (OffDly)</i>].	60 s (0 - 300 s)	871
L8-12 (04B8)	Ambient Temperature Setting	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the ambient temperature of the area where the drive is installed.	40 °C (-10 to 50 °C)	871
L8-15 (04BB)	oL2 Characteristics Sel atLowSpd	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects whether the drive overload capability is reduced at low speeds to prevent <i>oL2 [Drive Overloaded]</i> to protect the main circuit transistor in the drive during low speed operation (at 6 Hz or less). Note: Contact Yaskawa or your nearest sales representative for consultation before disabling this function at low speeds. Frequent operation of drives under conditions of high output current in low speed ranges may shorten the service life of the drive IGBT due to heat stress. 0 : Disabled 1 : Enabled	1 (0, 1)	871
L8-18 (04BE)	Software Current Limit Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Enables or disables the software current limit selection used to protect the main circuit transistor from significant current. 0 : Disabled 1 : Enabled	0 (0, 1)	872
L8-19 (04BF)	FreqReductRate DuringOH Pre- Alarm	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the ratio at which the frequency reference is derated when the <i>oH</i> alarm is output.	0.8 (0.1 to 0.9)	872
L8-20 (04C0)	CF SELECT	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the operation when <i>CF</i> faults are detected when $A1-02 = 4$ [<i>Control Method Selection = Advanced OpenLoop Vector Control</i>]. 0 : Disabled 1 : Coast to Stop 2 : Fast Stop	1 (0 - 2)	872
L8-27 (04DD)	Overcurrent Detection Gain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the PM motor overcurrent detection level as a percentage of the motor rated current value.	300.0% (0.0 - 400.0%)	873
L8-29 (04DF)	Current Unbalance Detect (LF2)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Enables or disables the detection of <i>LF2</i> . 0 : Disabled 1 : Enabled	1 (0, 1)	873
L8-31 (04E1)	LF2 Detection Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the LF2 [Output Current Imbalance] detection time.	3 (1 to 100)	873
L8-32 (04E2)	Cooling Fan Failure Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the drive operation when <i>FAn [Internal Agitating Fan Fault]</i> occurs. 0 : Ramp to Stop 1 : Coast to Stop 2 : Fast-Stop 3 : Alarm Only 4 : Run@L8-19 Rate	1 (0 to 4)	873
L8-35 (04EC)	Installation Method Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the type of drive installation. 0 : IP00/Open-Chassis enclosure 1 : Side-by-Side mounting 2 : IP20/UL Type 1/IP55 3 : Finless/Fin Ext	Determined by o2-04 (0 - 3)	874

No. (Hex.)	Name	Description	Default (Range)	Ref.
L8-38 (04EF)	Carrier Frequency Reduction	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the operation of the carrier frequency reduction function. The carrier frequency is reduced when the output current exceeds a specific level. 0 : Disabled 1 : Enabled below 6 Hz 2 : Enabled for the EntireSpeedRange	Determined by A1-02, C6-01, and o2-04 (0 - 2)	874
L8-40 (04F1)	CarrierFreqReduct Off DelayTime	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the time (off delay time) taken for the automatically reduced carrier frequency to return to the state before the reduction.	Determined by A1-02 (0.00 - 2.00 s)	875
L8-41 (04F2)	High Current Alarm Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Triggers an <i>HCA [Current Alarm]</i> when the output current exceeds 150% of the drive rated current. 0 : Disabled 1 : Enabled	0 (0, 1)	875
L8-51 (0471)	STPo I Detection Level	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the <i>STPo [Desynchronization Error]</i> on the basis of the output current.	0.0% (0.0 - 300.0%)	875
L8-52 (0472)	STPo Integration Level	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the detection level for <i>STPo [Desynchronization Error]</i> on the basis of the ACR integral value.	1.0 (0.1 - 2.0)	875
L8-53 (0473)	STPo Integration Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the time until <i>STPo</i> is detected after the value of L8-51 [<i>STPo I Detection Level</i>] is exceeded.	1.0 s (1.0 - 10.0 s)	875
L8-54 (0474)	STPo Id Diff Detection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Enables or disables the Id deviation detection function for <i>STPo [Desynchronization Error]</i>. 0 : Disabled 1 : Enabled	1 (0, 1)	875
L8-55 (045F)	InternalBraking TransistorProtect	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Enables or disables protection for the internal braking transistor. 0 : Disable 1 : Protection enabled	1 (0, 1)	876
L8-56 (047D)	Stl Act Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the time that the acceleration stall prevention function can continue to operate, after which the <i>STPo [Desynchronization Error]</i> is detected.	5000 ms (100 - 5000 ms)	876
L8-57 (047E)	Stl Retry Count	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the number of times the acceleration stall prevention function can operate until speeds match, after which the <i>STPo "Desynchronization Error"</i> is detected.	10 times (1 to 10 times)	876
L8-90 (0175)	STPo Detection Level	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV <i>STPo [Desynchronization Error]</i> is detected when the control fault reaches the detection level of L8-90.	Determined by A1-02 (0 to 5000 times)	876
L8-93 (073C)	LSO Detection Time at Low Speed	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the time before baseblock is executed after <i>LSO [LSO Fault]</i> is detected.	1.0 s (0.0 - 10.0 s)	877
L8-94 (073D)	LSO Detection Level at Low Speed	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the detection level for <i>LSO [LSO Fault]</i> as a percentage of E1-04 [<i>Maximum Output Frequency</i>].	3% (0 - 10%)	877
L8-95 (077F)	Average LSO Freq at Low Speed	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the average count of <i>LSO [LSO Fault]</i> detections.	10 times (1 to 50 times)	877

◆ L9: Drive Protection 2

No. (Hex.)	Name	Description	Default (Range)	Ref.
L9-16 (11DC)	FAn1 Detection Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the detection time for <i>FAn1</i> [Drive Cooling Fan Failure]. Do not change the value of this parameter unless absolutely necessary.	4.0 s (0.0 to 30.0 s)	877

10.12 n: Special Adjustment

◆ n1: Hunting Prevention

No. (Hex.)	Name	Description	Default (Range)	Ref.
n1-01 (0580)	Hunting Prevention Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Enables or disables the hunting prevention function. 0 : Disabled 1 : Enabled 2 : Enable (High carry)	1 (0 - 2)	878
n1-02 (0581)	Hunting Prevention Gain Setting	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Adjusts the behavior of the hunting prevention function. Normally there is no need to configure this setting.	1.00 (0.00 - 2.50)	878
n1-03 (0582)	Hunting Prevention Time Constant	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Adjusts the responsiveness (primary delay time constant) of the hunting prevention function. Normally there is no need to configure this setting.	Determined by o2-04 (0 - 500 ms)	878
n1-05 (0530)	Hunting Prevention Gain while in Rev	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Adjusts the behavior of the hunting prevention function. Used to adjust Reverse run. Normally there is no need to configure this setting.	0.00 (0.00 - 2.50)	878
n1-08 (1105)	Leak cur antivib	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the method of suppressing motor vibration caused by leakage current. Normally there is no need to configure this setting. 0 : Method1 1 : Method2	0 (0, 1)	879
n1-13 (1B59)	DC Bus Stabilization Control	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Enables or disables the oscillation suppression function for the DC bus voltage. 0 : Disabled 1 : Enabled	0 (0, 1)	879
n1-14 (1B5A)	DC Bus Stabilization Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV With a setting of <i>n1-13 = 1</i> [DC Bus Stabilization Control = Enabled], adjustments that address a lack of oscillation suppression capability with respect to the DC bus voltage can be made.	100.0 ms (50.0 - 500.0 ms)	879
n1-15 (0BF8)	Voltage Calibration Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the calibration method to be used for the suppression of torque/current ripple. 0 : Calibration Invalid 1 : Calibrate only 1 time 2 : Calibrate every time	1 (0 - 2)	879
n1-16 (0BFB)	High carry Hunt Prev Gain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Adjusts the behavior of the hunting prevention function. This is most effective when a high carrier frequency has been set. Normally there is no need to configure this setting.	0.50 (0.00 - 2.50)	880
n1-17 (0BFC)	Hunt Prev Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Adjusts the responsiveness of the hunting prevention function. Normally there is no need to configure this setting.	500 ms (0 - 1000 ms)	880

◆ n2: SpdFeedbackDetectControl(AFR)Tun

No. (Hex.)	Name	Description	Default (Range)	Ref.
n2-01 (0584)	SpdFeedbackDe tectCtr (AFR) Gain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV This parameter sets the gain of the AFR function as a magnification value. Normally there is no need to change this setting.	1.00 (0.00 - 10.00)	880
n2-02 (0585)	SpdFeedback DetCtr(AFR) TimeConst1	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV This parameter sets the time constant that determines the rate of change for the AFR function. Normally there is no need to change this setting.	50 ms (0 - 2000 ms)	880
n2-03 (0586)	SpdFeedback DetCtr(AFR) TimeConst2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV This parameter sets the time constant that determines the variation in speed of the AFR function. Use this parameter when performing speed searches or regeneration. Normally there is no need to change this setting.	750 ms (0 - 2000 ms)	881

◆ n3: High Slip Braking (HSB)

No. (Hex.)	Name	Description	Default (Range)	Ref.
n3-01 (0588)	HSB Deceleration Frequency Width	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the amount by which the output frequency is to be lowered during high-slip braking, as a percentage of <i>E1-04 [Maximum Output Frequency]</i> , which represents the 100% value.	5% (1 - 20%)	883
n3-02 (0589)	High-Slip Braking Current Limit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the maximum current output during high-slip braking as a percentage of <i>E2-01 [Motor Rated Current (FLA)]</i> , which represents the 100% value. Set the current suppression so that the drive's overload tolerance is not exceeded.	Determined by C6-01, L8-38 (0 - 200%)	883
n3-03 (058A)	HSB Dwell Time at Stop	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV This sets the dwell time, a period of time in which the motor has slowed down and runs at a steady speed, which occurs when the high-slip braking is nearing completion. For a predetermined amount of time only, the actual output frequency will be held at the minimum output frequency that was set for <i>E1-09</i> .	1.0 s (0.0 - 10.0 s)	883
n3-04 (058B)	High-Slip Braking Overload Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the time used for detection of <i>oL7 [Motor Overload Protection During High Slip Braking]</i> , which is implemented at times when for some reason the output frequency did not change during high-slip braking. Normally there is no need to configure this setting.	40 s (30 - 1200 s)	883
n3-13 (0531)	Overexcitation Deceleration Gain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The overexcitation level is determined by multiplying the gain set by this parameter with the V/f pattern output value during overexcitation deceleration.	1.10 (1.00 - 1.40)	883
n3-14 (0532)	HarmInj@ HiFlxBrk	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Enables or disables the function that injects harmonic signals during overexcitation deceleration. 0 : Disabled 1 : Enabled	0 (0, 1)	884
n3-21 (0579)	High- SlipSuppression Current Lvl	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV This sets, as a percentage value, the upper limit of the current suppressed at the time of overexcitation deceleration, taking the drive rated current as a value of 100%.	100% (0 - 150%)	884
n3-23 (057B)	Overexcitation Operation Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV This parameter selects the direction of motor rotation of motors for which overexcitation operation is applied. 0 : Enabled in both directions 1 : Enabled only when rotating FWD 2 : Enabled only when in REV	0 (0 - 2)	884

◆ n4: Observer

No. (Hex.)	Name	Description	Default (Range)	Ref.
n4-60 (1B80)	Motoring Low Speed Comp Gain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV This compensation gain improves the control characteristics for motoring loads in the low speed range.	100.0% (50.0 - 200.0%)	885
n4-61 (1B81)	Low Speed Comp Frequency Level	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Set a frequency that enables the settings for <i>n4-60 [Motoring Low Speed Comp Gain]</i>, <i>n4-62 [Low Speed compensation Gain]</i>. When the output frequency < <i>n4-61</i>, torque compensation is carried out in accordance with the settings for <i>n4-60</i>, <i>n4-62</i>. Normally there is no need to change this setting.	6.00 Hz (0.50 - 12.00 Hz)	885
n4-62 (1B82)	Regen Low Speed Comp Gain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV This compensation gain improves the control characteristics for regenerative loads in the low speed range.	100.0% (50.0 - 200.0%)	885
n4-63 (1B83)	SpdEstimationResponseForHighFreq	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV In high speed ranges, where the output frequency is $\geq$ <i>n4-67 [SwitchingFreq for Estimation gain]</i>, this adjusts the responsiveness of the speed estimation.	60.0 (0.1 - 150.0)	885
n4-64 (1B84)	SpdEstimationResponse forLowFreq	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV In low speed ranges, where $0 \leq$ the output frequency, which is < <i>n4-67 [SwitchingFreq for Estimation gain]</i>, this adjusts the responsiveness of the speed estimation.	60.0 (0.1 - 150.0)	886
n4-65 (1B85)	FluxEstimationResponseForHighFrq	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV In high speed ranges, where the output frequency is $\geq$ <i>n4-67 [SwitchingFreq for Estimation gain]</i>, this adjusts the responsiveness of the magnetic flux estimation. Normally there is no need to change this setting.	0.90 (0.50 - 1.50)	886
n4-66 (1B86)	FluxEstimationResponseForLowFreq	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV In low speed ranges, where $0 \leq$ the output frequency, which is < <i>n4-67 [SwitchingFreq for Estimation gain]</i>, this adjusts the responsiveness of the magnetic flux estimation. Normally there is no need to change this setting.	0.90 (0.50 - 1.50)	886
n4-67 (1B87)	SwitchingFreq for Estimation gain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Set the switching frequency for estimation gain for each of the following: <i>n4-63 [SpdEstimationResponseForHighFreq]</i>, <i>n4-64 [SpdEstimationResponse forLowFreq]</i>, <i>n4-65 [FluxEstimationResponseForHighFrq]</i>, and <i>n4-66 [FluxEstimationResponseForLowFreq]</i>. Normally there is no need to change this setting.	6.00 Hz (0.00 to E1-04)	886
n4-68 (1B88)	FilterTimeConst forSpdEstimation	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the primary delay time constant for the speed estimation value. Normally there is no need to change this setting.	0.001 s (0.001 - 0.010 s)	886
n4-69 (1B89)	Response of Flux loop	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Stabilizes motor vibrations through unified control of magnetic flux.	1.00 (0.00 - 60.00)	887
n4-70 (1B8A)	Speed Command Comp @ Low Freq	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Adjust this to improve stability when running at low speeds. Normally there is no need to change this setting.	0.60 Hz (0.00 - 1.50 Hz)	887
n4-72 (1B8C)	PG Mode	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Select whether an encoder option is to be connected or not when set to <i>A1-02 = 4 [Control Method Selection = Advanced OpenLoop Vector Control]</i>. 0 : WithOut PG 1 : With PG	0 (0, 1)	887
n4-73 (1B8D)	PGO ret ope	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV If the encoder is disconnected, this determines whether the drive is to restart in the WithOut PG mode, or is to restart in the With PG mode. 0 : WithOut PG 1 : With PG	0 (0, 1)	887
n4-74 (1B8E)	Limit of Flux loop	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the control level for flux loop control output.	160% (100 to 500%)	888

◆ n5: Feed Forward Control

No. (Hex.)	Name	Description	Default (Range)	Ref.
n5-01 (05B0)	Feedforward Control Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Enables and disables the Feedforward function. 0 : Disabled 1 : Enabled	0 (0, 1)	889
n5-02 (05B1)	Motor Acceleration Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Set the time required for the motor to accelerate from the stopped state to the maximum frequency when using a single motor at the rated torque. The motor acceleration time is automatically set by Inertia Tuning.	Determined by C6-01, E5-01, and o2-04 (0.001 - 10.000 s)	889
n5-03 (05B2)	Feedforward Control Gain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV This parameter sets the ratio between load inertia and motor inertia. The Feedforward Control Gain value is automatically set by Inertia Tuning.	1.00 (0.00 - 100.00)	890
n5-04 (05B3) RUN	Spd Response F	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the response frequency for the speed reference in increments of 0.01 Hz. Normally there is no need to configure this setting.	Determined by A1-02 (0.00 - 500.00 Hz)	890

◆ n6: Online Tuning

No. (Hex.)	Name	Description	Default (Range)	Ref.
n6-01 (0570)	Online Tuning Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the type of motor data Online Tuning uses for OLV control. 0 : Disabled 1 : Line-to-line resistance tuning 2 : VoltageAdjustm	0 (0 - 2)	891
n6-05 (05C7)	Online Tuning Gain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the compensation gain when set to $n6-01 = 2$ [Online Tuning Selection = VoltageAdjustm]. Normally there is no need to configure this setting.	1.0 (0.1 - 50.0)	891
n6-11 (1B56)	online resister tuning	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Adjusts the responsiveness for online resister tuning. To be enabled, the value should be set to approximately 1.000. This is disabled if the value is set to 0.	0.000 (0.000 - 1.000)	891

◆ n7: EZ Drive

No. (Hex.)	Name	Description	Default (Range)	Ref.
n7-01 (3111)	Flux Estimation Cut-off Freq	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the oscillation suppression gain for the low speed range.	1.0 (0.1 - 10.0)	891
n7-05 (3115)	Torque Control Response Gain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the response gain relative to load changes.	100 (10 - 1000)	891
n7-07 (3117)	PLL response 1	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the speed calculation gain during normal operation. Normally there is no need to change this setting.	15.0 Hz (1.0 - 50.0 Hz)	892
n7-08 (3118)	PLL response 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the speed calculation gain during a speed search.	25.0 Hz (1.0 - 50.0 Hz)	892

No. (Hex.)	Name	Description	Default (Range)	Ref.
n7-10 (311A)	Sensorless Switchover StartSpeed	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV The speed range within which pull-in current commands are enabled is set as a proportion relative to the rated frequency.	10.0% (0.0 - 100.0%)	892
n7-17 (3122)	Resistance TemperatureCorrec tion	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the function for compensating for changes in the motor resistance value caused by temperature fluctuations. 0 : Invalid 1 : Valid (Only 1 time) 2 : Valid (Every time)	1 (0 - 2)	892

◆ n8: PM Motor Control Tuning

No. (Hex.)	Name	Description	Default (Range)	Ref.
n8-01 (0540)	Init Rotor Position Est Current	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets, as a percentage, the Initial Rotor Position Estimated Current, taking the <i>E5-03 [Motor Rated Current (FLA)]</i> as the 100% value. Normally there is no need to change this setting.	50% (0 - 100%)	892
n8-02 (0541)	Pole Attraction Current	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the current at the time of polar attraction as a percentage of the motor rated current, which is deemed to be 100%. Normally there is no need to change this setting.	80% (0 - 150%)	893
n8-03 (0542)	Current Starting Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the amount for the Current Starting Time, which is used when carrying out Z Pulse Offset Tuning. Normally there is no need to change this setting.	1.5 s (1.5 - 5.0 s)	893
n8-04 (0543)	Polar Attraction Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the amount for the Polar Attraction Time, which is used when carrying out Z Pulse Offset Tuning. Normally there is no need to change this setting.	1.5 s (1.5 - 5.0 s)	893
n8-11 (054A)	Induction Volt Estimation Gain 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the gain for speed estimation. Normally there is no need to change this setting.	Determined by n8-72 (0.0 - 1000.0)	893
n8-14 (054D)	Polarity Compensation Gain 3	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the gain for speed estimation. Normally there is no need to change this setting.	1.000 (0.000 - 10.000)	894
n8-15 (054E)	Polarity Compensation Gain 4	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the gain for speed estimation. Normally there is no need to change this setting.	0.500 (0.000 - 10.000)	894
n8-21 (0554)	Motor Ke Gain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the gain for speed estimation. Normally there is no need to change this setting.	0.90 (0.80 - 1.00)	894
n8-35 (0562)	InitRotorPosition Detect Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects how the rotor position is detected at start. Note: Select a value of 0 if using SPM motors. Values between 0 to 2 can be selected if using IPM motors. 0 : Pull-In 1 : High frequency injection 2 : Pulse injection	Determined by A1-02 (0 - 2)	894
n8-36 (0563)	InjectionSignal FreqForInductTurn	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the injection frequency for high frequency injection.	500 Hz (200 - 5000 Hz)	894
n8-37 (0564)	High Freq Injection Amplitude	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Considering 200 V to be the 100% value with a 200 V class unit, and 400 V to be 100% with a 400 V class unit, set the high frequency injection amplitude as a percentage value. Normally there is no need to change this setting.	20.0% (0.0 - 50.0%)	895
n8-41 (0568)	HFI Overlap Pole Detection Pgain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the speed estimation response for high frequency injection. Normally there is no need to change this setting.	3.0 (1.0 - 100.0)	895

No. (Hex.)	Name	Description	Default (Range)	Ref.
n8-42 (0569)	HFI Overlap Pole Detection iTime	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the oscillation suppression gain of the speed estimation for high frequency injection. Normally there is no need to change this setting.	1.0 (0.1 - 5.0)	895
n8-45 (0538)	Spd Feedback Detect Control Gain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the internal speed feedback detection reduction unit gain as a magnification value. Normally there is no need to change this setting.	0.80 (0.00 - 10.00)	895
n8-47 (053A)	Pull-InCurCompensationTime Const	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the time constant used to match the pull-in current reference value with the actual current value. Normally there is no need to change this setting.	5.0 s (0.0 - 100.0 s)	895
n8-48 (053B)	Pull-In Current (for PM Motors)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV On the basis that parameter E5-03 [Motor Rated Current (FLA)] is the 100% value, this parameter sets the d-axis current that flows to the motor during run at constant speed as a percentage.	30% (20 - 200%)	896
n8-49 (053C)	d-Axis Cur forHighEfficiency Cont	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets, in terms of a percentage, the d-axis current to be supplied to the motor to run it at a uniform speed with a heavy load. Considers E5-03 [Motor Rated Current] to be 100%. Normally there is no need to change this setting.	Determined by E5-01 (-200.0 - 0.0%)	896
n8-51 (053E)	Accel / Decel Pull-In Current	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets, as a percentage, the pull-in current allowed to flow during acceleration/deceleration, taking the motor rated current as a value of 100%.	Determined by A1-02 (0 - 200%)	896
n8-54 (056D)	VoltErrorCompensationTime Const	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the time constant used when compensating for voltage errors.	1.00 s (0.00 - 10.00 s)	896
n8-55 (056E)	Load Inertia	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV This parameter sets the ratio between motor inertia and machine inertia. 0 : Below 1:10 1 : Between 1:10 and 1:30 2 : Between 1:30 and 1:50 3 : Beyond 1:50	0 (0 - 3)	897
n8-57 (0574)	High Frequency Injection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects whether to perform a high frequency injection to detect motor speed. 0 : Disabled 1 : Enabled	0 (0, 1)	897
n8-62 (057D)	Output Voltage Limit (for PM)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV To prevent saturation of the output voltage, set the output voltage limit. Normally there is no need to configure this setting.	200 V Class: 200.0 V, 400 V Class: 400.0 V (200 V Class: 0.0 - 230.0 V, 400 V Class: 0.0 - 460.0 V)	898
n8-65 (065C)	SpdFdbkDetectCtrl GainduringOVSup	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the gain of internal speed feedback detection suppression while the overvoltage suppression function is working, as a magnification value. Normally there is no need to configure this setting.	1.50 (0.00 - 10.00)	898
n8-69 (065D)	Speed Calculation Gain	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the gain used for speed estimation. Normally there is no need to change this setting.	1.00 (0.00 - 20.00)	898
n8-72 (0655)	Speed Estimation Method Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the speed estimation method. Normally there is no need to change this setting. 0 : Conventional method 1 : A1000 method	1 (0, 1)	898
n8-74 (05C3)	LghtLoadCurLvl 1	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Set n8-48 [Pull-In Current (for PM Motors)] to the level of the load current (q-axis current) to be applied.	30% (0 - 255%)	898

No. (Hex.)	Name	Description	Default (Range)	Ref.
n8-75 (05C4)	LghtLoadCurLvl 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Set <i>n8-78 [MedLoad Id Level]</i> to the level of the load current (q-axis current) to be applied.	50% (0 - 255%)	898
n8-77 (05CE)	IPM HiEffCtrLev2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Set <i>n8-49 [d-Axis Cur for High Efficiency Cont]</i> to the level of the load current (q-axis current) to be applied.	90% (0 - 255%)	899
n8-78 (05F4)	MedLoad Id Level	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the level of the pull-in current for midrange loads.	0% (0 - 255%)	899
n8-84 (02D3)	InitPolarityEstimationTimeoutCur	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets, as a percentage, the current for processing an estimation of the initial motor magnetic pole, assuming that the <i>E5-03 [PM Motor Rated Current (FLA)]</i> is the 100% value.	100% (0 - 150%)	899
n8-94 (012D)	Selection of Recognition Criteria	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Choose the criteria for recognizing changes in speed or load. Normally there is no need to change this setting. 0 : Softstarter 1 : Speed feedback	Determined by d5-01 (0, 1)	899
n8-95 (012E)	Observer Estimation TimeConstant	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the time constant of the filter used with respect to the recognition criteria value for speed and load changes. Normally there is no need to change this setting.	30 ms (0 - 100 ms)	900

10.13 o: Keypad-Related Settings

◆ o1: Keypad Display Selection

No. (Hex.)	Name	Description	Default Setting (Setting Range)	Ref.
o1-01 (0500) RUN	Drive Mode Unit Monitor Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configure the <i>U</i> monitor that appears for the Drive Mode. This parameter is only enabled for LED keypads. Note: <i>U2</i> monitor [Fault Trace] and <i>U3</i> Monitor [Fault History] cannot be selected.	106 (104 - 855)	902
o1-02 (0501) RUN	User Monitor Select afterPowerUp	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the monitor item initially displayed when the drive is energized. Refer to "U: Monitors" for more information on monitor items that can be displayed. This parameter is only enabled for LED keypads. 1 : Frequency reference (U1-01) 2 : Direction 3 : Output frequency (U1-02) 4 : Output current (U1-03) 5 : UserSelect Monitor(set by o1-01)	1 (1 - 5)	902
o1-03 (0502)	Keypad Display Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the display units for the frequency reference and output frequency. 0 : 0.01 Hz 1 : 0.01% (100% = E1-04) 2 : r/min 3 : User-selected units	Determined by A1-02 (0 - 3)	902
o1-04 (0503)	V/f Pattern Display Unit	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Select the setting unit of parameters that configure the V/f pattern frequency. 0 : Hz 1 : r/min	Determined by A1-02 (0, 1)	903
o1-05 (0504) RUN	LCD Contrast adjustment	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the LCD display contrast.	3 (0 - 5)	903
o1-10 (0520)	User-Set Display Units Max Value	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the value displayed for the maximum output frequency.	Determined by o1-03 (1 - 60000)	903
o1-11 (0521)	User-SetDisplayUnits Dec Display	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the number of decimal places for frequency reference and monitor values. 0 : No Dec (XXXXXX) 1 : 1 Dec (XXXX.X) 2 : 2 Dec (XXX.XX) 3 : 3 Dec (XX.XXX)	Determined by o1-03 (0 - 3)	904
o1-24 to o1-35 (11AD to 11B8) RUN	1st to 12th Monitor Settings	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Up to a maximum of 12 desired monitors can be selected as user monitors. This parameter is only enabled for LED keypads.	o1-24: 101 o1-25: 102 o1-26: 103 o1-27 to o1-35: 000 (000, 101 - 825)	904
o1-36 (11B9) RUN	LCD backlight adjustment	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the brightness of the LCD keypad backlight.	3 (1 - 5)	904

No. (Hex.)	Name	Description	Default Setting (Setting Range)	Ref.
o1-37 (11BA) RUN	LCD backlight ON/OFF Selection	V/f CL-V/f OLV CLV AOLV OLVP/AOLV/PM CLV/PM EZOLV Selects the operation of the LCD backlight automatic shut off function. 0 : OFF 1 : ON	1 (0, 1)	904
o1-38 (11BB) RUN	Time to turn off LCD backlight	V/f CL-V/f OLV CLV AOLV OLVP/AOLV/PM CLV/PM EZOLV Configures the time at which the LCD backlight automatically turns off.	60 s (10 - 300 s)	905
o1-39 (11BC)	Initial setup display selection	V/f CL-V/f OLV CLV AOLV OLVP/AOLV/PM CLV/PM EZOLV Selects whether or not to display the LCD keypad initial setup screen every time the power is turned on. This parameter is only enabled for LED keypads. 0 : No 1 : Yes	1 (0, 1)	905
o1-40 (11BD) RUN	Home display selection	V/f CL-V/f OLV CLV AOLV OLVP/AOLV/PM CLV/PM EZOLV Selects the monitor display mode used to display the Home screen. This parameter is only enabled for LED keypads. 0 : Custom Monitor 1 : Bar Graph 2 : Line Graph / XY Plot	0 (0 - 2)	905
o1-41 (11C1) RUN	1st Monitor Min Setting	V/f CL-V/f OLV CLV AOLV OLVP/AOLV/PM CLV/PM EZOLV Selects the horizontal axis used to display the monitor configured with o1-24 as a bar graph. This parameter is only enabled for LED keypads. 0 : + - Area (- o1-42 ~ o1-42) 1 : + Area (0 ~ o1-42) 2 : - Area (- o1-42 ~ 0)	0 (0 - 2)	905
o1-42 (11C2) RUN	1st Monitor Max Setting	V/f CL-V/f OLV CLV AOLV OLVP/AOLV/PM CLV/PM EZOLV Configures the setting value for the horizontal axis used to display the monitor configured with o1-24 as a bar graph. This parameter is only enabled for LED keypads.	100.0% (0.0 - 100.0%)	906
o1-43 (1131) RUN	2nd Monitor Min Setting	V/f CL-V/f OLV CLV AOLV OLVP/AOLV/PM CLV/PM EZOLV Selects the horizontal axis used to display the monitor configured with o1-25 as a bar graph. This parameter is only enabled for LED keypads. 0 : + - Area (- o1-46 ~ o1-46) 1 : + Area (0 ~ o1-46) 2 : - Area (- o1-46 ~ 0)	0 (0 - 2)	906
o1-44 (11C4) RUN	2nd Monitor Max Setting	V/f CL-V/f OLV CLV AOLV OLVP/AOLV/PM CLV/PM EZOLV Configures the setting value for the horizontal axis used to display the monitor configured with o1-25 as a bar graph. This parameter is only enabled for LED keypads.	100.0% (0.0 - 100.0%)	906
o1-45 (11C5) RUN	3rd Monitor Min Setting	V/f CL-V/f OLV CLV AOLV OLVP/AOLV/PM CLV/PM EZOLV Selects the horizontal axis used to display the monitor configured with o1-26 as a bar graph. This parameter is only enabled for LED keypads. 0 : + - Area (- o1-46 ~ o1-46) 1 : + Area (0 ~ o1-46) 2 : - Area (- o1-46 ~ 0)	0 (0 - 2)	906
o1-46 (11C6) RUN	3rd Monitor Max Setting	V/f CL-V/f OLV CLV AOLV OLVP/AOLV/PM CLV/PM EZOLV Configures the setting value for the horizontal axis used to display the monitor configured with o1-26 as a bar graph. This parameter is only enabled for LED keypads.	100.0% (0.0 - 100.0%)	906
o1-55 (11EE) RUN	Analog Gauge Min Setting	V/f CL-V/f OLV CLV AOLV OLVP/AOLV/PM CLV/PM EZOLV Selects the region used to display the monitor configured with o1-24 [1st Monitor Setting] as an analog meter. This parameter is only enabled for LED keypads.	1 (1)	906
o1-56 (11EF) RUN	Analog Gauge Area Setting	V/f CL-V/f OLV CLV AOLV OLVP/AOLV/PM CLV/PM EZOLV Configures the setting value used when displaying the monitor configured with o1-24 [1st Monitor Setting] as an analog meter. This parameter is only enabled for LED keypads.	100.0% (0.0 - 100.0%)	907

◆ o2: Keypad Operation

No. (Hex.)	Name	Description	Default (Range)	Ref.
o2-01 (0505)	LO/RE Key Function Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects whether to enable or disable switching between local and remote modes via LO/RE. 0 : Disabled 1 : Enabled	1 (0, 1)	907
o2-02 (0506)	STOP Key Function Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects whether or not to enable functionality of the STOP on the keypad when the Run command source for the drive is set to REMOTE (external) and not assigned to the keypad. 0 : Disabled 1 : Enabled	1 (0, 1)	907
o2-03 (0507)	User Parameter Default Value	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Changed parameter setting values are stored as the user parameter default settings used when the drive is initialized. 0 : No change 1 : Set defaults 2 : Clear all	0 (0 - 2)	908
o2-04 (0508)	Drive Model Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the Drive Model code for the corresponding Drive Model. This parameter must be configured when control boards are replaced.	Determined by the drive (-)	908
o2-05 (0509)	Freq Ref Setting Method Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects whether or not the  must be pressed to change the frequency reference value with the keypad when in Drive Mode. 0 : Disabled 1 : Enabled	0 (0, 1)	908
o2-06 (050A)	Ope Select @Keypad is Disconnect	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects whether or not to stop the drive when the keypad connection cable is disconnected from the drive or damaged while the keypad is the Run command source. 0 : Disabled 1 : Enabled	0 (0, 1)	909
o2-07 (0527)	MotorDirect@Po wUpWhenUsing Keypad	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the direction of motor rotation when the power is turned on when the keypad is the Run command source. 0 : Forward 1 : Reverse	0 (0, 1)	909
o2-09 (050D)	Factory use	-	-	-
o2-23 (11F8)	Lost Detection of Ext. Power 24V	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects whether or not to provide warning when the backup external 24-V power supply turns off while the main circuit power supply is supplied. 0 : Disabled 1 : Enabled	0 (0, 1)	909
o2-24 (11FE)	LED Light Function Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the option to enable or disable the LED status rings and LED lamps on the keypad. 0 : Both Enable 1 : LED Status Ring Disable 2 : Keypad LED Light Disable	0 (0 - 2)	910

◆ o3: Copy Function

No. (Hex.)	Name	Description	Default (Range)	Ref.
o3-01 (0515)	Copy Function Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Drive parameters can be saved and copied to another drive using the keypad. 0 : Copy select 1 : Drive → Keypad Backup 2 : Keypad → Drive Restore 3 : Keypad ↔ Drive (compare)	0 (0 - 3)	910
o3-02 (0516)	Copy Allowed Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the enabled/disabled status of backup when <i>o3-01 = 1</i> [<i>Copy Function Selection = Drive → Keypad Backup</i>]. 0 : Disabled 1 : Enabled	0 (0, 1)	910
o3-04 (0B3E)	Select Backup/Restore Location	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the storage region for drive parameters when backing up and restoring parameters. This parameter is only enabled for LED keypads. 0 : Memory Location 0 1 : Memory Location 1 2 : Memory Location 2 3 : Memory Location 3	0 (0 - 3)	910
o3-05 (0BDA)	Select items to Backup/Restore	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the parameters that are backed up, restored, and referenced. This parameter is only enabled for LED keypads. 0 : Standard Parameters 1 : Standard + DWEZ Parameters	0 (0, 1)	911
o3-06 (0BDE)	Auto Parameter Backup Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects whether to enable or disable the automatic parameter backup function. This parameter is only enabled for LED keypads. 0 : Disabled 1 : Enabled	1 (0, 1)	911
o3-07 (0BDF)	Period setting of auto backup	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the interval at which the automatic parameter backup function saves parameters from the drive to the keypad. Note: This parameter is only enabled for LED keypads. 0 : every 10 minutes 1 : every 30 minutes 2 : every 60 minutes 3 : every 12 hours	1 (0 - 3)	911

◆ o4: Maintenance Mon Settings

No. (Hex.)	Name	Description	Default (Range)	Ref.
o4-01 (050B)	Cumulative Operation TimeSetting	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the initial value of the cumulative drive operation time in units of 10 hours.	0 h (0 - 9999 h)	912
o4-02 (050C)	Cumulative Operation Time Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the condition used to count the cumulative operation time. 0 : Logs power-on time 1 : Running Time	0 (0, 1)	912
o4-03 (050E)	CoolingFan OperationTime Setting	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the value from which to start the cumulative drive cooling fan operation time in 10-hour units.	0 h (0 - 9999 h)	912
o4-05 (051D)	Capacitor Maintenance Setting	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV U4-05 [<i>Capacitor Maintenance</i>] monitor values can be overwritten.	0% (0 - 150%)	912

10.13 o: Keypad-Related Settings

No. (Hex.)	Name	Description	Default (Range)	Ref.
o4-07 (0523)	DCBusPreChargeRelayMaintenanceSetting	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV U4-06 [SchgBypassRelayMaint] monitor values can be overwritten.	0% (0 - 150%)	912
o4-09 (0525)	IGBT Maintenance Setting	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV U4-07 [IGBT Maintenance] monitor values can be overwritten.	0% (0 - 150%)	913
o4-11 (0510)	U2, U3 Initialization	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Resets the records of Monitors U2-xx [Fault Trace] and U3-xx [Fault History]. 0 : No Reset 1 : Reset	0 (0, 1)	913
o4-12 (0512)	kWh Monitor Initialization	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Resets the monitor values for U4-10 [kWh, Lower 4 Digits] and U4-11 [kWh, Upper 5 Digits]. 0 : No Reset 1 : Reset	0 (0, 1)	913
o4-13 (0528)	NumOfRunCommands Counter Initial	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Resets the monitor values for U4-02 [Num of Run Commands], U4-24 [No of Travels(L)], and U4-25 [No of Travels(H)]. 0 : No Reset 1 : Reset	0 (0, 1)	913
o4-22 (154F)	Time Format	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the time display format. This parameter is only enabled for LED keypads. 0 : 24 hour clock 1 : 12 hour EA clock 2 : 12 hour JP clock	0 (0 - 2)	914
o4-23 (1550)	Date Format	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the date display format. This parameter is only enabled for LED keypads. 0 : YYYY/MM/DD(2016/01/31) 1 : DD/MM/YYYY(31/01/2016) 2 : MM/DD/YYYY(01/31/2016)	0 (0 - 2)	914
o4-24 (310F) RUN	bAT Detection Operation Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the operation performed when bAT (low keypad battery) faults are detected. 0 : Disabled 1 : Enabled	0 (0, 1)	914

◆ o5: Log Function

No. (Hex.)	Name	Description	Default (Range)	Ref.
o5-01 (1551) RUN	Log Start/Stop Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Starts or stops the data log function. This parameter is only enabled for LED keypads. 0 : OFF 1 : ON	0 (0, 1)	917
o5-02 (1552) RUN	Log Sampling Interval	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the data log sampling cycle. This parameter is only enabled for LED keypads.	1000 ms (100 - 60000 ms)	917
o5-03 (1553) RUN	Log Monitor Data 1	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the data log monitor. This parameter is only enabled for LED keypads.	101 (000,101 - 855)	917
o5-04 (1554) RUN	Log Monitor Data 2	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the data log monitor. This parameter is only enabled for LED keypads.	102 (000,101 - 855)	918

No. (Hex.)	Name	Description	Default (Range)	Ref.
o5-05 (1555) RUN	Log Monitor Data 3	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the data log monitor. This parameter is only enabled for LED keypads.	103 (000,101 - 855)	918
o5-06 (1556) RUN	Log Monitor Data 4	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the data log monitor. This parameter is only enabled for LED keypads.	107 (000,101 - 855)	918
o5-07 (1557) RUN	Log Monitor Data 5	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the data log monitor. This parameter is only enabled for LED keypads.	108 (000,101 - 855)	918
o5-08 (1558) RUN	Log Monitor Data 6	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the data log monitor. This parameter is only enabled for LED keypads.	000 (000,101 - 855)	918
o5-09 (1559) RUN	Log Monitor Data 7	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the data log monitor. This parameter is only enabled for LED keypads.	000 (000,101 - 855)	919
o5-10 (155A) RUN	Log Monitor Data 8	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the data log monitor. This parameter is only enabled for LED keypads.	000 (000,101 - 855)	919
o5-11 (155B) RUN	Log Monitor Data 9	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the data log monitor. This parameter is only enabled for LED keypads.	000 (000,101 - 855)	919
o5-12 (155C) RUN	Log Monitor Data 10	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Configures the data log monitor. This parameter is only enabled for LED keypads.	000 (000,101 - 855)	919

10.14 q: DriveWorksEZ Parameters

◆ q1-01 to q8-40: Reserved for DriveWorksEZ

No. (Hex.)	Name	Description	Default (Range)
q1-01 to q8-40: (1600 to 17E7)	Reserved for DriveWorksEZ	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV This parameters are reserved for use with DriveWorksEZ.	Refer to the DriveWorksEZ Online Manual.

Note:

qx-xx parameters are reserved for use with DriveWorksEZ. Refer to the DriveWorksEZ Online Manual for more information on these parameters.

10.15 r: DWEZ Connection 1-20

◆ r1-01 to r1-40: DriveWorksEZ Connection Parameters 1 to 20 (Upper / Lower)

No. (Hex.)	Name	Description	Default (Range)
r1-01 to r1-40: (1840 - 1867)	DriveWorksEZ Connection Parameters 1 to 20 (Upper / Lower)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV DriveWorksEZ Connection Parameters 1 to 20 (Upper / Lower)	0 (0 - FFFFH)

Note:

r1-xx parameters are reserved for use with DriveWorksEZ. Refer to the DriveWorksEZ Online Manual for more information on these parameters.

10.16 T: Motor Tuning

◆ T0: Tuning Mode Selection

No. (Hex.)	Name	Description	Default (Range)	Ref.
T0-00 (1197)	Tuning Mode Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the type of Auto-Tuning to be used. 0 : Motor Parameter Tuning 1 : Control Tuning	0 (0, 1)	920

◆ T1: Induction Motor Auto-Tuning

No. (Hex.)	Name	Description	Default (Range)	Ref.
T1-00 (0700)	Motor 1/Motor 2 Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the motor to be tuned when motor 1/2 switching is enabled. This parameter can only be set via the keypad not via external input terminals. Note: This parameter can be set when $H1-xx = 16$ [Motor 2 Selection] is ON and is not displayed when $H1-xx = 16$ is OFF. 1 : Motor 1 (sets E3-xx, E4-xx) 2 : Motor 2 (sets E3-xx, E4-xx)	1 (1, 2)	920
T1-01 (0701)	Auto-Tuning Mode Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the type of Auto-Tuning to be used. 0 : Rotational Auto-Tuning 1 : Stationary Auto-Tuning 1 2 : StaTun for LinetoLine Resistance	Determined by A1-02 (Determined by A1-02)	921
T1-02 (0702)	Motor Rated Power	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the rated output power (kW) of the motor.	Determined by o2-04 and C6-01 (0.00 - 650.00 kW)	921
T1-03 (0703)	Motor Rated Voltage	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the rated voltage (V) of the motor. Enter the base speed voltage here for constant output motors.	Determined by o2-04 and C6-01 (200 V Class: 0.0 - 255.5 V, 400 V Class: 0.0 - 511.0 V)	921
T1-04 (0704)	Motor Rated Current	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the rated current (A) of the motor.	Determined by o2-04 (10% to 200% of the drive rated current)	921
T1-05 (0705)	Motor Base Frequency	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the base frequency of the motor (Hz).	50.0 Hz (0.0 - 590.0 Hz)	921
T1-06 (0706)	Number of Motor Poles	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the number of motor poles.	4 (2 - 48)	922
T1-07 (0707)	Motor Base Speed	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the motor base speed for performing auto tuning (min^{-1} (r/min)).	1450 min^{-1} (r/min) (0 - 24000 min^{-1} (r/min))	922
T1-08 (0708)	PG Number of PulsesPerRevolution	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the number of PG (pulse generator, encoder) pulses.	1024 ppr (0 - 60,000 ppr)	922
T1-09 (0709)	Motor No-Load Current	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the no-load current of the motor.	- (0A to T1-04; max. of 2999.9)	922

No. (Hex.)	Name	Description	Default (Range)	Ref.
T1-10 (070A)	Motor Rated Slip	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the motor rated slip.	- (0.00 - 20.00 Hz)	922
T1-11 (070B)	Motor Iron Loss	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the iron loss information for determining the energy-saving coefficient.	Determined by E2-11 or E4-11 (0 - 65535 W)	922

◆ T2: PM Motor Auto-Tuning

No. (Hex.)	Name	Description	Default (Range)	Ref.
T2-01 (0750)	PM Motor Auto-Tuning Mode Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the type of Auto-Tuning for PM motors to be used. 0 : PM Motor Parameter Settings 1 : PM Stationary Auto-Tuning 2 : PM StaTun for Stator Resistance 3 : Z Pulse Offset Tuning 4 : PM Rotational Auto-Tuning	0 (Determined by A1-02)	923
T2-02 (0751)	PM Motor Code Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV If the drive is operating a Yaskawa PM motor from the SMRA, SSR1, or SST4 series, enter the PM motor code in accordance with the rotation speed and motor output.	Determined by A1-02 and o2-04 (0000 - FFFF)	923
T2-03 (0752)	PM Motor Type	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the type of PM motor the drive will operate. 0 : IPM Motor 1 : SPM Motor	1 (0, 1)	923
T2-04 (0730)	PM Motor Rated Power	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the rated output power (kW) of a PM motor.	Determined by o2-04 and C6-01 (0.00 - 650.00 kW)	924
T2-05 (0732)	PM Motor Rated Voltage	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the rated voltage (V) of the motor.	200 V Class: 200.0 V, 400 V Class: 400.0 V (200 V Class: 0.0 - 255.0 V, 400 V Class: 0.0 - 510.0 V)	924
T2-06 (0733)	PM Motor Rated Current	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the rated current (A) of the motor.	Determined by o2-04 (10% to 200% of the drive rated current)	924
T2-07 (0753)	PM Motor Base Frequency	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the base frequency of the motor (Hz).	87.5 Hz (0.0 - 590.0 Hz)	924
T2-08 (0734)	Number of PM Motor Poles	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the number of motor poles.	6 (2 - 48)	924
T2-09 (0731)	PM Motor Base Speed	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the motor base speed (min ⁻¹ (r/min)).	1750 min ⁻¹ (r/min) (0 - 34500 min ⁻¹ (r/min))	924
T2-10 (0754)	PM Motor Stator Resistance	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the stator resistance per phase of the motor. Note: Do not confuse this parameter with line-to-line resistance.	Determined by T2-02 (0.000 - 65.000 Ω)	924
T2-11 (0735)	PM Motor d-Axis Inductance	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the d-Axis inductance of the motor on a per phase basis.	Determined by T2-02 (0.00 - 600.00 mH)	925
T2-12 (0736)	PM Motor q-Axis Inductance	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the q-Axis inductance of the motor on a per phase basis.	Determined by T2-02 (0.00 - 600.00 mH)	925

No. (Hex.)	Name	Description	Default (Range)	Ref.
T2-13 (0755)	InducedVoltage Const Unit Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the unit used for setting the induced voltage constant. 0 : mV/(r/min) 1 : mV/(rad/sec)	1 (0, 1)	925
T2-14 (0737)	PM Motor Induced Voltage Const	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the motor induced voltage constant (K _e).	Determined by T2-13 (0.0 - 2000.0)	925
T2-15 (0756)	Pull-InCurrentLv forPM Motor Tun	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the level of the pull-in current as a percentage, with 100% representing the motor rated current. Normally there is no need to configure this setting.	30% (0 - 120%)	925
T2-16 (0738)	PGNumOfPulses/ Rev forPMMotor Tun	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the number of PG (pulse generator, encoder) pulses.	1024 ppr (1 - 15000 ppr)	925
T2-17 (0757)	Encoder Z-Pulse Offset	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV If the encoder Z-pulse offset ($\Delta\theta$) (pulse generator, encoder) is listed on the motor nameplate, set it in units of 0.1°.	0.0° (-180.0 - +180.0°)	925

◆ T3: ASR and Inertia Tuning

No. (Hex.)	Name	Description	Default (Range)	Ref.
T3-00 (1198)	Control Loop Tuning Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the Control Auto-Tuning method to be used. 0 : Inertia Tuning 1 : ASR Tuning	0 (0 - 1)	926
T3-01 (0760)	Test Signal Frequency	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Normally there is no need to change this setting. Sets the frequency of the test signal applied to the motor during Inertia Tuning.	3.0 Hz (0.1 - 20.0 Hz)	926
T3-02 (0761)	Test Signal Amplitude	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the amplitude of the test signal applied to the motor during Inertia Tuning. Normally there is no need to change this setting.	0.5 rad (0.1 - 10.0 rad)	926
T3-03 (0762)	Motor Inertia	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the inertia of the motor. This value is used to determine the load inertia using the test signal response.	Determined by o2-04, C6-01, and E5-01 (0.0001 - 6.0000 kgm ²)	926
T3-04 (0763)	System Response Frequency	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV C5-01 [ASR Proportional Gain 1] is automatically calculated and set using the load inertia value derived by the Inertia Tuning process.	10.0 Hz (0.1 - 50.0 Hz)	926

◆ T4: EZ Tuning

No. (Hex.)	Name	Description	Default (Range)	Ref.
T4-01 (3130)	EZ Tuning Mode Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the Auto-Tuning method used for EZ Open Loop Vector Control. 0 : Motor constant setting Auto-Tuning 1 : Stationary Auto-Tuning for Line-to-Line Resistance	0 (0, 1)	927
T4-02 (3131)	Motor Type Selection	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Selects the type of motor. 0 : IM 1 : PM 2 : SynRM	0 (0, 1, 2)	927

No. (Hex.)	Name	Description	Default (Range)	Ref.
T4-03 (3132)	Motor Max Revolutions	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the motor max revolutions (min ⁻¹).	- ((40 to 120 Hz) × 60 × 2 / E9-08)	927
T4-04 (3133)	Motor Rated Revolutions	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets rated rotation speed of the motor (min ⁻¹).	- ((40 to 120 Hz) × 60 × 2 / E9-08)	927
T4-05 (3134)	Motor Rated Frequency	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the motor rated frequency (Hz).	Determined by E9-01 and o2-04 (40.0 - 120.0 Hz)	928
T4-06 (3135)	Motor Rated Voltage	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the rated voltage (V) of the motor.	200 V Class: 200.0 V, 400 V Class: 400.0 V (200 V Class: 0.0 - 255.0 V, 400 V Class: 0.0 - 510.0 V)	928
T4-07 (3136)	Motor Rated Current	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the rated current (A) of the motor.	Determined by o2-04 and C6-01 (10% to 200% of the drive rated current)	928
T4-08 (3137)	Motor Rated Capacity	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the motor rated capacity in 0.01 kW units.	Determined by E9-10 (0.10 - 650.00 kW)	928
T4-09 (3138)	Number of Motor Poles	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the number of motor poles.	Determined by E9-01 (2 - 48)	928
T4-10 (3139)	Motor Rated Slip	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the amount of motor rated slip in Hz units.	Determined by o2-04 (0.000 - 20.000 Hz)	928
T4-11 (313A)	Motor Line Resistance	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Sets the line-to-line resistance for motor stator windings in Ω units.	Determined by E9-01, o2-04, and o2-09. (0.000 - 65.000 Ω)	929

10.17 U: Monitors

◆ U1: Operation Status Monitor

No. (Hex.)	Name	Description	MFAO Signal Level	Ref.
U1-01 (0040)	Frequency reference	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the frequency reference value. Display units are determined by <i>o1-03 [Keypad Display Selection]</i>. Unit: 0.01 Hz	10 V = Max. frequency (-10 V to +10 V)	-
U1-02 (0041)	Output frequency	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the current output frequency. Display units are determined by <i>o1-03 [Keypad Display Selection]</i>. Unit: 0.01 Hz	10 V = Max. frequency (-10 V to +10 V)	-
U1-03 (0042)	Output current	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the current output current. The value of <i>U1-03</i> is displayed in ampere (A) on the keypad. When viewing via MEMOBUS/Modbus communications, the current is "8192 = drive rated current (A)." Current can be calculated from the monitor value present at MEMOBUS/Modbus communications using "Numerals being displayed / 8192 × drive rated current (A)." Unit: Determined by the drive model. - Models 2004 to 2042, 4002 to 4023: 0.01 A - Models 2056 to 2415, 4031 to 4726: 0.1 A	10 V = Drive rated current	-
U1-04 (0043)	Control method	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the control method assigned to the drive. 0 : V/f Control 1 : Closed Loop V/f Control 2 : Open Loop Vector Control 3 : Closed Loop Vector Control 4 : Advanced OpenLoop Vector Control 5 : PM Open Loop Vector Control 6 : PM Advanced Open Loop Vector 7 : PM Closed Loop Vector Control 8 : EZ Open Loop Vector Control	No signal output available	-
U1-05 (0044)	Motor speed	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the motor speed that is currently detected. Display units are determined by <i>o1-03 [Keypad Display Selection]</i>. Unit: 0.01 Hz	10 V = Max. frequency (-10 V to +10 V)	-
U1-06 (0045)	OutVoltage Reference	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the output voltage reference. Unit: 0.1 V	200 V class: 10 V = 200 Vrms 400 V class: 10 V = 400 Vrms	-
U1-07 (0046)	DC bus voltage	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the DC bus voltage. Unit: 1 V	200 V class: 10 V = 200 V 400 V class: 10 V = 400 V	-
U1-08 (0047)	Output Power	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the output power (this value is calculated internally). The signal level of the analog output changes according to the <i>A1-02 [Control Method Selection]</i> setting. <i>A1-02</i> = 0, 1 [<i>V/f Control</i>]: Drive capacity (kW) <i>A1-02</i> = 2 to 8 [<i>Vector Control</i>]: Motor Rated Power (kW) [E2-11] Unit: Determined by the maximum applicable motor output. The maximum applicable motor output is determined by the drive capacity and <i>C6-01 [Normal / Heavy Duty Selection]</i>. - Less than 11 kW (15 HP): 0.01 kW - Less than 11 kW (15 HP): 0.1 kW	10 V: Drive capacity (motor rated power) kW (-10 V to +10 V)	-

No. (Hex.)	Name	Description	MFAO Signal Level	Ref.
U1-09 (0048)	Torque Reference	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the internal torque reference value. Unit: 0.1%	10 V = Motor rated torque (-10 V to +10 V)	-
U1-10 (0049)	Input Terminal Status	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the status of the multi-function input terminal using 1 (ON) and 0 (OFF). For example, <i>U1-10 = 00000011</i> is displayed when terminals S1 and S2 are ON. bit 0 : Terminal S1 (MFDI 1) bit 1 : Terminal S2 (MFDI 2) bit 2 : Terminal S3 (MFDI 3) bit 3 : Terminal S4 (MFDI 4) bit 4 : Terminal S5 (MFDI 5) bit 5 : Terminal S6 (MFDI 6) bit 6 : Terminal S7 (MFDI 7) bit 7 : Terminal S8 (MFDI 8)	No signal output available	-
U1-11 (004A)	Output Terminal Status	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the status of the multi-function output terminal using 1 (ON) and 0 (OFF). For example, <i>U1-11 = 00000011</i> is displayed when terminals M1 and M3 are ON. bit 0 : Terminals M1-M2 bit 1 : Terminals M3-M4 bit 2 : Terminals M5-M6 bit 3 : Not used (normal value of 0). bit 4 : Not used (normal value of 0). bit 5 : Not used (normal value of 0). bit 6 : Not used (normal value of 0). bit 7 : Fault relay MA/MB-MC	No signal output available	-
U1-12 (004B)	Drive Status	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the drive status using 1 (ON) and 0 (OFF). For example, <i>U1-12 = 00000101</i> is displayed during run with the Reverse run command. bit 0 : During run bit 1 : During zero-speed bit 2 : During reverse bit 3 : During fault reset signal input bit 4 : During speed agreement bit 5 : Drive ready bit 6 : During minor fault detection bit 7 : During fault detection	No signal output available	-
U1-13 (004E)	Terminal A1 Input Lv	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the signal level of terminal A1. Unit: 0.1%	10 V = 100% (-10 V to +10 V)	-
U1-14 (004F)	Terminal A2 Input Lv	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the signal level of terminal A2. Unit: 0.1%	10 V = 100% (-10 V to +10 V)	-
U1-15 (0050)	Terminal A3 Input Lv	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the signal level of terminal A3. Unit: 0.1%	0 V = 100% (-10 V to +10 V)	-
U1-16 (0053)	Output Freq afterSFS	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the output frequency after soft start. Displays the frequency with acceleration and deceleration times and S-curves. Display units are determined by <i>o1-03 [Keypad Display Selection]</i>. Unit: 0.01 Hz	10 V = Max. frequency (-10 V to +10 V)	-

No. (Hex.)	Name	Description	MFAO Signal Level	Ref.
U1-17 (0058)	DI-A3 Input Status	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the reference value input from the DI-A3 option card. Displays in hexadecimal as determined by the F3-01 [Digital Input Function Selection] setting. 3FFFF: Set (1 bit) + Sign (1 bit) + 16 bit	No signal output available	-
U1-18 (0061)	oPE Fault Parameter	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays a parameter number that caused the oPE02 [Parameter Range Setting Error] or oPE08 [Parameter Selection Error].	No signal output available	-
U1-19 (0066)	MEMOBUS/Modbus Error Code	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the contents of the MEMOBUS/Modbus communication error using 1 (error) and 0 (no error). For example, U1-19 = 00000001 is displayed when a CRC error occurs. bit 0 : CRC Error bit 1 : Data Length Error bit 2 : Not used (normal value of 0). bit 3 : Parity Error bit 4 : Overrun Error bit 5 : Framing Error bit 6 : Timed Out bit 7 : Not used (normal value of 0).	No signal output available	-
U1-21 (0077)	AI-A3 Term V1 Monitor	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the input voltage to terminal V1 on analog input option card AI-A3. Unit: 0.1%	10 V = 100% (-10 V to +10 V)	-
U1-22 (072A)	AI-A3 Term V2 Monitor	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the input voltage to terminal V2 on analog input option card AI-A3. Unit: 0.1%	10 V = 100% (-10 V to +10 V)	-
U1-23 (072B)	AI-A3 Term V3 Monitor	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the input voltage to terminal V3 on analog input option card AI-A3. Unit: 0.1%	10 V = 100% (-10 V to +10 V)	-
U1-24 (007D)	Input Pulse Monitor	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the frequency to pulse train input terminal RP. Unit: 1 Hz	Determined by H6-02	-
U1-25 (004D)	SoftwareNumber Flash	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the FLASH ID.	No signal output available	-
U1-26 (005B)	SoftwareNumber ROM	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the ROM ID.	No signal output available	-
U1-50 (1199)	Virtual Ai	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the virtual analog input value.	Determined by H7-40	-
U1-91 (154E)	Output Voltage	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the drive internal output voltage reference. Unit: 0.1 V	200 V class: 10 V = 200 Vrms 400 V class: 10 V = 400 Vrms	-

◆ U2: Fault Trace

No. (Hex.)	Name	Description	MFAO Signal Level	Ref.
U2-01 (0080)	Current Fault	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the current fault.	No signal output available	-
U2-02 (0081)	Previous Fault	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the latest fault.	No signal output available	-

No. (Hex.)	Name	Description	MFAO Signal Level	Ref.
U2-03 (0082)	FreqRef at Pre Fault	V/f CL-V/f OLV CLV AOLV OLVP/AOLV/PM CLVP/PM EZOLV Displays the frequency reference at the latest fault. The current frequency reference can be monitored by <i>U1-01 [Freq Reference]</i>. Unit: 0.01 Hz	No signal output available	-
U2-04 (0083)	OutFreq at Pre Fault	V/f CL-V/f OLV CLV AOLV OLVP/AOLV/PM CLVP/PM EZOLV Displays the output frequency at the latest fault. The current output frequency can be monitored by <i>U1-02 [Output Frequency]</i>. Unit: 0.01 Hz	No signal output available	-
U2-05 (0084)	OutCurr at Pre Fault	V/f CL-V/f OLV CLV AOLV OLVP/AOLV/PM CLVP/PM EZOLV Displays the output current at the latest fault. The current output current can be monitored by <i>U1-03 [Output Current]</i>. The value of <i>U1-03</i> is displayed in ampere (A) on the keypad. When viewing via MEMOBUS/Modbus communications, the current is "8192 = drive rated current (A)." Current can be calculated from the monitor value present at MEMOBUS/Modbus communications using "Numerals being displayed / 8192 × drive rated current (A)." Unit: Determined by the drive model. • Models 2004 to 2042, 4002 to 4023: 0.01 A • Models 2056 to 2415, 4031 to 4726: 0.1 A	No signal output available	-
U2-06 (0085)	MotorSpd at PreFault	V/f CL-V/f OLV CLV AOLV OLVP/AOLV/PM CLVP/PM EZOLV Displays the motor speed at the latest fault. The current motor speed can be monitored by <i>U1-05 [Motor Speed]</i>. Unit: 0.01 Hz	No signal output available	-
U2-07 (0086)	OutVolt at Pre Fault	V/f CL-V/f OLV CLV AOLV OLVP/AOLV/PM CLVP/PM EZOLV Displays the output voltage reference at the latest fault. The current output voltage reference can be monitored by <i>U1-06 [OutVoltage Reference]</i>. Unit: 0.1 V	No signal output available	-
U2-08 (0087)	DCBusVolt atPreFault	V/f CL-V/f OLV CLV AOLV OLVP/AOLV/PM CLVP/PM EZOLV Displays the DC bus voltage at the latest fault. The current DC bus voltage can be monitored by <i>U1-07 [DC Bus Voltage]</i>. Unit: 1 V	No signal output available	-
U2-09 (0088)	Out Pow at Pre Fault	V/f CL-V/f OLV CLV AOLV OLVP/AOLV/PM CLVP/PM EZOLV Displays the output power at the latest fault. The current output power can be monitored by <i>U1-08 [Output Power]</i>. Unit: 0.1 kW	No signal output available	-
U2-10 (0089)	Torque Ref at Previous Fault	V/f CL-V/f OLV CLV AOLV OLVP/AOLV/PM CLVP/PM EZOLV Displays the torque reference at the latest fault as a percentage of the motor rated torque. The current torque reference can be monitored by <i>U1-09 [Torque Reference]</i>. Unit: 0.1%	No signal output available	-

No. (Hex.)	Name	Description	MFAO Signal Level	Ref.
U2-11 (008A)	In Term Status at Previous Fault	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the status of the multi-function digital input terminals at the latest fault using 1 (ON) and 0 (OFF). The status of the current MFDI terminals can be monitored by <i>U1-10 [Input Terminal Status]</i>. For example, <i>U2-11 = 00000011</i> is displayed when terminals S1 and S2 are ON. bit 0 : Terminal S1 bit 1 : Terminal S2 bit 2 : Terminal S3 bit 3 : Terminal S4 bit 4 : Terminal S5 bit 5 : Terminal S6 bit 6 : Terminal S7 bit 7 : Terminal S8	No signal output available	-
U2-12 (008B)	OutTerm Status at Previous Fault	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the status of the multi-function digital output terminal at the latest fault using 1 (ON) and 0 (OFF). The current status of the MFDO terminals can be monitored by <i>U1-11 [Output Terminal Status]</i>. For example, <i>U2-12 = 00000011</i> is displayed when terminals M1 and M3 are ON. bit 0 : Terminals M1-M2 bit 1 : Terminals M3-M4 bit 2 : Terminals M5-M6 bit 3 : Not used (normal value of 0). bit 4 : Not used (normal value of 0). bit 5 : Not used (normal value of 0). bit 6 : Not used (normal value of 0). bit 7 : Fault relay MA/MB-MC	No signal output available	-
U2-13 (008C)	DriveOpe Status at Previous Fault	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the operation status of the drive at the latest fault using 1 (ON) and 0 (OFF). The current drive status can be monitored by <i>U1-12 [Drive Status]</i>. For example, <i>U2-13 = 00000001</i> is displayed during run. bit 0 : During run bit 1 : During zero-speed bit 2 : During reverse bit 3 : During fault reset signal input bit 4 : During speed agreement bit 5 : Drive ready bit 6 : During minor fault detection bit 7 : During fault detection	No signal output available	-
U2-14 (008D)	CumOpeTimeatPre Fault	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the cumulative operation time of the drive at the latest fault. The current cumulative operation time can be monitored by <i>U4-01 [Cumulative Ope Time]</i>. Unit: 1 h	No signal output available	-
U2-15 (07E0)	RunSpd SFS atPreFault	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the output frequency after soft start at the latest fault. The current output frequency after soft start can be monitored by <i>U1-16 [Output Freq afterSFS]</i>. Unit: 0.01 Hz	No signal output available	-
U2-16 (07E1)	Motor qCur atPreFault	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the q-axis current of the motor at the latest fault. The current q-axis current of the motor can be monitored by <i>U6-01 [MotorSecondary CurIq]</i>. Unit: 0.1 %	No signal output available	-

No. (Hex.)	Name	Description	MFAO Signal Level	Ref.
U2-17 (07E2)	Motor dCur atPreFalt	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the d-axis current of the motor at the latest fault. The current d-axis current of the motor can be monitored by <i>U6-02 [Motor Excit Cur Id]</i>. Unit: 0.1 %	No signal output available	-
U2-19 (07EC)	RotorDev at PreFault	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the amount of control axis deviation ($\Delta\theta$) at the latest fault. The current amount of control axis deviation ($\Delta\theta$) can be monitored by <i>U6-10 [ContAxisDeviation $\Delta\theta$]</i>. Unit: 0.1 °	No signal output available	-
U2-20 (008E)	HeatsinkTmpatPreFalt	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the heatsink temperature at the latest fault. The current temperature of the heatsink can be monitored by <i>U4-08 [Heatsink Temperature]</i>. Unit: 1 °C	No signal output available	-
U2-21 (1166)	STO Det Sts	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Monitors conditions to detect <i>STPo [Desynchronization Error]</i> faults. The bit for each condition is displayed as ON or OFF. bit 0 : Excessive current bit 1 : Induced voltage deviation bit 2 : d-axis current deviation bit 3 : Motor lock at startup bit 4 : Acceleration stall continue bit 5 : Acceleration stall repeat bit 6 : Not used (normal value of 0). bit 7 : Not used (normal value of 0).	No signal output available	-

◆ U3: Fault History

No. (Hex.)	Name	Description	MFAO Signal Level	Ref.
U3-01 to U3-10 (0090 - 0093) (0804 - 0809)	1st to 10th MostRecent Fault	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the fault history of the 1st to 10th most recent faults. Note: The fault history of <i>U3-01 to U3-04 [1st to 4th MostRecent Fault]</i> is saved to two kinds of registers simultaneously for the MEMOBUS/Modbus communications.	No signal output available	-
U3-11 to U3-20 (0094 - 0097, 080E - 0813)	Elapsed Time 1 to 10	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the cumulative operation time at which the 1st to 10th most recent faults occurred. Unit: 1 h Note: The cumulative operation time of <i>U3-11 to U3-14 [Elapsed Time 1 to 4]</i> is saved to two kinds of registers simultaneously for the MEMOBUS/Modbus communications.	No signal output available	-

◆ U4: Maintenance Monitors

No. (Hex.)	Name	Description	MFAO Signal Level	Ref.
U4-01 (004C)	Cumulative Ope Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the cumulative operation time of the drive. The value for the cumulative operation time counter can be reset in <i>o4-01 [Cumulative Operation TimeSetting]</i>. Select the cumulative operation times from the following with <i>o4-02 [Cumulative Operation Time Select]</i>. - The time when the power supply is energized up to when it is de-energized - The time at which the Run command is turned ON The maximum number displayed is 99999. This value will reset to 0 and start counting again after reaching 99999. Unit: 1 h Note: The MEMOBUS/Modbus communication data is displayed in 10 h units. If data in 1 h units are required, refer to register number 0099H.	No signal output available	-
U4-02 (0075)	Num of Run Commands	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the number of times the Run command is entered. Parameter <i>o4-13 [NumOfRunCommands Counter Initial]</i> can be used to reset this monitor. The maximum number displayed is 65535. This value will reset to 0 and start counting again after reaching 65535. Unit: 1	No signal output available	-
U4-03 (0067)	Cooling Fan Ope Time	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the cumulative operation time of the cooling fan. Parameter <i>o4-03 [CoolingFan OperationTime Setting]</i> can be used to reset this monitor. The maximum number displayed is 99999. This value will reset to 0 and start counting again after reaching 99999. Unit: 1 h Note: The MEMOBUS/Modbus communication data is displayed in 10 h units. If data in 1 h units are required, refer to register number 009BH.	No signal output available	-
U4-04 (007E)	Cool Fan Maintenance	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the cumulative operation time of the cooling fan as a percentage of the replacement life of the cooling fan. Parameter <i>o4-03 [CoolingFan OperationTime Setting]</i> can be used to reset this monitor. Unit: 1% Note: Replace the cooling fan when this monitor reaches 90%.	No signal output available	-
U4-05 (007C)	CapacitorMaintenance	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the usage time of the electrolytic capacitor for the main and control circuits as a percentage of the replacement life of the electrolytic capacitor. Parameter <i>o4-05 [Capacitor Maintenance Setting]</i> can be used to reset this monitor. Unit: 1% Note: Replace the electrolytic capacitor when this monitor reaches 90%.	No signal output available	-
U4-06 (07D6)	SChgBypassRelay Maint	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the usage time of the soft charge bypass relay as a percentage of the replacement life of the soft charge bypass relay. Parameter <i>o4-07 [DCBusPreChargeRelayMainteSetting]</i> can be used to reset this monitor. Unit: 1% Note: Replace the drive when this monitor reaches 90%.	No signal output available	-

No. (Hex.)	Name	Description	MFAO Signal Level	Ref.
U4-07 (07D7)	IGBT Maintenance	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the usage time of the IGBT as a percentage of the replacement life of the IGBT. Parameter <i>o4-09 [IGBT Maintenance Setting]</i> can be used to reset this monitor. Unit: 1% Note: Replace the drive when this monitor reaches 90%.	No signal output available	-
U4-08 (0068)	Heatsink Temperature	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the heatsink temperature of the drive. Unit: 1 °C	10 V: 100 °C	-
U4-09 (005E)	LED Check	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Lights all segments of the LEDs on the keypad and LED Status Ring light to verify that the display is working properly. Note: The internal status of the drive cannot be determined when the board for the LED Status Ring is damaged. Do not rely on the LED Status Ring alone to determine the status of the drive and motors. - Set <i>o2-24 = 0 [LED Light Function Selection = Both Enable]</i>. - With <i>U4-09</i> displayed, press . All LEDs on the keypad and LED Status Ring will glow. Note: When Safety input 2 CH is open (STo), READY will flash.	No signal output available	-
U4-10 (005C)	kWh, Lower 4 Digits	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the lower 4 digits of the watt hour value for the drive. Unit: 1 kWh Note: The watt hour is displayed in 9 digits. Parameter <i>U4-11 [kWh, Upper 5 Digits]</i> displays the upper 5 digits and <i>U4-10</i> displays the lower 4 digits. Example for 12345678.9 kWh: <i>U4-10</i>: 678.9 kWh <i>U4-11</i>: 12345 MWh	No signal output available	-
U4-11 (005D)	kWh, Upper 5 Digits	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the upper 5 digits of the watt hour value for the drive. Unit: 1 MWh Note: The watt hour is displayed in 9 digits. Parameter <i>U4-11</i> displays the upper 5 digits and <i>U4-10 [kWh, Lower 4 Digits]</i> displays the lower 4 digits. Example for 12345678.9 kWh: <i>U4-10</i>: 678.9 kWh <i>U4-11</i>: 12345 MWh	No signal output available	-
U4-13 (07CF)	Peak Hold Current	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the hold value of the peak value (rms) for the drive output current. The drive output frequency at the time the output current is held can be monitored by <i>U4-14 [PeakHold Output Freq]</i>. The peak hold current will be cleared at the next startup and restart of the power supply. The drive keeps the value that was under hold during baseblock (during stop). The value of <i>U4-13</i> appears in amperes (A) on the keypad. When viewing via MEMOBUS/Modbus communications, the current is "8192 = drive rated current (A)." Current can be calculated from the monitor value present at MEMOBUS/Modbus communications using "Numerals being displayed / 8192 × drive rated current (A)." Unit: Determined by the drive model. - Models 2004 to 2042, 4002 to 4023: 0.01 A - Models 2056 to 2415, 4031 to 4726: 0.1 A	No signal output available	-

No. (Hex.)	Name	Description	MFAO Signal Level	Ref.
U4-14 (07D0)	PeakHold Output Freq	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the output frequency at which the peak value (rms) of the drive output current is held. The peak hold current can be monitored by U4-13 [Peak Hold Current]. The peak hold output frequency will be cleared at the next startup and restart of the power supply. The drive keeps the value that was under hold during baseblock (during stop). Unit: 0.01 Hz	No signal output available	-
U4-16 (07D8)	MotorOLEstimate (oL1)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the integrated value of oL1 [Motor Overload] as a percentage of oL1 detection level. Unit: 0.1%	10 V: 100%	-
U4-18 (07DA)	Freq Ref Source Sel	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the frequency reference source that is currently selected. The frequency reference source is presented as XY-nn in the following manner. X: External Reference 1/2 Selection [H1-xx = 2] selection status 1: b1-01 [Frequency Reference Selection 1] 2: b1-15 [Frequency Reference Selection 2] Y-nn: Frequency reference source 0-01: Keypad (d1-01 [Reference 1]) 1-00: Analog input (unassigned) 1-01: MFAI terminal A1 1-02: MFAI terminal A2 1-03: MFAI terminal A3 2-02 to 2-17: Multi-step speed reference (d1-02 to d1-17 [Reference 2 to 16, Jog Reference]) 3-01: MEMOBUS/Modbus communications 4-01: Communication option card 5-01: Pulse train input 7-01: DriveWorksEZ 9-01: Up/Down command	No signal output available	-
U4-19 (07DB)	Freq Ref from Comm	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the frequency reference to the drive from the MEMOBUS/Modbus communications as a decimal number. Unit: 0.01%	No signal output available	-
U4-20 (07DC)	Option FreqReference	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the frequency reference to the drive from the communication option as a decimal number.	No signal output available	-

No. (Hex.)	Name	Description	MFAO Signal Level	Ref.
U4-21 (07DD)	RunCom Source Select	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the Run command source that is currently selected. The Run command source is presented as XY-nn in the following manner. X: <i>External Reference 1/2 Selection [H1-xx = 2]</i> selection status 1: <i>b1-02 [Run Command Selection 1]</i> 2: <i>b1-16 [Run Command Selection 2]</i> Y: Run command source 0: Keypad 1: Control circuit terminal 3: MEMOBUS/Modbus communications 4: Communication option card 7: DriveWorksEZ nn: Run command limit status data 00: No limit status. 01: The Run command was left ON when the drive stopped in the Programming Mode. 02: The Run command was left ON when switching from LOCAL Mode to REMOTE Mode. 03: The Run command is in standby after the drive was energized until the soft charge bypass contactor turns ON. Note: When the soft charge bypass contactor does not turn ON after 10 s have passed, <i>Uv1 [DC Bus Undervoltage]</i> or <i>Uv [Undervoltage]</i> is detected. 04: Restart after run stop is prohibited. 05: Fast stop has been executed using the MFDI terminal. Or, the motor has ramped to stop by pressing the STOP key on the keypad. 06: <i>b1-17 = 0 [Run Command at Power Up = Disregard existing RUN command]</i> is set. 07: During baseblock while coast to stop with timer. 08: Frequency reference is below <i>E1-09 [Minimum Output Frequency]</i> during baseblock. 09: Waiting for the Enter command from PLC.	No signal output available	-
U4-22 (07DE)	MEMOBUS Comm Ref	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the operation signal (register No. 0001H) from MEMOBUS/Modbus communications to the drive as a four-digit hexadecimal number (zero suppress). The operation signal is displayed in the following way. bit 0 : Forward run/Stop bit 1 : Reverse run/Stop bit 2 : External fault bit 3 : Fault Reset bit 4 : Multi-function input 1 bit 5 : Multi-function input 2 bit 6 : Multi-function input 3 bit 7 : Multi-function input 4 bit 8 : Multi-function input 5 bit 9 : Multi-function input 6 bit A : Multi-function input 7 bit B : Multi-function input 8 bit C : Not used (normal value of 0). bit D : Not used (normal value of 0). bit E : Not used (normal value of 0). bit F : Not used (normal value of 0).	No signal output available	-

No. (Hex.)	Name	Description	MFAO Signal Level	Ref.
U4-23 (07DF)	Comm Option Card Ref	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the operation signal (register No. 0001H) from communication option to the drive as a four-digit hexadecimal number. The operation signal is displayed in the following way. bit 0 : Forward run/Stop bit 1 : Reverse run/Stop bit 2 : External fault bit 3 : Fault Reset bit 4 : Multi-function input 1 bit 5 : Multi-function input 2 bit 6 : Multi-function input 3 bit 7 : Multi-function input 4 bit 8 : Multi-function input 5 bit 9 : Multi-function input 6 bit A : Multi-function input 7 bit B : Multi-function input 8 bit C : Not used (normal value of 0). bit D : Not used (normal value of 0). bit E : Not used (normal value of 0). bit F : Not used (normal value of 0).	No signal output available	-
U4-24 (07E6)	No of Travels(L)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the lower 4 digits of the drive run count. Note: The drive run count appears as an 8-digit number. The upper 4 digits of U4-25 [No of Travels(H)] and the lower 4 digits of U4-24 appears.	No signal output available	-
U4-25 (07E7)	No of Travels(H)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the upper 4 digits of the drive run count. Note: The drive run count appears as an 8-digit number. The upper 4 digits of U4-25 and the lower 4 digits of U4-24 [No of Travels(L)] appears.	No signal output available	-

◆ U5: PID Monitors

No. (Hex.)	Name	Description	MFAO Signal Level	Ref.
U5-01 (0057)	PID Feedback	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the PID control feedback value. Display units are set by b5-20 [PID Setpoint Scaling]. Unit: 0.01%	10 V: Maximum frequency (-10 V to +10 V)	-
U5-02 (0063)	PID Input	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the deviation between the PID setpoint and PID feedback (the amount of PID input) as a percentage of the maximum output frequency. Unit: 0.01%	10 V: Maximum frequency (-10 V to +10 V)	-
U5-03 (0064)	PID Output	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the PID control output as a percentage of the maximum output frequency. Unit: 0.01%	10 V: Maximum frequency (-10 V to +10 V)	-
U5-04 (0065)	PID Setpoint	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the PID setpoint. Display units are set by b5-20 [PID Setpoint Scaling]. Unit: 0.01%	10 V: Maximum frequency (-10 V to +10 V)	-

No. (Hex.)	Name	Description	MFAO Signal Level	Ref.
U5-05 (07D2)	PID DifferentialFdbk	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the PID differential feedback value as a percentage of the maximum output frequency. This parameter is available when <i>H3-02, H3-10, or H3-06 = 16</i> [<i>MFAI Function Select = Differential PID Feedback</i>] is set. Unit: 0.01%	10 V: Maximum frequency (-10 V to +10 V)	-
U5-06 (07D3)	PID AdjustedFeedback	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the final feedback value after subtracting <i>U5-05</i> [<i>PID DifferentialFdbk</i>] from <i>U5-01</i> [<i>PID Feedback</i>]. Unit: 0.01% Note: When <i>H3-02, H3-10, H3-16 = 16</i> [<i>MFAI = Differential PID Feedback</i>] is not set, <i>U5-01</i> [<i>PID Feedback</i>] and <i>U5-06</i> will be the same value.	10 V: Maximum frequency (-10 V to +10 V)	-
U5-21 (0872)	AutoCalEnSav Coef Ki	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the energy-saving coefficient Ki value for PM. Unit: 0.01	No signal output available	-
U5-22 (0873)	AutoCalEnSav Coef Kt	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the energy-saving coefficient Kt value for PM. Unit: 0.01	No signal output available	-
U5-99 (1599)	PID Setpoint Command	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the PID setpoint command. Display units are set by <i>b5-20</i> [<i>PID Setpoint Scaling</i>]. Unit: 0.01%	10 V: Maximum frequency (-10 V to +10 V)	-

◆ U6: Operation Status Monitors

No. (Hex.)	Name	Description	MFAO Signal Level	Ref.
U6-01 (0051)	MotorSecondary CurIq	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the value calculated for the motor secondary current as a percentage of the motor rated secondary current. (q axis) Unit: 0.1%	10 V: Motor secondary rated current (-10 V to +10 V)	-
U6-02 (0052)	Motor Excit Cur Id	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the value calculated for the motor excitation current as a percentage of the motor rated secondary current. (d axis) Unit: 0.1%	10 V: Motor secondary rated current (-10 V to +10 V)	-
U6-03 (0054)	ASR Input	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the ASR input value as a percentage of the maximum frequency. Unit: 0.01%	10 V: Maximum frequency (-10 V to +10 V)	-
U6-04 (0055)	ASR Output	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the ASR output value as a percentage of the motor rated secondary current. Unit: 0.01%	10 V: Motor secondary rated current (-10 V to +10 V)	-
U6-05 (0059)	OutVolt Reference Vq	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the drive internal voltage reference for motor secondary current control. (q axis) Unit: 0.1 V	200 V class: 10 V = 200 Vrms 400 V class: 10 V = 400 Vrms (-10 V to +10 V)	-
U6-06 (005A)	OutVolt Reference Vd	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the drive internal voltage reference for motor excitation current control. (d axis) Unit: 0.1 V	200 V class: 10 V = 200 Vrms 400 V class: 10 V = 400 Vrms (-10 V to +10 V)	-

No. (Hex.)	Name	Description	MFAO Signal Level	Ref.
U6-07 (005F)	q-Axis ACR Output	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the output value for current control relative to motor secondary current. (q axis) Unit: 0.1%	200 V class: 10 V = 200 Vrms 400 V class: 10 V = 400 Vrms (-10 V to +10 V)	-
U6-08 (0060)	d-Axis ACR Output	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the output value for current control relative to motor excitation current. (d axis) Unit: 0.1%	200 V class: 10 V = 200 Vrms 400 V class: 10 V = 400 Vrms (-10 V to +10 V)	-
U6-09 (07C0)	AdvPhase Compens $\Delta\theta$	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the data on forward phase compensation for the calculation results of the amount of control axis deviation. Unit: 1 °	10 V: 180° (-10 V to +10 V)	-
U6-10 (07C1)	ContAxisDeviation $\Delta\theta$	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the amount of deviation between the $\gamma\delta$ -Axis used for motor control and the actual dq-Axis. Unit: 0.1 °	10 V: 180° (-10 V to +10 V)	-
U6-13 (07CA)	FluxPosDetect sensor	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the value of the flux position detection. Unit: 0.1 °	10 V: 180° (-10 V to +10 V)	-
U6-14 (07CB)	FluxPosEstimOb server	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the value of the flux position estimation. Unit: 0.1 °	10 V: 180° (-10 V to +10 V)	-
U6-17 (07D1)	Total Current Value for Detection of Direction of Motor Rotation	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the accumulated time of direction of motor rotation detections for Speed Estimation Speed Searches. This is used to adjust the value of <i>b3-26 [Direction Determining Level]</i> . Note: Upper and lower limits are set to values of ± 32767 .	No signal output available	-
U6-18 (07CD)	SpdDetectPG1 Counter	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the number of pulses for speed detection (PG1). Unit: 1 pulse	10 V: 65536	-
U6-19 (07E5)	SpdDetectPG2 Counter	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the number of pulses for speed detection (PG2). Unit: 1 pulse	10 V: 65536	-
U6-20 (07D4)	Frequency Ref Bias (Up/Down 2)	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the bias value used to adjust the frequency reference. Unit: 0.1%	10 V: Maximum Frequency	-
U6-21 (07D5)	Offset Frequency	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the total value of <i>d7-01 to d7-03 [Offset Frequency 1 to 3]</i> selected with <i>Offset frequency 1 to 3 [H1-xx = 44 to 46]</i> . Unit: 0.1%	10 V: Maximum Frequency	-
U6-22 (0062)	ZeroServoPulse Move	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays how far the rotor has moved from its last position when Zero Servo is available. Displays a number that is 4 times the number of PG pulses. Unit: 1 pulse	10 V: Number of pulses per revolution (-10 V to +10 V)	-
U6-25 (006B)	Fdbk Control Output	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the primary delay filter input value of the ASR (speed control loop). Unit: 0.01%	10 V: Motor secondary rated current (-10 V to +10 V)	-
U6-26 (006C)	Feed Fwd Cont Output	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the Feed Forward control output. Unit: 0.01%	10 V: Motor secondary rated current (-10 V to +10 V)	-

No. (Hex.)	Name	Description	MFAO Signal Level	Ref.
U6-27 (006D)	FF Estimate SPD	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the feed forward estimated speed. Unit: 0.01%	10 V = Maximum frequency (-10 V to +10 V)	-
U6-36 (0720)	Comm Errors-HOST	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Counts the number of inter-CPU communication errors. This count is reset to 0 when the power to the drive is turned off.	No signal output available	-
U6-37 (0721)	Comm Errors-Sensor	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Counts the number of inter-CPU communication errors. This count is reset to 0 when the power to the drive is turned off.	No signal output available	-
U6-48 (072E)	ASIC Comm Errors	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Counts the number of inter-ASIC communication errors detected by the ASIC. This count is reset to 0 when the power to the drive is turned off.	No signal output available	-
U6-57 (07C4)	PoleDis IdDifVal	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the deviation from the integrated current when determining the polarity. Unit: 1 Note: If the deviation from the integrated current is lower than 819, then increase the value set to <i>n8-84</i> [<i>InitPolarityEstimationTimeoutCur</i>]. <i>U6-57 = 8192</i> is equivalent to the motor rated current.	No signal output available	-
U6-80 to U6-83 (07B0 to 07B3)	OPT IP ADR1 to 4	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the currently available local IP Address. • <i>U6-80</i> : 1st octet • <i>U6-81</i> : 2nd octet • <i>U6-82</i> : 3rd octet • <i>U6-83</i> : 4th octet	No signal output available	-
U6-84 to U6-87 (07B4 to 07B7)	Online Subnets 1 to 4	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the currently available subnet mask. • <i>U6-84</i> : 1st octet • <i>U6-85</i> : 2nd octet • <i>U6-86</i> : 3rd octet • <i>U6-87</i> : 4th octet	No signal output available	-
U6-88 to U6-91 (07B8 to 07F1)	Online Gateways 1 to 4	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the currently available gateway address. • <i>U6-88</i> : 1st octet • <i>U6-89</i> : 2nd octet • <i>U6-90</i> : 3rd octet • <i>U6-91</i> : 4th octet	No signal output available	-
U6-92 (07F2)	Online Speed	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the currently available communications speed. 10: 10 Mbps 100: 100 Mbps	No signal output available	-
U6-93 (07F3)	Online Duplex	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the currently available Duplex setting.	No signal output available	-
U6-98 (07F8)	First Fault	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the contents of previous fault from communication options (DeviceNet, Modbus TCP/IP, EtherNet/IP).	No signal output available	-
U6-99 (07F9)	Current Fault	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the contents of current faults from communication options (DeviceNet, Modbus TCP/IP, EtherNet/IP).	No signal output available	-

◆ U8: DriveWorksEZ Monitors

No. (Hex.)	Name	Description	MFAO signal level	Ref.
U8-01 to U8-10 (1950 - 1959)	DriveWorksEZ User Monitors 1 to 10	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays DWEZ Monitors 1 to 10. Unit: 0.01%	10 V = 100%	-
U8-11 to U8-13 (195A - 195C)	DWEZ Versions 1 to 3	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays DriveWorks EZ Versions 1 to 3.	No signal output available	-
U8-18 (1961)	DriveWorksEZ Platform Ver	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays the DriveWorksEZ platform version.	No signal output available	-
U8-21 to U8-25 (1964 - 1968)	DriveWorksEZ User Monitors 21 to 25	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays DriveWorksEZ User Monitors 21 to 25. 0.01%	10 V = 100%	-
U8-31 to U8-40 (196E - 1977)	DriveWorksEZ User Monitors 31 to 40	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays DriveWorksEZ User Monitors 31 to 40. 0.01%	10 V = 100%	-
U8-51 to U8-55 (1982 - 1986)	DriveWorksEZ User Monitors 51 to 55	V/f CL-V/f OLV CLV AOLV OLV/PM AOLV/PM CLV/PM EZOLV Displays DriveWorksEZ User Monitors 51 to 55. 0.01%	10 V = 100%	-